



Hik Device Gateway API

Developer Guide

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Chapter 1 Overview

1.1 Introduction

The Hik Device Gateway API based on a text protocol in RESTful style provides a uniform integration scheme for Hikvision products, which makes up for the integration difficulties caused by multiple open protocols, improves integration efficiency and the compatibility of Hikvision devices (e.g., body cameras, portable speed domes, mobile devices, alarm devices, access control devices, etc.) and the third-party platform. This manual mainly introduces the communication and security mechanism, gateway operations, operations (e.g., live view, playback, video search, etc.) of devices that added to Hik Device Gateway, alarm/event subscription, access control, and so on.



Note

REST (REpresentational State Transfer) is a protocol design method (named as RESTful) which abstracts all information as the resources. The abstracted resources are marked by the uniform identifies, i.e., URI (Uniform Resource Identifiers), for simple and extendable management.

1.2 Product Scopes

Table 1-1 Supported Product Type and Model

Product Name	Model
Body Camera	DS-MH2211
	DS-MH2311
	DS-MCW405
Mobile DVR	DS-MP7508/GW(1T)
	DS-MP7508/GW
	DS-MP5604
NVR	DS-7608NI-I2
Mobile NVR	DS-MP7608HN
Camera	DS-2CD2125FWD-IS
	DS-2XS6D22E-LGL
Portable Speed Dome	DS-MH6171I
Dome Camera	DS-2DF7225IX-AEL
4G Solar Camera	DS-2XS6A25G0-I/CH20S40

Product Name	Model
ISUP5.0 Facial Recognition Terminal	DS-K1T642MFW
	DS-K1T671TMFW
	DS-K5604A-ZV
	DS-K5671-3XF/ZU
ISUP5.0 Access Control Terminal	DS-K1T8003EF
	DS-K1A8503EF-B
	DS-K1T804EF

1.3 Update History

Summary of Changes in Version 1.7.0_February, 2024

1. Access devices support the stream getting API: [/ISAPI/System/streamMedia?format=json&devIndex=<uuid>](#) .
2. Extended the time synchronization parameters message [JSON_Time](#) (related URL: [/ISAPI/System/time?format=json&devIndex=<uuid>](#)):
added the NTP mode to field **timeMode** (time synchronization mode).
3. Added the API for importing face pictures to the face picture library: [/ISAPI/Intelligent/FDLib/pictureUpload?format=json&devIndex=<uuid>](#) .
4. Added the event of uploading people counting statistics:
[JSON_EventNotificationAlert_peopleCounting](#) .
5. Added the event of device event transmission: [JSON_EventNotificationAlert_devXmlEvent](#) .

Summary of Changes in Version 1.6.1_January, 2023

1. Extended the message about search conditions of added devices [JSON_SearchDescription](#) (related URL: POST [/ISAPI/ContentMgmt/DeviceMgmt/deviceList?format=json](#)):
added two fields **EhomeParams** (search by device number) and **devIndex** (search by device UUID).
2. Extended the message about device search results [JSON_SearchResult](#) (related URL: POST [/ISAPI/ContentMgmt/DeviceMgmt/deviceList?format=json](#)):
added three fields **devVersion** (device version), **devSerial** (device serial number), and **offlineHint** (offline reason).
3. Extended message about results of search for history access control events [JSON_AcsEvent](#) (related URL: POST [/ISAPI/AccessControl/AcsEvent?format=json&devIndex=<uuid>](#)):
added one field **deviceSerial** (device serial number).
4. Extended detailed messages about different types of events (see [Event Types](#) for details):

added one field **deviceSerial** (device serial number).

5. Added error codes (from 0X4000A000 to 0X4000A022, 0x30006010, 0x40001275, and 0x400011E9), see **Status Codes** for details.

Summary of Changes in Version 1.6.0_July, 2022

New document.

Chapter 2 Security Authentication

When communicating via Hik Device Gateway SDK, the digest of the session must be authenticated.

Note

- The authentication must be based on *HTTP Authentication: Basic and Digest Access Authentication*, see <https://tools.ietf.org/html/rfc2617> for details.
 - The request session must contain authentication information, otherwise, device will return 401 error code.
 - For login authentication, the available URL is GET .
-

The message digest, which contains user name, password, specific nonce value, HTTP or RTSP operation methods, and request URL, is generated by the MD5 algorithm, see the calculation rules below.

qop=Undefined

Digest=MD5(MD5(A1):<nonce>:MD5(A2))

qop="auth:"

Digest=MD5(MD5(A1):<nonce>:<nc>:<cnonce>:<qop>:MD5(A2))

qop="auth-int:"

Digest=MD5(MD5(A1):<nonce>:<nc>:<cnonce>:<qop>:MD5(A2))

Note

- The **qop** is a value for determining whether the authentication is required.
 - A1 and A2 are two data blocks required for digest calculation.
A1: Data block about security, which contains user name, password, security domain, random number, and so on. If the digest calculation algorithm is MD5, A1=<user>:<realm>:<password>; if the algorithm is MD5-sess, A1=MD5(<user>:<realm>:<password>):<nonce>:<cnonce>.
A2: Data block about message, such as URL, repeated requests, message body, and so on, it helps to prevent repeated, and realize the resource/message tamper-proof. If the **qop** is not defined or it is "auth:", A2=<request-method>:<uri-directive-value>; if the **qop** is "auth-int:", A2=<request-method>:<uri-directive-value>:MD5(<request-entity-body>).
 - The **nonce** is the random number generated by service, the following generation formula is suggested: nonce = BASE64(time-stamp MD5(time-stamp ":" ETag ":" private-key)). The **time-stamp** in the formula is the time stamp generated by service or the unique serial No.; the **ETag** is the value of HTTP ETag header in the request message; the **private-key** is the data that only known by service.
-

If authentication fail, the device will return the **JSON ResponseStatus AuthenticationFailed** message, and the remaining authentication attempts will also be returned. If the remaining attempts is 0, the user will be locked at the next authentication attempt.

Chapter 3 API Description

3.1 Operation Method

The resource operation methods of Device Gateway protocol are same as those of HTTP (Hyper Text Transport Protocol) and RTSP (Real Time Streaming Protocol), see details in the following table.

**Note**

The RTSP operation methods are mainly used to get the real-time stream for live view, two-way audio, and playback in this manual. For details about HTTP, HTTPS, and RTSP, please refer to <https://tools.ietf.org/html/rfc2612> , <https://tools.ietf.org/html/rfc2818> and <https://tools.ietf.org/html/rfc2326> .

Table 3-1 HTTP Operation Method

Method	Description
POST	Create resources. This method is only available for adding resource that does not exist before.
GET	Retrieve resources. This method cannot change the system status, only return data as the response to the requester.
PUT	Update resources. This method is usually for update the resource that already exists, but it can also be used to create the resource if the specific resource does not exist.
DELETE	Delete resources.

Table 3-2 RTSP Operation Method

Method	Description
OPTIONS	Get the supported RTSP operation methods. See the request and response message format below when interacting between client software and server. <pre>OPTIONS %s RTSP/1.0\r\n //Request URL CSeq:%u\r\n //Command No. User-Agent:%s\r\n //Client software name \r\n /*Succeeded*/ RTSP/1.0 200 OK\r\n //Succeeded CSeq: %u\r\n //Command No. Public: %s\r\n //Supported operation methods Date:%s\r\n //Date and time</pre>

Method	Description
	<pre> \r\n /*Failed*/ RTSP/1.0 4XX/5XX %s\r\n //Failed CSeq: %u\r\n //Command No. Date: %s\r\n //Date and time \r\n </pre>
DESCRIBE	Transfer basic information by SDP (Session Description Protocol, see https://tools.ietf.org/html/rfc2327) files, such as URL with SETUP command and so on. See the request and response message format below when interacting between client software and server. <pre> DESCRIBE %s RTSP/1.0\r\n //URL CSeq: %u\r\n //Command No. Accept: application/sdp\r\n //The SDP description is accepted Authorization: %s\r\n //Authentication information User-Agent: %s\r\n //Client software name \r\n /*Succeeded*/ RTSP/1.0 200 OK\r\n //Succeeded or failed CSeq: %u\r\n //Command No. Content-Type: application/sdp\r\n //The SDP description exists behind the command Content-Length: %d\r\n //The length of contents behind the command \r\n [content] //SDP description /*Failed*/ RTSP/1.0 4XX/5XX %s\r\n //Failed CSeq: %u\r\n //Command No. \r\n </pre>
SETUP	Interact the session information, such as transmission mode, port number, and so on. See the request and response message format below when interacting between client software and server. <pre> SETUP %s RTSP/1.0\r\n //URL CSeq: %u\r\n //Command No. Authorization: %s\r\n //Authentication information Session: %s\r\n //Session ID is only returned at the even number of times Transport: %s\r\n //Transmission protocol User-Agent: %s\r\n //Client software name \r\n /*Succeeded*/ RTSP/1.0 200 OK \r\n //Succeeded CSeq: %u\r\n //Command No. Session: %s\r\n //Session ID </pre>

Method	Description
	Transport: s% //Transmission method /*Failed*/ RTSP/1.0 4XX/5XX %s\r\n //Failed CSeq: %u\r\n //Command No. \r\n
PLAY	Start the stream transmission. See the request and response message format below when interacting between client software and server. PLAY %s RTSP/1.0\r\n //URL CSeq: %u\r\n //Command No. Authorization: %s\r\n //Authentication information Session: %s\r\n //Session ID Range: npt=%f-%f\r\n //Determine the play range User-Agent: %s\r\n //Client software name \r\n /*Succeeded*/ RTSP/1.0 200 OK \r\n //Succeeded CSeq: %u\r\n Session: %s\r\n RTP-Info: %s /*Failed*/ RTSP/1.0 4XX/5XX %s\r\n //Failed CSeq: %u\r\n //Command No. Session: %s\r\n \r\n
PAUSE	Pause the stream transmission.
TEARDOWN	Stop the stream transmission. See the request and response message format below when interacting between client software and server. TEARDOWN %s RTSP/1.0\r\n //URL CSeq: %u\r\n //Command No. Authorization: %s\r\n //Authentication information Session: %s\r\n //Session ID User-Agent: %s\r\n //Client software name \r\n /*Succeeded*/ RTSP/1.0 200 OK \r\n //Succeeded CSeq: %u\r\n Session: %s\r\n \r\n /*Failed*/

Method	Description
	RTSP/1.0 4XX/5XX %s\r\n //Failed CSeq: %u\r\n //Command No. Session:%s\r\n\r\n

3.2 URL Format

URL (Uniform Resource Locator) is a further class of URIs, it can identify a resource and locate the resource by describing its primary access mechanism.

The format of URL is defined as the follows: {protocol}://{host}:{port}{abs_path}?{query}.

protocol

Protocol types, i.e., HTTP (version 1.1), HTTPS, RTSP (version 1.0).

host

Host name, IP address, or the FQDN (Fully Qualified Domain Name) of network devices.

port

Port number of host service for listening the connection status of TCP (Transmission Control Protocol, see <https://tools.ietf.org/html/rfc793>) or UDP (User Datagram Protocol, see <https://tools.ietf.org/html/rfc768>). If this field is not configured, the default port number will be adopted. For HTTP, the default port number is 80, for HTTPS, the default port number is 443, and for RTSP, the default port number is 554.

abs_path

Resource URI: /ServiceName/ResourceType/resource. Here, the **ServiceName** is ISAPI; the **ResourceType** is predefined with upper camel case according to different functions; the **resource** is defined with lower camel case and can be extended in actual applications. E.g., /ISAPI/System/deviceInfo/deviceName.

query

Strings for describing resources information, including related parameters. The parameter names and values must be listed as the following format in this field: ?p1=v1&p2=v2&...&pn=vn.



Note

- To locate the connected device, when operating lower-level device via the URL, the **query** field should be filled as ?devIndex=uuid&p1=v1&p2=v2&...&pn=vn. The uuid (or guid) is a 32-byte (128 bits) random number, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
- For message in JSON format, the **query** field should be filled as ?format=json&p1=v1&p2=v2&...&pn=vn. For details about message format, refer to the next

section below. E.g.,[http://10.17.132.22/ISAPI/System/time?
format=json&devIndex=550e8400e29b41d4a716446655440000](http://10.17.132.22/ISAPI/System/time?format=json&devIndex=550e8400e29b41d4a716446655440000).

3.3 Message Format

The request and response messages generated among the interaction of Hik Device Gateway, devices, and system are data in JSON format.

Note

The message format here is only available for URLs based on HTTP.

- The leaf node (without any sub node) in the message is named by lower camel case, while the non-leaf node in the message is named by upper camel case.
 - To communicate by the messages in JSON format, the devices must support the public specifications in <http://www.ecma-international.org/publications/files/ECMA-ST/ECMA-404.pdf> and HTTP with version 1.1.
-

Note

JSON is a lightweight data format which is a subset of JavaScript language and is small, fast, and easy to be parsed.

- Generally, for configuration information, the **Content-Type** of message is "application/json", see the example below:

```
//Request message
GET /ISAPI/System/deviceInfo?format=json HTTP/1.1
...

//Response message
HTTP/1.1 200 OK
...
Content-Type: application/json
...
"DeviceStatus":""
...
```

For data (e.g., firmware, configuration files), the **Content-Type** of message is "application/octet-stream", see the example below:

```
//Request message
POST /ISAPI/System/streamMedia?format=json&devIndex=550e8400e29b41d4a716446655440000 HTTP/1.1
...
Content-Type: application/octet-stream
...
[proprietary configuration file data content]

//Response message
HTTP/1.1 200 OK
...
```

```
Content-Type: application/json
```

```
...
```

```
"ResponseStatus":""
```

```
...
```

3.4 Others

Time Format

The time format in Hik Device Gateway APIs adopts ISO8601 standard (see details in <http://www.w3.org/TR/NOTE-datetime-970915>), that is, YYYY-MM-DDThh:mm:ss.sTZD (e.g., 2017-08-16T20:17:06+08:00, 2017-08-16T20:09:06Z).

Error Processing

During the integration applications of Hik Device Gateway, when the error of URL based on HTTP occurs, the ***JSON_ResponseStatus*** message which contains error code will be returned. If the error of URL based on RTSP occurs, the corresponding status code will directly be returned, for details, refer to <https://tools.ietf.org/html/rfc2326> .



Note

For batch operations, if some operations fail, both the ***JSON_ResponseStatus*** message and failure details message (see details in actual batch operations, such as adding devices or deleting devices in a batch) will be returned.

Chapter 4 Typical Applications

4.1 Add Devices to Device Gateway

You can add a device or multiple devices to Hik Device Gateway by the device information (i.e., device ID and device key). After adding devices, you can also get the added device list.

Steps

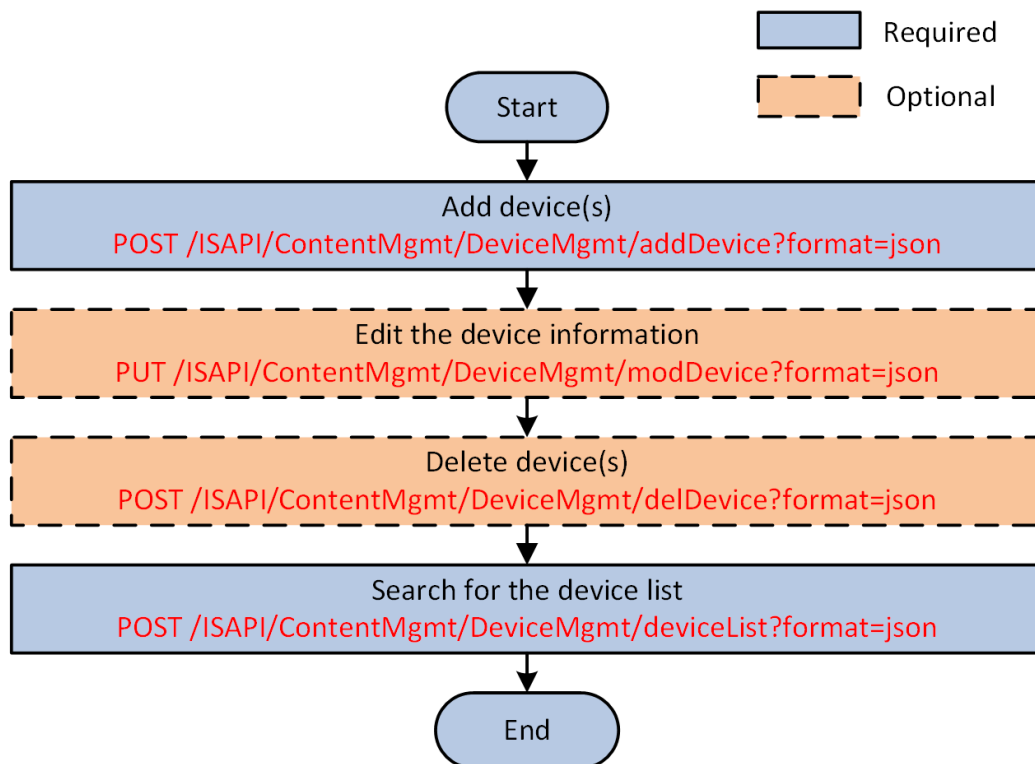


Figure 4-1 API Calling Flow of Adding Devices to Device Gateway

1. Call [/ISAPI/ContentMgmt/DeviceMgmt/addDevice?format=json](#) by POST method to add device(s) to Device Gateway.
2. **Optional:** Call [/ISAPI/ContentMgmt/DeviceMgmt/modDevice?format=json](#) by PUT method to edit the device information.
3. **Optional:** Call [/ISAPI/ContentMgmt/DeviceMgmt/delDevice?format=json](#) by POST to batch delete devices from Device Gateway.
4. Call [/ISAPI/ContentMgmt/DeviceMgmt/deviceList?format=json](#) by POST method to get the list of added devices.

4.2 Video and Audio

To realize the live view, playback, or two-way audio, you should get the stream URL from the device by sending request via Hik Device Gateway, and then start the live view, playback, or two-way audio according to the obtained stream URL by RTSP operation methods. You can also control PTZ, record videos manually, and download video files. Before starting live view, you can also configure the channel, OSD, and image parameters for devices.

4.2.1 Start Live View

The live view function helps to remotely get the real-time video or audio of monitoring area and know the status of human, vehicle, object, and so on.

Steps

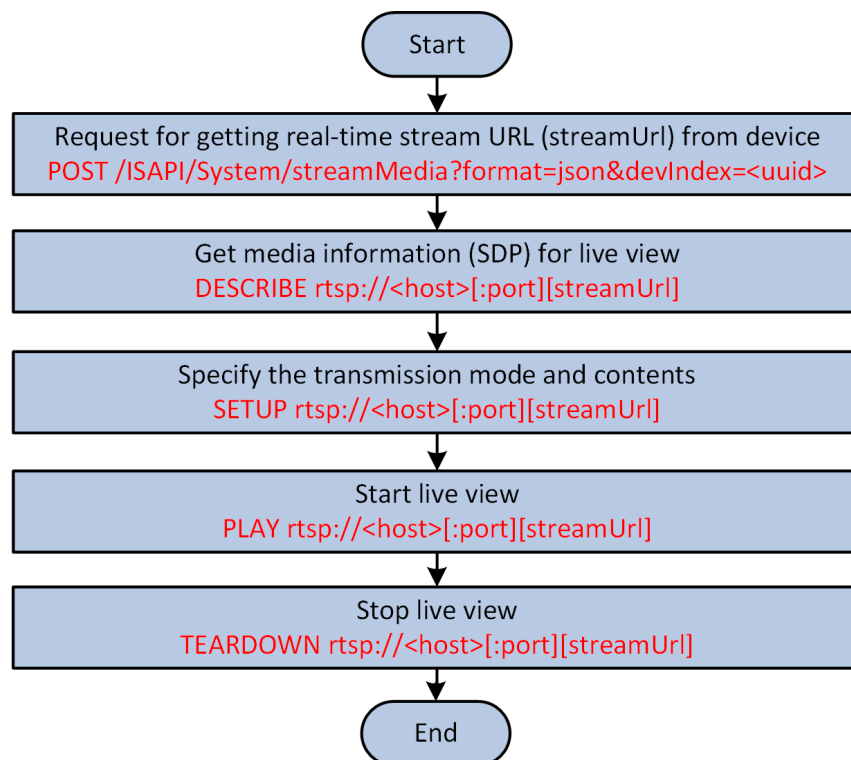


Figure 4-2 Programming Flow of Starting Live View

1. Call `/ISAPI/System/streamMedia?format=json&devIndex=<uuid>` by POST method to request for getting real-time stream URL from the device.
A URL which contains real-time stream data is returned to Hik Device Gateway for live view.
2. Call `rtsp://{ipAddress}:{port}{streamUrl}` by DESCRIBE method of RTSP to get media information (SDP) for live view.

3. Call ***rtsp://{ipAddress}:{port}{streamUrl}*** by SETUP method of RTSP to specify the transmission mode and contents.
4. Call ***rtsp://{ipAddress}:{port}{streamUrl}*** by PLAY method of RTSP to start live view.
5. Call ***rtsp://{ipAddress}:{port}{streamUrl}*** by TEARDOWN method of RTSP to stop live view.

4.2.2 Start Playback

You can play back the videos remotely by file or by time.

Before You Start

Make sure there are video files stored in the device.

Steps



Note

Only devices added via ISUP5.0 support playing back videos by time.

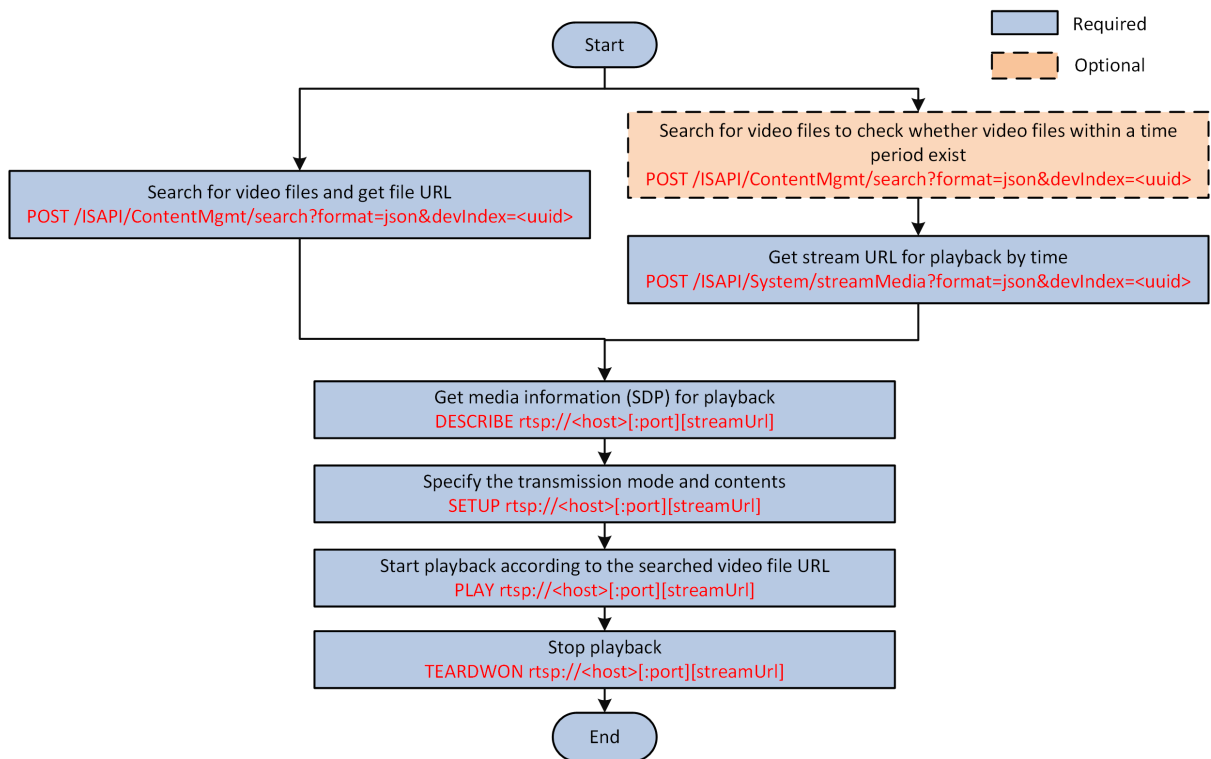


Figure 4-3 Programming Flow of Starting Playback

1. Get the video file URL or the stream URL for playback.
 - Search for video files to get the video file URL: POST ***/ISAPI/ContentMgmt/search?format=json&devIndex=<uuid>*** .
 - Get the stream URL for playback by time: POST ***/ISAPI/System/streamMedia?format=json&devIndex=<uuid>*** .

Note

You can search for video files to check whether video files within a time period exist before getting the stream URL.

The URL of searched video file (**playbackURI**) or stream URL (**URL**) will be returned for playback.

2. Call **rtsp://{ipAddress}:{port}{streamUrl}** by DESCRIBE method of RTSP to get media information (SDP) for playback.
3. Call **rtsp://{ipAddress}:{port}{streamUrl}** by SETUP method of RTSP to specify the transmission mode and contents.
4. Call **rtsp://{ipAddress}:{port}{streamUrl}** by PLAY method of RTSP to start playback according to the searched video file URL.
5. Call **rtsp://{ipAddress}:{port}{streamUrl}** by TEARDWON method of RTSP to stop playback.

4.2.3 Download Video Files

You can download video files to the local PC by file or by time.

Before You Start

Make sure there are video files stored in the device.

Steps

Note

Only devices added via ISUP5.0 support downloading video files by time.

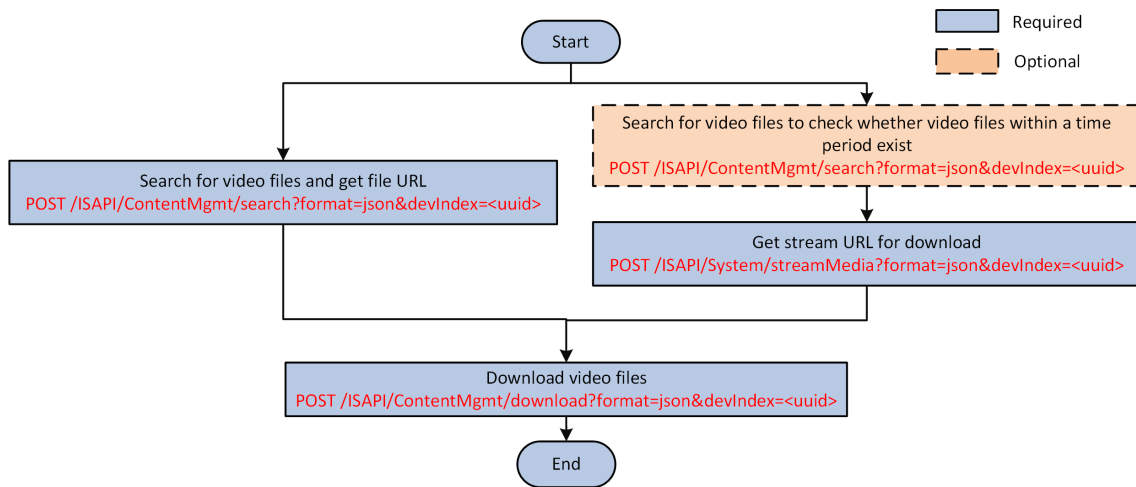


Figure 4-4 Programming Flow of Downloading Video Files

1. Get the video file URL or the stream URL for download.
 - Search for video files to get the video file URL: POST **/ISAPI/ContentMgmt/search?format=json&devIndex=<uuid>** .

- Get the stream URL for download by time: POST ***/ISAPI/System/streamMedia?format=json&devIndex=<uuid>*** .



Note

You can search for video files to check whether video files within a time period exist before getting the stream URL.

The URL of searched video file (**playbackURI**) or stream URL (**URL**) will be returned for download.

2. Call ***/ISAPI/ContentMgmt/download?format=json&devIndex=<uuid>*** by POST method to download video files to the local PC.

4.2.4 Start Two-Way Audio

The two-way audio based on Hik Device Gateway realizes the audio sending or receiving between the platform/system and the body cameras. During the law enforcement, if the exception occurs, the police officer can talk with the monitoring center to report the exception information for timely solution after starting two-way audio between the platform/system and body cameras.

Steps

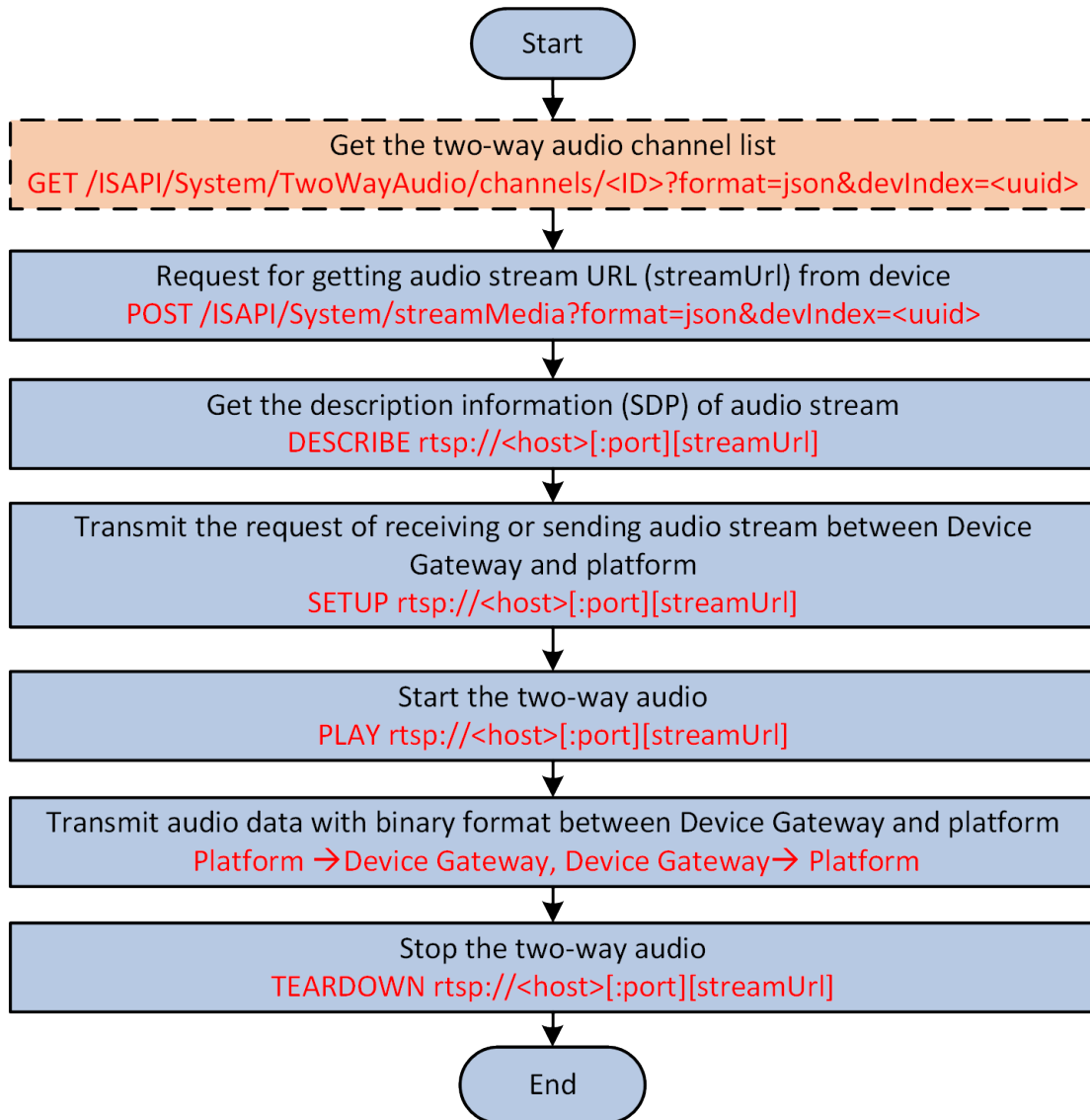


Figure 4-5 Programming Flow of Starting Two-way Audio

1. **Optional:** Call `/ISAPI/System/TwoWayAudio/channels/<ID>?format=json&devIndex=<uuid>` by GET method to get the information of a specific two-way audio channel.
2. Call `/ISAPI/System/streamMedia?format=json&devIndex=<uuid>` by POST method to request for getting the audio stream URL from device.
A URL which contains real-time audio stream data is returned to Hik Device Gateway for two-way audio.
3. Call `rtsp://{ipAddress}:{port}{streamUrl}` by DESCRIBE method of RTSP to get the description information (SDP) of audio stream.

Example

See the request format example of DESCRIBE method below:

```
DESCRIBE
rtsp://127.0.0.1:554/dac/talk/8F0B7279-6519-45C3-8EEF-07942E3497701/TCP RTSP/1.0
CSeq:1
Accept:application/sdp
Authorization: Digest username="admin", realm="DS-GWAS0101(27709)",
nonce="00000000ef38d86800000000026c2596", uri="rtsp://127.0.0.1:554/dac/talk/
8F0B7279-6519-45C3-8EEF-07942E3497701/TCP", response="63f8624c5845986141592b4b0d014aaa"
User-Agent:NKPlayer-1.00.00.081112
```

See the response format example of DESCRIBE method below:

```
RTSP/1.0 200 OK
CSeq: 1
Server: Media
Content-Type: application/sdp
Content-Length: 478

v=0
o=- 10001832 10001832 IN IP4 0.0.0.0
s=MediaSession
c=IN IP4 0.0.0.0
a=range:npt=now-
a=sendrecv
a=control:rtsp://127.0.0.1:554/dac/talk/8F0B7279-6519-45C3-8EEF-07942E3497701/TCP
t=0 0
m=audio 0 RTP/AVP 98
c=IN IP4 0.0.0.0
a=rtpmap:98 G7221/16000
a=control:rtsp://127.0.0.1:554/dac/talk/8F0B7279-6519-45C3-8EEF-07942E3497701/TCP/trackID=0
a=range:npt=now-
a=Media_header:MEDIAINFO=494D4B480101000000000007221011000003E8000007D00000000000000000000
000000000000;
```

4. Call **rtsp://{ipAddress}:{port}{streamUrl}** by SETUP method of RTSP to transmit request of receiving or sending audio stream between Hik Device Gateway and platform.

Example

See the request format example of SETUP method below:

```

SETUP
rtsp://127.0.0.1:554/dac/talk/8F0B7279-6519-45C3-8EEF-07942E3497701/TCP/trackID=0 RTSP/1.0
CSeq:2
Authorization: Digest username="admin", realm="DS-GWAS0101(27709)",
nonce="00000000ef38d8680000000026c2596", uri="rtsp://127.0.0.1:554/dac/talk/
8F0B7279-6519-45C3-8EEF-07942E3497701/TCP", response="950c955884761ed4f619c75a1a71dae4"
Transport:RTP/AVP/TCP;unicast;interleaved=0-1;ssrc=0
User-Agent:NKPlayer-1.00.00.081112

```

See the response format example of SETUP method below:

```
RTSP/1.0 200 OK
CSeq: 2
Transport: RTP/AVP/TCP;unicast;interleaved=0-1
Session: 10001832;timeout=20
Server: Media
```

5. Call **rtsp://{ipAddress}:{port}{streamUrl}** by PLAY method of RTSP to start the two-way audio.

Example

See the request format example of PLAY method below:

```
PLAY
rtsp://127.0.0.1:554/dac/talk/8F0B7279-6519-45C3-8EEF-07942E3497701/TCP RTSP/1.0
CSeq:3
Authorization: Digest username="admin", realm="DS-GWAS0101(27709)",
nonce="00000000ef38d8680000000026c2596", uri="rtsp://127.0.0.1:554/dac/talk/
8F0B7279-6519-45C3-8EEF-07942E3497701/TCP", response="8f5fd2b6ff63c68d86d53cb744a56107"
Session:0001832;timeout=20

Range:npt=0.000000-0.000000
User-Agent:NKPlayer-1.00.00.081112
```

See the response format example of PLAY method below:

```
RTSP/1.0 200 OK
CSeq: 3
Session: 10001832
Server: Media
```

6. Transmit the audio data with binary format between Hik Device Gateway and platform.



Note

The transmission format of audio data is RTP, and the supported audio type is G.711a (monaural, 16-bit wide, 8KHz sampling rate, 64Kbps bit rate), G.711u (monaural, 16-bit wide, 8KHz sampling rate, 64Kbps bit rate), G.7221 (monaural, 16-bit wide, 16KHz sampling rate, 16Kbps bit rate), and G.726 (monaural, 16-bit wide, 8KHz sampling rate, 16Kbps bit rate).

7. Call **rtsp://{ipAddress}:{port}{streamUrl}** by TEARDOWN to stop the two-way audio.

Example

See the request format example of TEARDOWN method below:

```
TEARDOWN
rtsp://127.0.0.1:554/dac/talk/8F0B7279-6519-45C3-8EEF-07942E3497701/TCP RTSP/1.0
CSeq:4
Authorization: Digest username="admin", realm="DS-GWAS0101(27709)",
nonce="00000000ef38d8680000000026c2596", uri="rtsp://127.0.0.1:554/dac/talk/
8F0B7279-6519-45C3-8EEF-07942E3497701/TCP", response="23a6f0e53da8f6e99ed17d9325bf82f4"
Session:0001832;timeout=20

User-Agent:NKPlayer-1.00.00.081112
```

See the response format example of TEARDOWN method below:

```
RTSP/1.0 200 OK
CSeq: 4
Session: 10001832
Server: Media
```

4.2.5 Control PTZ

PTZ control is to realize the pan, tilt, and zoom functions of cameras. You can also set and call presets, start auto-scan, perform focus in or out, adjust iris, enable auxiliary functions, or perform 3D positioning.

Steps

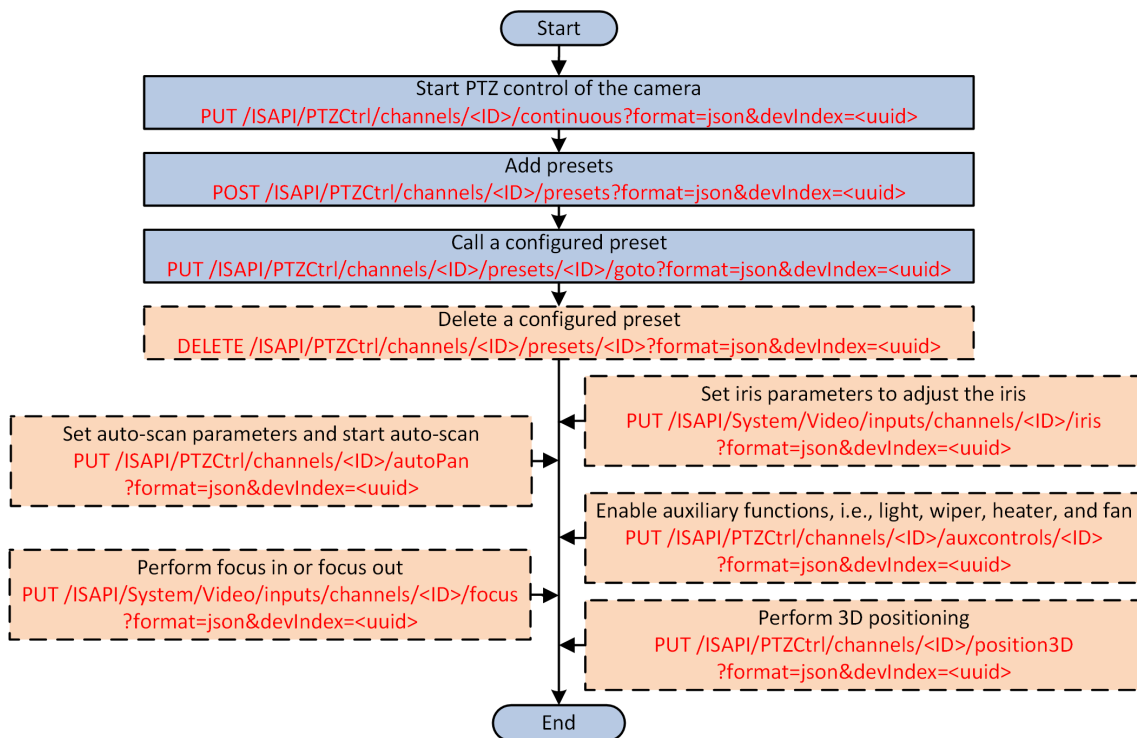


Figure 4-6 Programming Flow of Controlling PTZ

1. Call `/ISAPI/PTZCtrl/channels/<ID>/continuous?format=json&devIndex=<uuid>` by PUT method to start PTZ control of the camera.
2. Call `/ISAPI/PTZCtrl/channels/<ID>/presets?format=json&devIndex=<uuid>` by POST method to add presets.
3. Call `/ISAPI/PTZCtrl/channels/<ID>/presets/<ID>/goto?format=json&devIndex=<uuid>` by PUT method to call a configured preset.
4. **Optional:** Call `/ISAPI/PTZCtrl/channels/<ID>/presets/<ID>?format=json&devIndex=<uuid>` by DELETE to delete a configured preset.
5. **Optional:** Perform the following operation(s) by the corresponding URI(s) after starting PTZ control.

Start Auto-Scan	PUT <u>/ISAPI/PTZCtrl/channels/<ID>/autoPan?format=json&devIndex=<uuid></u> .
Focus In/Out	PUT <u>/ISAPI/System/Video/inputs/channels/<ID>/focus?format=json&devIndex=<uuid></u>
Adjust Iris	PUT <u>/ISAPI/System/Video/inputs/channels/<ID>/iris?format=json&devIndex=<uuid></u>
Enable Auxiliary Function	PUT <u>/ISAPI/PTZCtrl/channels/<ID>/auxcontrols/<ID>?format=json&devIndex=<uuid></u>
 Note The auxiliary functions include light, wiper, heater, and fan.	
3D Positioning	PUT <u>/ISAPI/PTZCtrl/channels/<ID>/position3D?format=json&devIndex=<uuid></u>

4.3 Access Control

Access Control is the selective restriction of access to a place or other resources. The access control applications integrated via Device Gateway in this manual take the person as the management and control unit, which indicates that a person will be linked with fingerprints, cards, and other attributes.

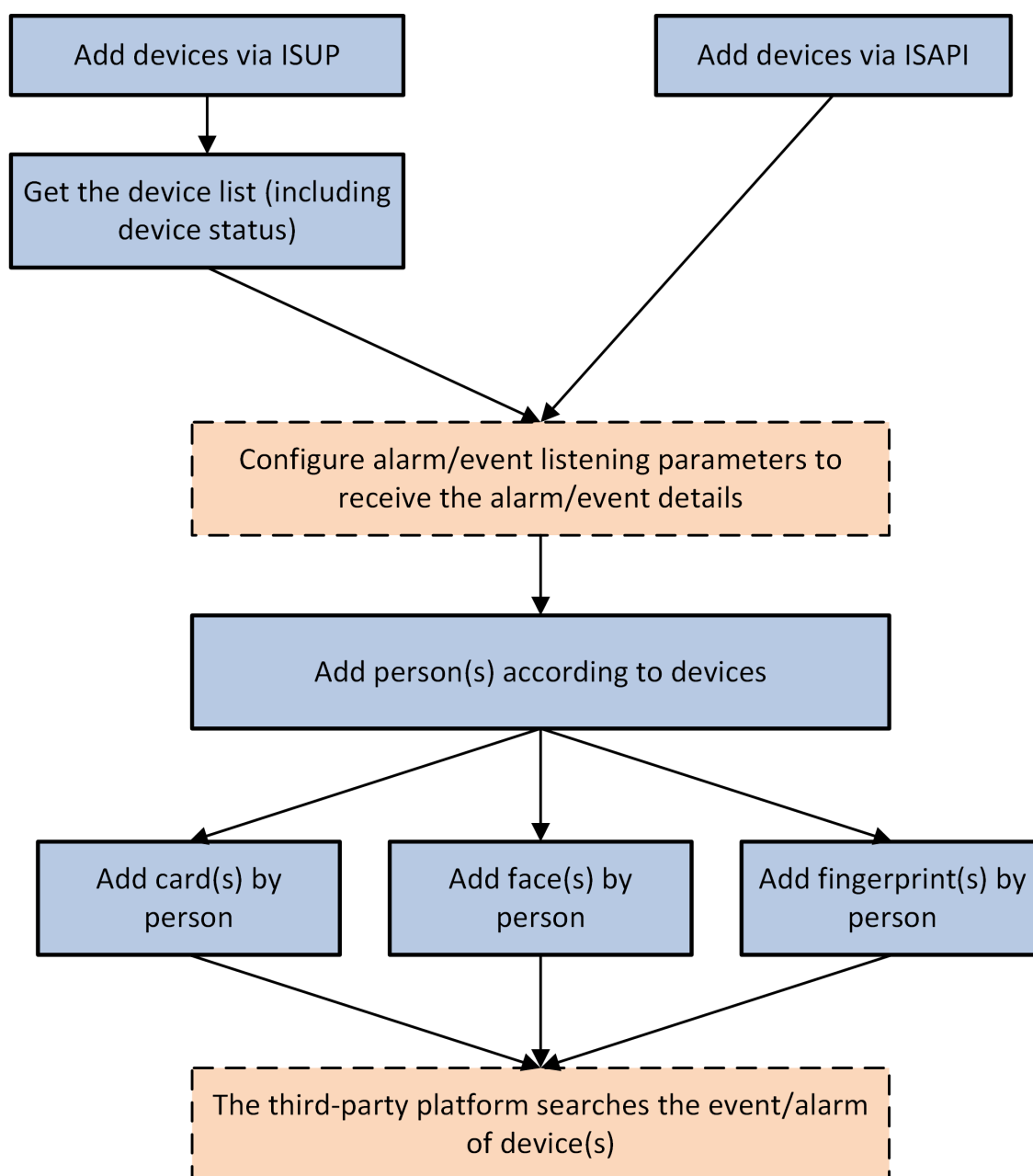
Note

We have provided a sample code file (AccessControl.csproj) of access control in the current directory of this document for reference.

Function	URI
Get or set face recognition mode	GET or PUT <u>/ISAPI/AccessControl/FaceRecognizeMode?format=json&devIndex=<uuid></u>
Search for history access control events	POST <u>/ISAPI/AccessControl/AcsEvent?format=json&devIndex=<uuid></u>

4.3.1 Overall Workflow

The overall workflow for the access control devices is shown below. You can get a introductory understanding of the basic workflow for integration.



Note

- You can add access control devices to Hik Device Gateway via ISUP and ISAPI.
- If the device is added via ISUP, you should get the device list and check the device online/offline status for further operations.
- Two methods are provided for receiving the device event/alarm:
Transmit the event/alarm information from devices to the third-party directly in real time, which requires the listening service of third-party platform.

Search the event/alarm information regularly, which does not require the listening service of third-party platform.

- The person is a base unit, in other words, you should add person(s) and apply person information before applying the card information, face information, and fingerprint information.

4.3.2 Manage Person Information

A person is a basic unit, which can be linked with multiple cards and fingerprints, for access control. So, before starting any other operations, you should add persons and apply the person information (e.g., person ID, name, organization, permissions, and so on) to access control devices.

Steps

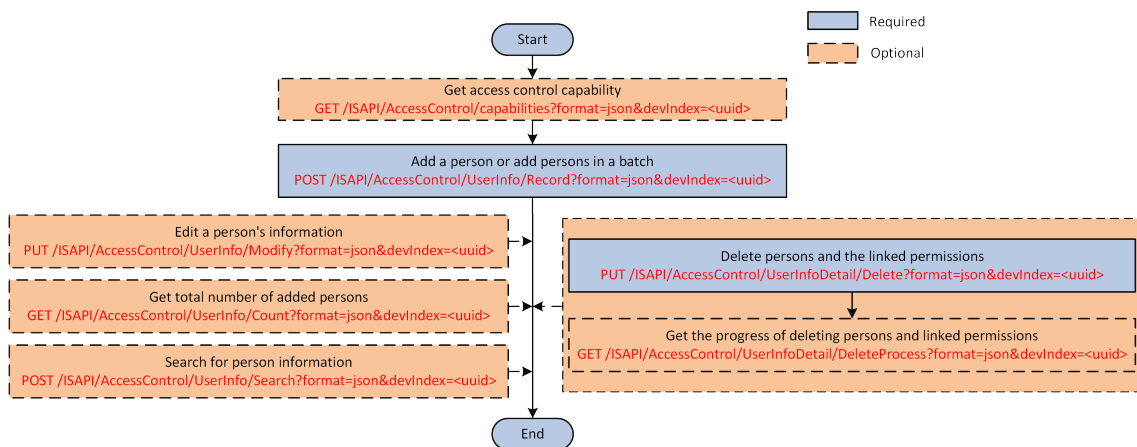


Figure 4-7 API Calling Flow of Managing Person Information

1. **Optional:** Call [/ISAPI/AccessControl/capabilities?format=json&devIndex=<uuid>](#) by GET method to get access control capability.
2. Call [/ISAPI/AccessControl/UserInfo/Record?format=json&devIndex=<uuid>](#) by POST method to add a person or add persons in a batch.
3. **Optional:** Perform the following operation(s) after adding person.

Edit a person's information	PUT <u>/ISAPI/AccessControl/UserInfo/Modify?format=json&devIndex=<uuid></u>
Get the total number of added persons	GET <u>/ISAPI/AccessControl/UserInfo/Count?format=json&devIndex=<uuid></u>
Search for the person information	POST <u>/ISAPI/AccessControl/UserInfo/Search?format=json&devIndex=<uuid></u>

Delete persons and linked permissions

- Call [/ISAPI/AccessControl/UserInfoDetail/Delete?format=json&devIndex=<uuid>](#) by PUT method to start deleting the persons and the linked permissions.
- Call [/ISAPI/AccessControl/UserInfoDetail/DeleteProcess?format=json&devIndex=<uuid>](#) by GET method to get the deleting progress.

4.3.3 Manage Card Information

If a person wants to access by card, you should add card and link the card with the person for getting the access permissions, and then apply card information (e.g., card No., card type, and so on) to access control device.

Before You Start

Make sure you have added persons according to [Manage Person Information](#).

Steps

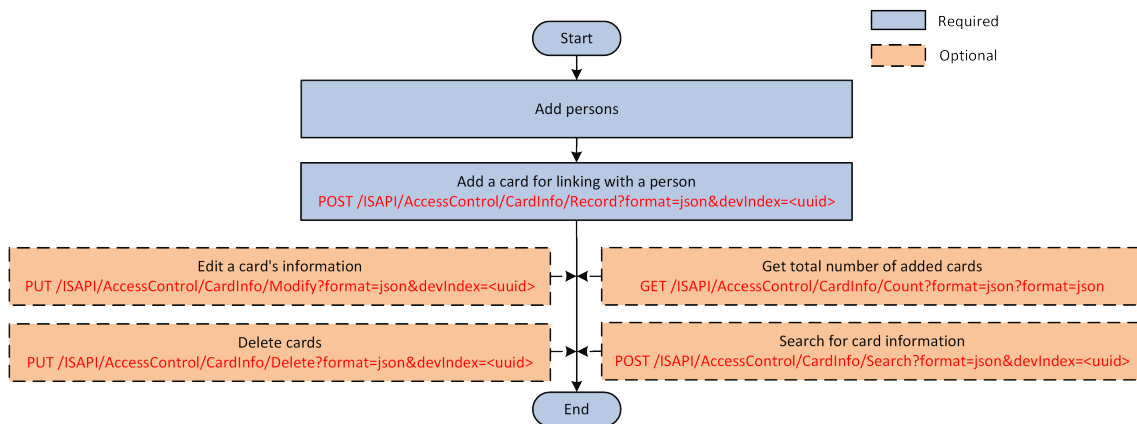


Figure 4-8 API Calling Flow of Managing Card Information

- Call [/ISAPI/AccessControl/CardInfo/Record?format=json&devIndex=<uuid>](#) by POST method to add a card for linking with a person.
- Optional:** Perform the following operation(s) after adding a card.

Edit a card's information

PUT [/ISAPI/AccessControl/CardInfo/Modify?format=json&devIndex=<uuid>](#)

Delete cards

PUT [/ISAPI/AccessControl/CardInfo/Delete?format=json&devIndex=<uuid>](#)

Get the total number of added cards

GET [/ISAPI/AccessControl/CardInfo/Count?format=json?format=json](#)

Search for the card information

POST [/ISAPI/AccessControl/CardInfo/Search?format=json&devIndex=<uuid>](#)

4.3.4 Manage Face Information

If a person wants to access by face, you should apply face records (including face record ID, information about the person in the picture, and so on) for getting the access permissions.

Before You Start

Make sure you have added persons according to ***Manage Person Information***.

Steps

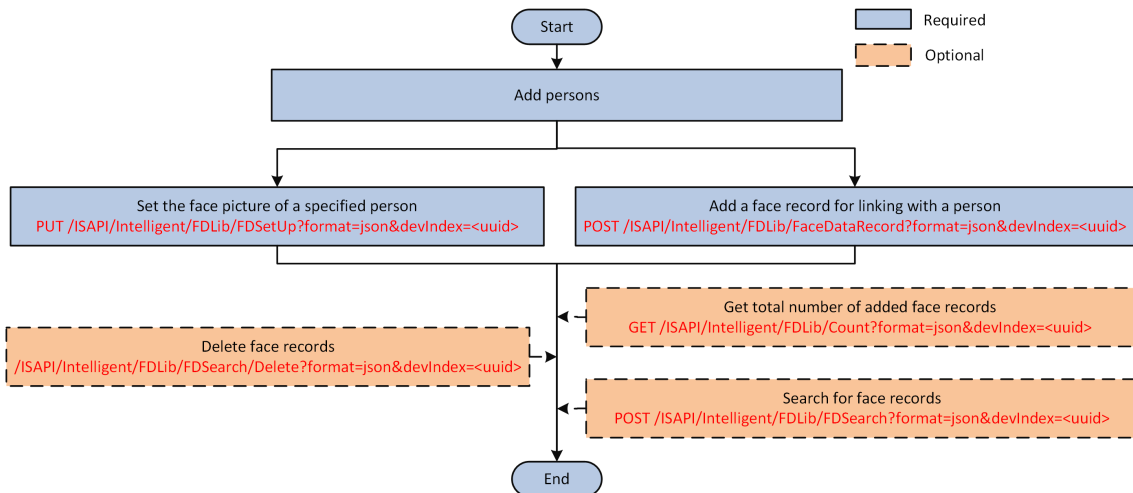


Figure 4-9 API Calling Flow of Managing Face Information

1. Add a face record or set the face picture of a specified person.
 - Add a face record for linking with a person: POST ***/ISAPI/Intelligent/FDLib/FaceDataRecord?format=json&devIndex=<uuid>***.
 - Set the face picture of a specified person: PUT ***/ISAPI/Intelligent/FDLib/FDSetUp?format=json&devIndex=<uuid>***.
2. **Optional:** Perform the following operation(s) after adding a face record.

Delete face records	PUT <u><i>/ISAPI/Intelligent/FDLib/FDSearch/Delete?format=json&devIndex=<uuid></i></u>
Get the total number of added face records	GET <u><i>/ISAPI/Intelligent/FDLib/Count?format=json&devIndex=<uuid></i></u>
Search for face records	POST <u><i>/ISAPI/Intelligent/FDLib/FDSearch?format=json&devIndex=<uuid></i></u>

4.3.5 Manage Fingerprint Information

If a person wants to access by fingerprint, you should collect the fingerprint data via the fingerprint recorder first, and then apply the fingerprint parameters (e.g., fingerprint data, ID, type, and so on)

to the fingerprint and card reader to link the fingerprint with the person for getting the access permissions.

Before You Start

Make sure you have added persons according to ***Manage Person Information*** .

Steps

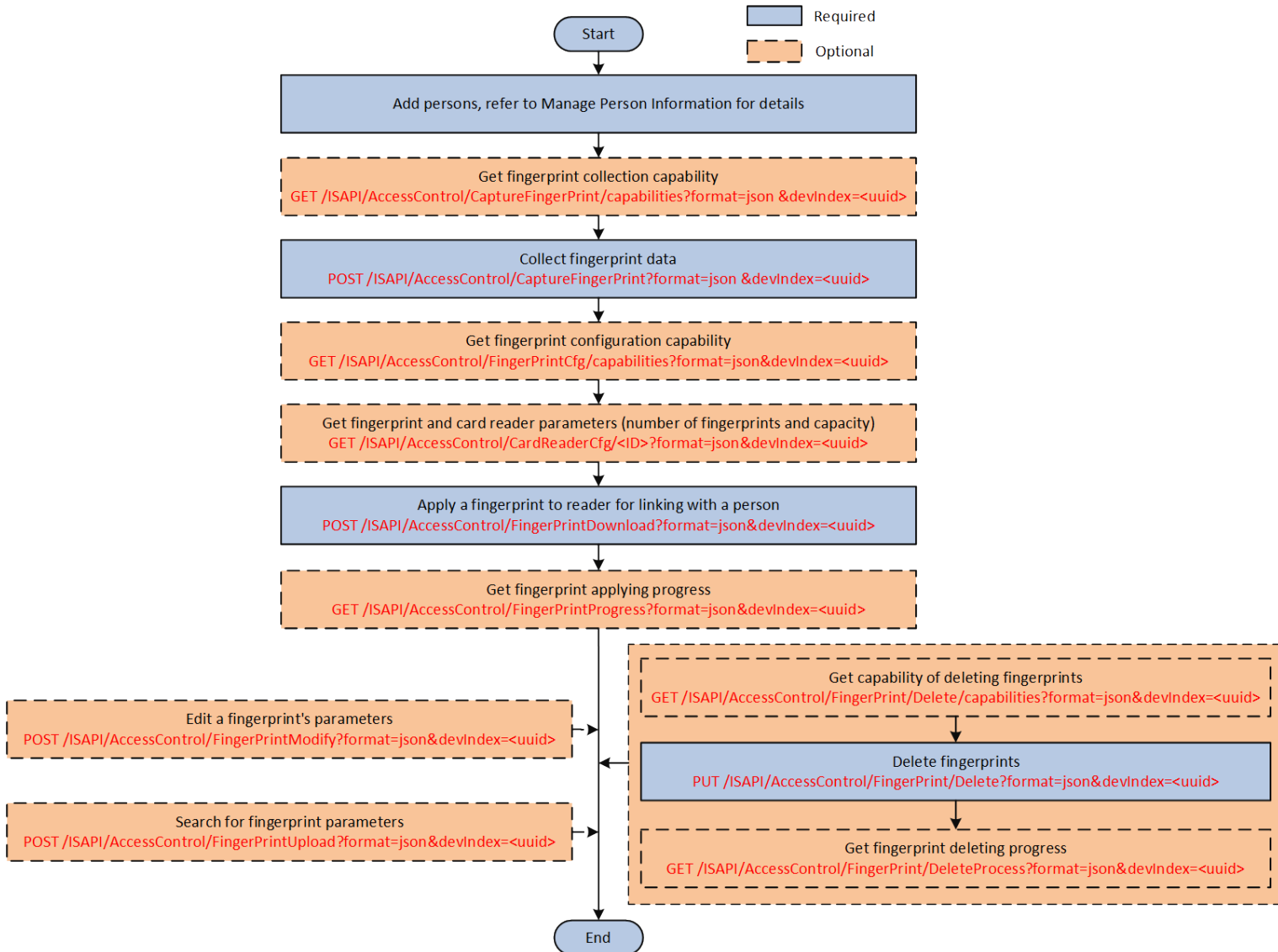


Figure 4-10 API Calling Flow of Managing Fingerprint Information

1. **Optional:** Call ***/ISAPI/AccessControl/CaptureFingerPrint/capabilities?format=json&devIndex=<uuid>*** by GET method to get fingerprint collection capability.
2. Call ***/ISAPI/AccessControl/CaptureFingerPrint?format=json&devIndex=<uuid>*** by POST method to collect fingerprint data.

3. **Optional:** Call [/ISAPI/AccessControl/FingerPrintCfg/capabilities?format=json&devIndex=<uuid>](#) by GET method to get fingerprint configuration capability.
4. **Optional:** Call [/ISAPI/AccessControl/CardReaderCfg/<ID>?format=json&devIndex=<uuid>](#) by GET method to get fingerprint and card reader parameters (including number of fingerprints and capacity).
5. Call [/ISAPI/AccessControl/FingerPrintDownload?format=json&devIndex=<uuid>](#) by POST method to apply a fingerprint's parameters to reader for linking with a person.
6. **Optional:** Call [/ISAPI/AccessControl/FingerPrintProgress?format=json&devIndex=<uuid>](#) by GET method to get fingerprint applying progress.
7. **Optional:** Perform the following operation(s) after applying fingerprints.

Edit a fingerprint's parameters	POST <u>/ISAPI/AccessControl/FingerPrintModify?format=json&devIndex=<uuid></u>
Search for fingerprint parameters	POST <u>/ISAPI/AccessControl/FingerPrintUpload?format=json&devIndex=<uuid></u>
Delete fingerprints	a. Call <u>/ISAPI/AccessControl/FingerPrint/Delete/capabilities?format=json&devIndex=<uuid></u> by GET method to get fingerprint deleting capability. b. Call <u>/ISAPI/AccessControl/FingerPrint/Delete?format=json&devIndex=<uuid></u> by PUT method to start deleting fingerprints. c. Call <u>/ISAPI/AccessControl/FingerPrint/DeleteProcess?format=json&devIndex=<uuid></u> by GET method to get the fingerprint deleting progress.

4.3.6 Remotely Control Door

You can remotely control the status of doors via the Hik Device Gateway.

Steps

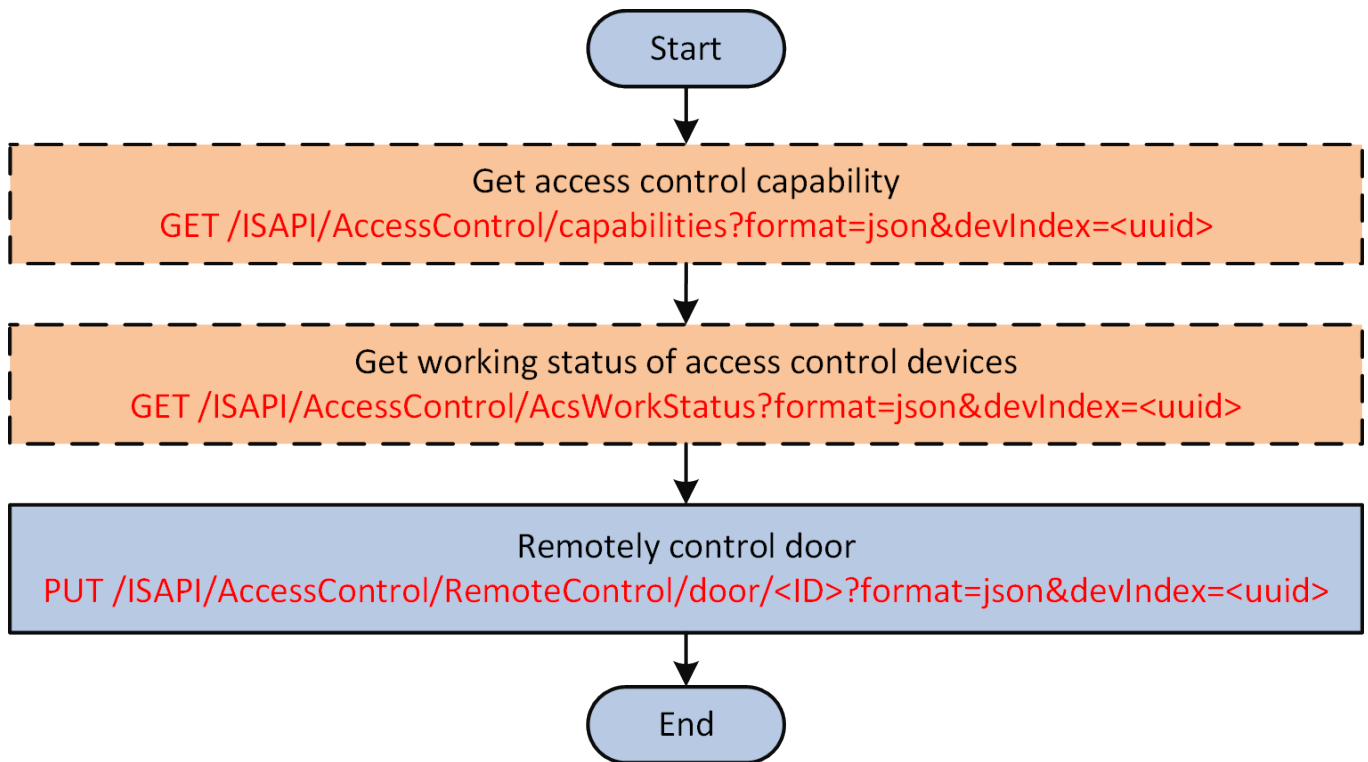


Figure 4-11 API Calling Flow of Remotely Control Door

1. **Optional:** Call `/ISAPI/AccessControl/capabilities?format=json&devIndex=<uuid>` by GET method to get access control capability.
2. **Optional:** Call `/ISAPI/AccessControl/AcsWorkStatus?format=json&devIndex=<uuid>` by GET method to get the working status of access control devices.
3. Call `/ISAPI/AccessControl/RemoteControl/door/<ID>?format=json&devIndex=<uuid>` by PUT method to remotely control the door in remain open, remain closed, or normal status.

4.4 Alarm and Event

When the event occurred or alarm is triggered, you can receive the alarm and event information, or subscribe to alarm and event types to receive the corresponding information as needed.

Note

Directly receiving mode and subscription mode cannot be used at the same time.

4.4.1 Receive Event and Alarm Directly

For devices that support uploading events/alarms in HTTP listening mode, the directly receiving mode is recommended. You can directly receive the event/alarm from devices after configuring the listening server address.

Note

The device must support uploading event and alarm information in listening mode.

1. Call [/ISAPI/Event/notification/httpHosts?format=json&devIndex=<uuid>](#) by POST method to set parameters of listening server for device.
2. When event occurred or alarm is triggered, the device automatically uploads alarm and event information to the configured listening server.

4.4.2 Subscribe to Event and Alarm

Before You Start

Make sure that you have enabled the alarm forwarding function on web client of Device Gateway.

Steps

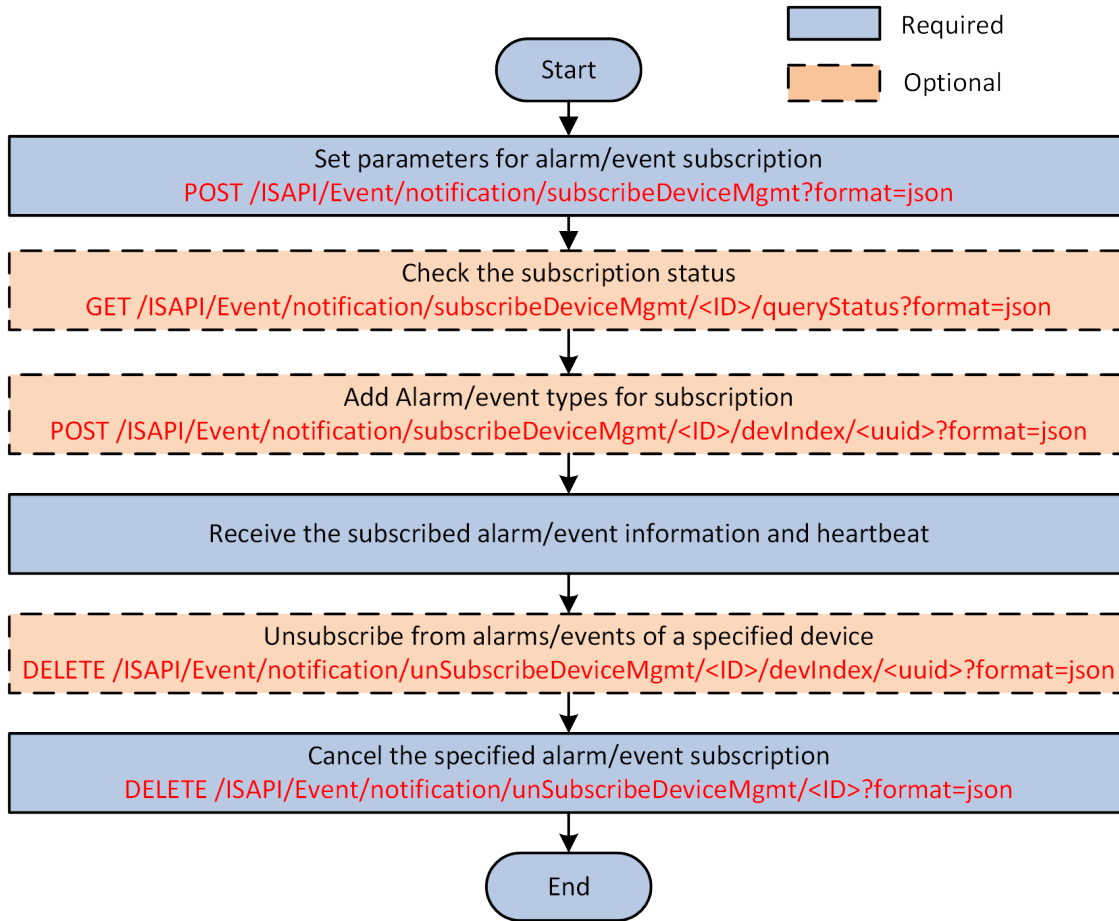


Figure 4-12 API Calling Flow of Subscribing to Event and Alarm

1. Call [/ISAPI/Event/notification/subscribeDeviceMgmt?format=json](#) by POST method to set parameters of subscribing to alarms/events.
The subscription ID is returned.
2. **Optional:** Call [/ISAPI/Event/notification/subscribeDeviceMgmt/<ID>/queryStatus?format=json](#) by GET method to check the subscription status.



Note

This step is not supported when the subscription mode (**eventMode**) is "all".

3. **Optional:** Call [/ISAPI/Event/notification/subscribeDeviceMgmt/<ID>/devIndex/<uuid>?format=json](#) by POST method to add alarm and event types for subscription.



Note

This step is not supported when the subscription mode (**eventMode**) is "all".

4. When the alarm is triggered or event occurred, receive the alarm/event information that are subscribed to and heartbeat from the messages **JSON_EventNotificationAlert_AlarmEventInfo** and **JSON_EventNotificationAlert_HeartbeatInfo** uploaded by device.

Note

- If receiving alarm/event information exception or receiving heartbeat timed out (the default heartbeat interval is 10 seconds, and the suggested timeout is 30 seconds), you should call **/ISAPI/Event/notification/subscribeDeviceMgmt?format=json** by POST to rebuild the link.
 - The alarm/event information messages vary with different alarm or event types, refer to common alarm or event information message in **JSON_EventNotificationAlert_AlarmEventInfo**.
 - The device ID (**devIndex**) in the event/alarm message is consistent with that returned when getting device list (call **/ISAPI/ContentMgmt/DeviceMgmt/deviceList?format=json** by POST method).
5. **Optional:** Call **/ISAPI/Event/notification/unSubscribeDeviceMgmt/<ID>/devIndex/<uuid>?format=json** by DELETE method to unsubscribe from alarms/events of a specified device.

Note

This step is not supported when the subscription mode (**eventMode**) is "all".

6. Call **/ISAPI/Event/notification/unSubscribeDeviceMgmt/<ID>?format=json** by DELETE to cancel subscribing to alarms/events.

Example

Message Example of Subscribing to Alarm/Event

Request Message

```
POST /ISAPI/Event/notification/subscribeDeviceMgmt?format=json HTTP/1.1
```

```
Host: 127.0.0.1:80
```

```
X-Forwarded-For: 127.0.0.1
```

```
X-Real-Port: 57946
```

```
Connection: Keep-Alive
```

```
Content-Length: 63
```

```
Content-Type: application/x-www-form-urlencoded
```

```
Authorization: Digest username="admin",
```

```
realm="DS-GWAS0101(6419)",
```

```
nonce="4e546c6c596a55355a5749364e5752684f4445344d57593d",
```

```
uri="/ISAPI/Event/notification/subscribeDeviceMgmt",
```

```
cnonce="7e867c78d9874aa904b489a6791aaef",
```

```
nc=00000001,
```

```
qop="auth",
```

```
response="fa7d3d95762b6e10c8a18f09f1f61e81"
```

```
{
  "SubscribeDeviceMgmt":{
    "eventMode":"all",
    "defenceMode":"all"
  }
}
```

Response Message

```
HTTP/1.1 200 OK
Server: nginx
Date: Wed, 26 Dec 2018 11:38:38 GMT
Content-Type: multipart/mixed; boundary=boundary
Connection: close
MIME-Version: 1.0

--boundary
Content-Type: application/json; charset="UTF-8"
Content-Length: 39

{"SubscribeDeviceMgmtRsp":{"id":5939}}
--boundary
Content-Type: application/json; charset="UTF-8"
Content-Length: 415

{
  "EventNotificationAlert":{
    "channelID": "1",
    "dateTime": "2018-01-21T12:50:39+08:00",
    "activePostCount": 1,
    "eventType": "devStatusChanged",
    "eventState": "active",
    "eventDescription": "device status about online or offline changed",
    "devIndex":"2cd6716d-767f-4756-ac55-50276a5e3b4a",
    "channelName": "Device1",
    "status":"online",
  }
}
--boundary
Content-Type: application/json; charset="UTF-8"
Content-Length: 183

{
  "EventNotificationAlert":{
    "dateTime":"2018-12-26T19:39:21+08:00",
    "activePostCount":1,
    "eventType":"heartBeat",
    "eventState":"active",
    "eventDescription":"Heart Beat"
  }
}
```

Note

We have also provided a sample code file (SubscribeAlarm.csproj) of alarm/event subscription in the current directory of this document for reference.

Chapter 5 API Reference

5.1 Device Gateway Operations and Maintenance

Function	API
Add device(s) to Device Gateway	<u>/ISAPI/ContentMgmt/DeviceMgmt/addDevice?format=json</u>
Batch delete devices from Hik Device Gateway	<u>/ISAPI/ContentMgmt/DeviceMgmt/delDevice?format=json</u>
Search for the added device list	<u>/ISAPI/ContentMgmt/DeviceMgmt/deviceList?format=json</u>
Edit the added device information	<u>/ISAPI/ContentMgmt/DeviceMgmt/modDevice?format=json</u>
Get or set Hik Device Gateway parameters	<u>/ISAPI/System/deviceInfo?format=json</u>
Reboot Hik Device Gateway	<u>/ISAPI/System/reboot?format=json</u>
Get or set time synchronization parameters of Hik Device Gateway	<u>/ISAPI/System/time?format=json</u>

5.1.1 /ISAPI/ContentMgmt/DeviceMgmt/addDevice?format=json

Add device(s).

Request URI Definition

Table 5-1 POST /ISAPI/ContentMgmt/DeviceMgmt/addDevice?format=json

Method	POST
Description	Add device(s).
Query	security: the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode. iv: the initialization vector, and it is required when security is 1. format: determine the format of request or response message.

Request	<i><u>JSON_DeviceInList</u></i>
Response	Succeeded: <i><u>JSON_DeviceOutList</u></i> Failed: <i><u>JSON_ResponseStatus</u></i> and <i><u>JSON_DeviceOutList</u></i>

Remarks

- For bulk operations, "succeeded" only indicates that one, multiple, or all operations are succeeded, refer to the **Device** field in the ***JSON_DeviceOutList*** message for the detailed successful operation(s); "failed" indicates that all operations are failed, you can also refer to the **Device** field in the ***JSON_DeviceOutList*** message for the error code (i.e., 0x20000009 or 0x400000a0) or description (see ***Status Codes*** for details).
- The sensitive information such as "userName", "password", and "EhomeKey" should be encrypted.

5.1.2 /ISAPI/ContentMgmt/DeviceMgmt/delDevice?format=json

Batch delete devices from Hik Device Gateway.

Request URI Definition

Table 5-2 POST /ISAPI/ContentMgmt/DeviceMgmt/delDevice?format=json

Method	POST
Description	Batch delete devices from Hik Device Gateway.
Query	format : determine the format of request or response message.
Request	<i><u>JSON_DevIndexList</u></i>
Response	Succeeded: <i><u>JSON_DelDevList</u></i> Failed: <i><u>JSON_ResponseStatus</u></i> and <i><u>JSON_DelDevList</u></i>

Remarks

For bulk operations, "succeeded" only indicates that one, multiple, or all operations are succeeded, refer to the message ***JSON_DelDevList*** for the detailed successful operation(s); "failed" indicates that all operations are failed, you can also refer to the message ***JSON_DelDevList*** for the error code or description (see ***Status Codes*** for details).

5.1.3 /ISAPI/ContentMgmt/DeviceMgmt/delDevice?format=json&devIndex=<uuid>

Delete a device.

Request URI Definition

Table 5-3 DELETE /ISAPI/ContentMgmt/DeviceMgmt/delDevice?format=json&devIndex=<uuid>

Method	DELETE
Description	Delete a device.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	None
Response	<u><i>JSON ResponseStatus</i></u>

5.1.4 /ISAPI/ContentMgmt/DeviceMgmt/deviceList?format=json

Search for devices in the added device list.

Request URI Definition

Table 5-4 POST /ISAPI/ContentMgmt/DeviceMgmt/deviceList?format=json

Method	POST
Description	Search for devices in the added device list.
Query	format: determine the format of request or response message.
Request	<u><i>JSON SearchDescription</i></u>
Response	<u><i>JSON SearchResult</i></u>

5.1.5 /ISAPI/ContentMgmt/DeviceMgmt/modDevice?format=json

Edit the added device information.

Request URI Definition

Table 5-5 PUT /ISAPI/ContentMgmt/DeviceMgmt/modDevice?format=json

Method	PUT
Description	Edit the added device information.
Query	format : determine the format of request or response message. security : the version No. of encryption scheme. When security does not exist, it indicates that the data is not encrypted; when security is 1, it indicates that the nodes of sensitive information in the message are encrypted in AES128 CBC mode iv : the initialization vector, and it is required when security is 1.
Request	<u><i>JSON_Edit_DeviceInfo</i></u>
Response	<u><i>JSON_ResponseStatus</i></u>

5.1.6 /ISAPI/System/deviceInfo?format=json

Get or set Hik Device Gateway parameters.

Request URI Definition

Table 5-6 GET /ISAPI/System/deviceInfo?format=json

Method	GET
Description	Get the Hik Device Gateway parameters.
Query	format : determine the format of request or response message.
Request	None
Response	Succeeded: <u><i>JSON_DeviceInfo</i></u> Failed: <u><i>JSON_ResponseStatus</i></u>

Table 5-7 PUT /ISAPI/System/deviceInfo?format=json

Method	PUT
Description	Set the Hik Device Gateway parameters.
Query	format : determine the format of request or response message.

Request	<u><i>JSON_DeviceInfo</i></u>
Response	<u><i>JSON_ResponseStatus</i></u>

Remarks

For PUT operation method, the read-only fields in the *JSON_DeviceInfo* message are invalid.

5.1.7 /ISAPI/System/reboot?format=json

Reboot the Hik Device Gateway.

Request URI Definition

Table 5-8 PUT /ISAPI/System/reboot?format=json

Method	PUT
Description	Reboot the Hik Device Gateway.
Query	format : determine the format of request or response message.
Request	None
Response	<u><i>JSON_ResponseStatus</i></u>

5.1.8 /ISAPI/System/time?format=json

Get or set time synchronization parameters of Hik Device Gateway.

Request URI Definition

Table 5-9 GET /ISAPI/System/time?format=json

Method	GET
Description	Get time synchronization parameters of Hik Device Gateway.
Query	format : determine the format of request or response message.
Request	None
Response	Succeeded: <u><i>JSON_Time</i></u> Failed: <u><i>JSON_ResponseStatus</i></u>

Table 5-10 PUT /ISAPI/System/time?format=json

Method	PUT
Description	Set time synchronization parameters of Hik Device Gateway.
Query	format : determine the format of request or response message.
Request	<u>JSON Time</u>
Response	<u>JSON ResponseStatus</u>

5.2 General Operations of Devices

Function	API
Device parameters	<u>/ISAPI/System/deviceInfo?format=json&devIndex=<uuid></u>
Reboot the device	<u>/ISAPI/System/reboot?format=json&devIndex=<uuid></u>
Time synchronization parameters of devices	<u>/ISAPI/System/time?format=json&devIndex=<uuid></u>
Time zone parameters of devices	<u>/ISAPI/System/time/timeZone?devIndex=<uuid></u>
Trigger alarm output manually	<u>/ISAPI/System/IO/outputs/<ID>/trigger?format=json&devIndex=<uuid></u>
Get all network interfaces' information of the device	<u>/ISAPI/System/Network/interfaces?format=json&devIndex=<uuid></u>
Get or set the information of a specific network interface	<u>/ISAPI/System/Network/interfaces/<ID>?format=json&devIndex=<uuid></u>
Get or set IP address of a specific network interface	<u>/ISAPI/System/Network/interfaces/<ID>/ipAddress?format=json&devIndex=<uuid></u>
Upgrade the device via the upgrade packet downloaded from FTP server	<u>/ISAPI/System/upgradeEhome?format=json&devIndex=<uuid></u>
Get the alarm input and output information of a device	<u>/ISAPI/System/IO?format=json&devIndex=<uuid></u>
ISAPI URI transmission	<u>http://<ipAddress>:<port>/<ISAPIURI>?devIndex=<uuid></u>

5.2.1 /ISAPI/System/deviceInfo?format=json&devIndex=<uuid>

Get or set device parameters.

Request URI Definition

Table 5-11 GET /ISAPI/System/deviceInfo?format=json&devIndex=<uuid>

Method	GET
Description	Get the device parameters.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	None
Response	Succeeded: <u><i>JSON_DeviceInfo</i></u> Failed: <u><i>JSON_ResponseStatus</i></u>

Table 5-12 PUT /ISAPI/System/deviceInfo?format=json&devIndex=<uuid>

Method	PUT
Description	Set the device parameters.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	<u><i>JSON_DeviceInfo</i></u>
Response	<u><i>JSON_ResponseStatus</i></u>

Remarks

- In PUT operation, the read-only fields in the *JSON_DeviceInfo* message is invalid.
- The <uuid> in the URI refers to the device ID generated by operating system, see details in *URL Format* .

5.2.2 /ISAPI/System/reboot?format=json&devIndex=<uuid>

Reboot the device.

Request URI Definition

Table 5-13 PUT /ISAPI/System/reboot?format=json&devIndex=<uuid>

Method	PUT
Description	Reboot the device.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	None
Response	<u><i>JSON ResponseStatus</i></u>

5.2.3 /ISAPI/System/time?format=json&devIndex=<uuid>

Get or set time synchronization parameters of devices.

Request URI Definition

Table 5-14 GET /ISAPI/System/time?format=json&devIndex=<uuid>

Method	GET
Description	Get time synchronization parameters of devices.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	None
Response	Succeeded: <u><i>JSON Time</i></u> Failed: <u><i>JSON ResponseStatus</i></u>

Table 5-15 PUT /ISAPI/System/time?format=json&devIndex=<uuid>

Method	PUT
Description	Set time synchronization parameters of devices.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	<u><i>JSON Time</i></u>
Response	<u><i>JSON ResponseStatus</i></u>

5.2.4 /ISAPI/System/time/timeZone?devIndex=<uuid>

Get or set time zone parameters of devices.

Request URI Definition

Table 5-16 GET /ISAPI/System/time/timeZone?devIndex=<uuid>

Method	GET
Description	Get time zone parameters of devices.
Query	devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	None
Response	Succeeded: Time string (e.g., GST-8:00:00 and CST-8:00:00DST00:30:00,M4.1.0/02:00:00,M10.5.0/02:00:00, the later one is with daylight saving time) Failed: <u><i>JSON ResponseStatus</i></u>

Table 5-17 PUT /ISAPI/System/time/timeZone?devIndex=<uuid>

Method	PUT
Description	Set time zone parameters of devices.

Query	devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	Time string (e.g., GST-8:00:00 and CST-8:00:00DST00:30:00,M4.1.0/02:00:00,M10.5.0/02:00:00, the later one is with daylight saving time)
Response	<u>JSON_ResponseStatus</u>

Remarks

In the time string example with daylight saving time, the "DST00:30:00" refers to the offset of daylight saving time, the "M4.1.0/02:00:00" refers to the start daylight saving time (i.e., 02:00:00 on Sunday of the first week in April), and the "M10.5.0/02:00:00" is the end daylight saving time (i.e., 02:00:00 on Sunday of the fifth week in October).

5.2.5 /ISAPI/System/IO/outputs/<ID>/trigger?format=json&devIndex=<uuid>

Trigger alarm output manually.

Request URI Definition

Table 5-18 PUT /ISAPI/System/IO/outputs/<ID>/trigger?format=json&devIndex=<uuid>

Method	PUT
Description	Trigger alarm output manually.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	<u>JSON_IOPortData</u>
Response	<u>JSON_ResponseStatus</u>

Remarks

- The alarm output is toggled to a high or low signal.
- The **ID** in the URI is the alarm output No.

5.2.6 /ISAPI/System/Network/interfaces?format=json&devIndex=<uuid>

Get all network interfaces' information of the device.

Request URI Definition

Table 5-19 GET /ISAPI/System/Network/interfaces?format=json&devIndex=<uuid>

Method	GET
Description	Get all network interfaces' information of the device.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	None.
Response	Succeeded: <u><i>JSON_NetworkInterfaceList</i></u> Failed: <u><i>JSON_ResponseStatus</i></u>

5.2.7 /ISAPI/System/Network/interfaces/<ID>?format=json&devIndex=<uuid>

Get or set the information of a specific network interface.

Request URI Definition

Table 5-20 GET /ISAPI/System/Network/interfaces/<ID>?format=json&devIndex=<uuid>

Method	GET
Description	Get the information of a specific network interface.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	None.
Response	Succeeded: <u><i>JSON_NetworkInterface</i></u> Failed: <u><i>JSON_ResponseStatus</i></u>

Table 5-21 PUT /ISAPI/System/Network/interfaces/<ID>?format=json&devIndex=<uuid>

Method	PUT
Description	Set the information of a specific network interface.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	<u>JSON_NetworkInterface</u>
Response	<u>JSON_ResponseStatus</u>

Remarks

If the above operations are failed, the following error code may be returned: 0x60000004, 0x60000005, 0x60000006, 0x60000007, and 0x60000010, see the corresponding error descriptions in **Status Codes** .

5.2.8 /ISAPI/System/Network/interfaces/<ID>/ipAddress? format=json&devIndex=<uuid>

Get or set IP address of a specific network interface.

Request URI Definition

**Table 5-22 GET /ISAPI/System/Network/interfaces/<ID>/ipAddress?
format=json&devIndex=<uuid>**

Method	GET
Description	Get the IP address of a specific network interface.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	None
Response	Succeeded: <u>JSON_IPAddress</u> Failed: <u>JSON_ResponseStatus</u>

**Table 5-23 PUT /ISAPI/System/Network/interfaces/<ID>/ipAddress?
format=json&devIndex=<uuid>**

Method	PUT
Description	Set the IP address of a specific network interface.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	<u><i>JSON_IPAddress</i></u>
Response	<u><i>JSON_ResponseStatus</i></u>

Remarks

During the above operations, the following error codes may be returned: 0x60000004, 0x60000005, 0x60000006, 0x60000007, and 0x60000010, see the corresponding error descriptions in *Status Codes* .

5.2.9 /ISAPI/System/upgradeEhome?format=json&devIndex=<uuid>

Upgrade the device via the upgrade packet downloaded from FTP (File Transfer Protocol) server.

Request URI Definition

Table 5-24 PUT /ISAPI/System/upgradeEhome?format=json&devIndex=<uuid>

Method	PUT
Description	Upgrade the device via the upgrade packet downloaded from FTP (File Transfer Protocol) server.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	<u><i>JSON_UpgradeParams</i></u>
Response	<u><i>JSON_ResponseStatus</i></u>

Remarks

During the upgrade, the following error codes may be returned: 0x20000003, 0x20000006, 0x40001041, 0x2000000a, and 0x30000002. For the corresponding error descriptions, refer to **Status Codes** .

5.2.10 /ISAPI/System/IO?format=json&devIndex=<uuid>

Get alarm input and output information of device.

Request URI Definition

Table 5-25 GET /ISAPI/System/IO?format=json&devIndex=<uuid>

Method	GET
Description	Get alarm input and output information of device.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	None
Response	Succeeded: <u>JSON_IOPortList</u> Failed: <u>JSON_ResponseStatus</u>

5.2.11 http://<ipAddress>:<port>/<ISAPIURI>?devIndex=<uuid>

Transmit the device ISAPI URI to the device for implementing the corresponding function. This API is supported by devices added via ISUP5.0 and ISAPI only.

Request URI Definition

Table 5-26 http://<ipAddress>:<port>/<ISAPIURI>?devIndex=<uuid>

Method	Same as the method of ISAPI URI.
Description	Transmit the device ISAPI URI to the device for implementing the corresponding function.
Query	devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and

	generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	Same as the request of ISAPI URI.
Response	Same as the request of ISAPI URI.

ipAddress

The IP address of Hik Device Gateway.

port

The port number of Hik Device Gateway.

ISAPIURI

The ISAPI URI supported by the device.

Remarks

The device upgrade does not support transmission. You can call the URI **/ISAPI/System/upgradeEhome?format=json&devIndex=<uuid>** to upgrade the device via the upgrade packet downloaded from FTP (File Transfer Protocol) server.

5.2.12 /ISAPI/System/streamMedia?format=json&devIndex=<uuid>

Get the stream URL for live view, or playback, or video download.

Request URI Definition

Table 5-27 POST /ISAPI/System/streamMedia?format=json&devIndex=<uuid>

Method	POST
Description	Get the stream URL for live view, or playback, or video download.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	<u>JSON StreamInfo</u>
Response	<u>JSON MediaAccessInfo</u>

Remarks

- The stream URL is returned by the response message ***JSON_MediaAccessInfo*** , and you can get the stream via the stream URL.
- When the streaming method (**method**) is "playback", the returned stream URL can be used for playback and video download.
- When the streaming method (**method**) is "preview", it is supported by ISUP5.0 access control devices and ISUP5.0 zero-channels.

5.3 Video Device Operations

Streaming URL

Function	API
Get the stream URL to start live view or playback	<u><a href="/ISAPI/System/streamMedia?format=json&devIndex=<uuid>">/ISAPI/System/streamMedia?format=json&devIndex=<uuid></u>

Recording

Function	API
Search for video files	<u><a href="/ISAPI/ContentMgmt/search?format=json&devIndex=<uuid>">/ISAPI/ContentMgmt/search?format=json&devIndex=<uuid></u>
Download a video file	<u><a href="/ISAPI/ContentMgmt/download?format=json&devIndex=<uuid>">/ISAPI/ContentMgmt/download?format=json&devIndex=<uuid></u>
Get recording schedules or add a recording schedule	<u><a href="/ISAPI/ContentMgmt/record/tracks?format=json&devIndex=<uuid>">/ISAPI/ContentMgmt/record/tracks?format=json&devIndex=<uuid></u>
Set a recording schedule for a camera	<u><a href="/ISAPI/ContentMgmt/record/tracks/<ID>?format=json&devIndex=<uuid>">/ISAPI/ContentMgmt/record/tracks/<ID>?format=json&devIndex=<uuid></u>
Manually start the recording of a camera	<u><a href="/ISAPI/ContentMgmt/record/control/manual/start/tracks/<ID>?format=json&devIndex=<uuid>">/ISAPI/ContentMgmt/record/control/manual/start/tracks/<ID>?format=json&devIndex=<uuid></u>
Manually stop the recording of a camera	<u><a href="/ISAPI/ContentMgmt/record/control/manual/stop/tracks/<ID>?format=json&devIndex=<uuid>">/ISAPI/ContentMgmt/record/control/manual/stop/tracks/<ID>?format=json&devIndex=<uuid></u>

Two-Way Audio

Function	API
Get the information of a specific two-way audio channel	<u>/ISAPI/System/TwoWayAudio/channels/<ID>?format=json&devIndex=<uuid></u>
Operations about playing stream based on RTSP methods	<u>rtsp://{ipAddress}:{port}/{streamUrl}</u>

PTZ Control

Function	API
Enable or disable auxiliary functions of PTZ	<u>/ISAPI/PTZCtrl/channels/<ID>/auxcontrols/<ID>?format=json&devIndex=<uuid></u>
Control the PTZ of a specific camera, such as pan, tilt, and zoom	<u>/ISAPI/PTZCtrl/channels/<ID>/continuous?format=json&devIndex=<uuid></u>
Set the focus parameter of a specific camera to perform focus in or out	<u>/ISAPI/System/Video/inputs/channels/<ID>/focus?format=json&devIndex=<uuid></u>
Set the iris parameter for a specific camera to adjust the iris manually	<u>/ISAPI/System/Video/inputs/channels/<ID>/iris?format=json&devIndex=<uuid></u>
Set the auto-scan parameters for a specific camera	<u>/ISAPI/PTZCtrl/channels/<ID>/autoPan?format=json&devIndex=<uuid></u>
Add presets for a specific camera	<u>/ISAPI/PTZCtrl/channels/<ID>/presets?format=json&devIndex=<uuid></u>
Delete a configured preset of a specific camera	<u>/ISAPI/PTZCtrl/channels/<ID>/presets/<ID>?format=json&devIndex=<uuid></u>
Call a configured preset of a specific camera	<u>/ISAPI/PTZCtrl/channels/<ID>/presets/<ID>/goto?format=json&devIndex=<uuid></u>
Set the 3D positioning parameters for a specific camera	<u>/ISAPI/PTZCtrl/channels/<ID>/position3D?format=json&devIndex=<uuid></u>

Audio and Video Configurations of Channels

Function	API
Get or set streaming parameters of a camera	<u>/ISAPI/Streaming/channels/<ID>?format=json&devIndex=<uuid></u>
Batch get the video input parameters of multiple cameras	<u>/ISAPI/System/Video/inputs/channels?format=json&devIndex=<uuid></u>
Get or set video input parameters of a specific camera	<u>/ISAPI/System/Video/inputs/channels/<ID>?format=json&devIndex=<uuid></u>

OSD Configurations

Function	API
Get or set OSD parameters	<u>/ISAPI/System/Video/inputs/channels/<ID>/overlays?format=json&devIndex=<uuid></u>
Batch get or set overlay parameters of multiple texts for a specified camera	<u>/ISAPI/System/Video/inputs/channels/<ID>/overlays/text?format=json&devIndex=<uuid></u>
Get or set text overlay parameters of a channel	<u>/ISAPI/System/Video/inputs/channels/<ID>/overlays/text/<ID>?format=json&devIndex=<uuid></u>
Configure channel name overlay of a channel	<u>/ISAPI/System/Video/inputs/channels/<ID>/overlays/channelNameOverlay?format=json&devIndex=<uuid></u>
Get or set time overlay parameters of a channel	<u>/ISAPI/System/Video/inputs/channels/<ID>/overlays/dateTime?format=json&devIndex=<uuid></u>

Channel Color Configurations

Function	API
Get or set color parameters of a channel	<u>/ISAPI/Image/channels/<ID>/color?format=json&devIndex=<uuid></u>

Event Configurations

Function	API
Get or set alarm/event linkage actions	<u>/ISAPI/Event/triggers/<ID>?format=json&devIndex=<uuid></u>
Get or set arming schedules of motion detection alarm for multiple channels	<u>/ISAPI/Event/schedules/motionDetections?format=json&devIndex=<uuid></u>
Get or set the motion detection arming schedule of a specified channel	<u>/ISAPI/Event/schedules/motionDetections/VMD_video<channelID>?format=json&devIndex=<uuid></u>
Get or set motion detection alarm parameters of a specified video input channel	<u>/ISAPI/System/Video/inputs/channels/<ID>/motionDetection?format=json&devIndex=<uuid></u>
Get or set arming schedules of video tampering alarm for multiple channels	<u>/ISAPI/Event/schedules/tamperDetections?format=json&devIndex=<uuid></u>
Get or set the arming schedule of video tampering detection for a specified channel	<u>/ISAPI/Event/schedules/tamperDetections/tamperdetection_video<channelID>?format=json&devIndex=<uuid></u>
Get or set video tampering alarm parameters of a specified video input channel	<u>/ISAPI/System/Video/inputs/channels/<ID>/tamperDetection?format=json&devIndex=<uuid></u>
Get or set arming schedules of video loss alarm for multiple channels	<u>/ISAPI/Event/schedules/videolosses?format=json&devIndex=<uuid></u>
Get or set the arming schedule of video loss alarm of a specific channel	<u>/ISAPI/Event/schedules/videolosses/Videoloss_video<channelID>?format=json&devIndex=<uuid></u>

Capture Manually

Function	APIs
Manually capture pictures	<u>/ISAPI/Streaming/channels/<channelID>/picture?format=json&devIndex=<uuid></u>

Wake Up

Function	APIs
Wake up the device from sleep mode	<u>/ISAPI/System/wakeUp?format=json&devIndex=<uuid></u>

5.3.1 Recording

[/ISAPI/ContentMgmt/search?format=json&devIndex=<uuid>](#)

Search for video files.

Request URI Definition

Table 5-28 POST [/ISAPI/ContentMgmt/search?format=json&devIndex=<uuid>](#)

Method	POST
Description	Search for video files.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	<u>JSON_CMSearchDescription</u>
Response	<u>JSON_CMSearchResult</u>

[/ISAPI/ContentMgmt/download?format=json&devIndex=<uuid>](#)

Download a video file of the camera.

Request URI Definition

Table 5-29 POST [/ISAPI/ContentMgmt/download?format=json&devIndex=<uuid>](#)

Method	POST
Description	Download a video file of the camera.

Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	<u>JSON downloadRequest</u>
Response	Video file data

/ISAPI/ContentMgmt/record/tracks?format=json&devIndex=<uuid>

Get recording schedules or add a recording schedule.

Request URL Definition

Table 5-30 GET /ISAPI/ContentMgmt/record/tracks?format=json&devIndex=<uuid>

Method	GET
Description	Get the recording schedules of the camera.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	None.
Response	<u>JSON TrackList</u>

Table 5-31 POST /ISAPI/ContentMgmt/record/tracks?format=json&devIndex=<uuid>

Method	POST
Description	Add a recording schedule for the camera.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	<u>JSON Track</u>
Response	<u>JSON ResponseStatus</u>

/ISAPI/ContentMgmt/record/tracks/<ID>?format=json&devIndex=<uuid>

Set a recording schedule for a camera.

Request URI Definition

Table 5-32 PUT /ISAPI/ContentMgmt/record/tracks/<ID>?format=json&devIndex=<uuid>

Method	PUT
Description	Set a recording schedule for a camera.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	<i><u>JSON Track</u></i>
Response	<i><u>JSON ResponseStatus</u></i>

**/ISAPI/ContentMgmt/record/control/manual/start/tracks/<ID>?
format=json&devIndex=<uuid>**

Manually start the recording of the camera during live view.

Request URI Definition

**Table 5-33 POST /ISAPI/ContentMgmt/record/control/manual/start/tracks/<ID>?
format=json&devIndex=<uuid>**

Method	POST
Description	Manually start the recording of the camera during live view.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	None.
Response	<i><u>JSON ResponseStatus</u></i>

**/ISAPI/ContentMgmt/record/control/manual/stop/tracks/<ID>?
format=json&devIndex=<uuid>**

Manually stop the recording of the camera.

Request URI Definition

**Table 5-34 POST /ISAPI/ContentMgmt/record/control/manual/stop/tracks/<ID>?
format=json&devIndex=<uuid>**

Method	POST
Description	Manually stop the recording of the camera.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	None.
Response	<i><u>JSON ResponseStatus</u></i>

http://<ipAddress>:<port>/<videoUrl>

Download the video or picture files from alarm via the permission of admin user.

Request URI Definition

Table 5-35 GET http://<ipAddress>:<port>/<videoUrl>

Method	GET
Description	Download the video or picture files from alarm via the permission of admin user.
Query	None
Request	None
Response	Video Data

Remarks

- The <ipAddress> and <port> in the URL refer to the IP address and port No. of Hik Device Gateway; the <videoUrl> in the URL is the video or picture file URL uploaded in the alarm message.
- For access control events, the uploaded event and alarm message only contains picture file URL.

5.3.2 Two-Way Audio

/ISAPI/System/TwoWayAudio/channels/<ID>?format=json&devIndex=<uuid>

Get the information of a specific two-way audio channel.

Request URI Definition

Table 5-36 GET /ISAPI/System/TwoWayAudio/channels/<ID>?format=json&devIndex=<uuid>

Method	GET
Description	Get the information of a specific two-way audio channel.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	None
Response	<u>JSON TwoWayAudioChannel</u>

Remarks

The <ID> in the URI refers to the two-way audio channel ID.

rtsp://{ipAddress}:{port}{streamUrl}

Operations about playing stream based on RTSP methods.

Request URL Definition

Table 5-37 OPTIONS rtsp://{ipAddress}:{port}{streamUrl}

Method	OPTIONS
Description	Get the supported RTSP operation methods of this stream URL.

Table 5-38 DESCRIBE rtsp://{ipAddress}:{port}{streamUrl}

Method	DESCRIBE
Description	Get the media information (SDP) for playing stream.

Table 5-39 SETUP rtsp://{ipAddress}:{port}{streamUrl}

Method	SETUP
Description	Specify transmission mode and contents.

Table 5-40 PLAY rtsp://{ipAddress}:{port}{streamUrl}

Method	PLAY
Description	Start playing stream.

Table 5-41 TEARDOWN rtsp://{ipAddress}:{port}{streamUrl}

Method	TEARDOWN
Description	Stop playing stream.

Table 5-42 PAUSE rtsp://{ipAddress}:{port}{streamUrl}

Method	PAUSE
Description	Pause playing stream.

Remarks

- The {streamUrl} in the URL refers to the stream URL returned by POST [***/ISAPI/System/streamMedia?format=json&devIndex=<uuid>***](#) .
- For details about other members in the URL, RTSP, and its operation methods, please refer to [***API Description***](#) or [***https://tools.ietf.org/html/rfc2326***](https://tools.ietf.org/html/rfc2326) .

5.3.3 PTZ Control

/ISAPI/PTZCtrl/channels/<ID>/auxcontrols/<ID>?format=json&devIndex=<uuid>

Enable or disable the auxiliary functions, i.e., light, wiper, fan, and heater, of PTZ control.

Request URL Definition

Table 5-43 PUT /ISAPI/PTZCtrl/channels/<ID>/auxcontrols/<ID>?format=json&devIndex=<uuid>

Method	PUT
Description	Enable or disable the auxiliary functions, i.e., light, wiper, fan, and heater, of PTZ control.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	<i><u>JSON_PTZAux</u></i>
Response	<i><u>JSON_ResponseStatus</u></i>

Remarks

- Check whether the device supports auxiliary functions by the PTZ control capabilities.
- If there are multiple lights, you can call this URI to turn on or turn off all lights. But turning on one light only is not supported, so does other auxiliary functions.

/ISAPI/PTZCtrl/channels/<ID>/continuous?format=json&devIndex=<uuid>

Control the PTZ of a specific camera, such as pan, tilt, and zoom.

Request URI Definition

Table 5-44 PUT /ISAPI/PTZCtrl/channels/<ID>/continuous?format=json&devIndex=<uuid>

Method	PUT
Description	Control the PTZ of a specific camera, such as pan, tilt, and zoom.
Query	format: determine the format of request or response message.

	devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	<i>JSON_PTZData</i>
Response	<i>JSON_ResponseStatus</i>

/ISAPI/System/Video/inputs/channels/<ID>/focus?format=json&devIndex=<uuid>

Set focus parameters of a specific camera to perform focus in or out.

Request URI Definition

Table 5-45 PUT /ISAPI/System/Video/inputs/channels/<ID>/focus?format=json&devIndex=<uuid>

Method	PUT
Description	Set focus parameters of a specific camera to perform focus in or out.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	<i>JSON_FocusData</i>
Response	<i>JSON_ResponseStatus</i>

/ISAPI/System/Video/inputs/channels/<ID>/iris?format=json&devIndex=<uuid>

Set the iris parameter for a specific camera to adjust the iris manually.

Request URI Definition

Table 5-46 PUT /ISAPI/System/Video/inputs/channels/<ID>/iris?format=json&devIndex=<uuid>

Method	PUT
Description	Set the iris parameter for a specific camera to adjust the iris manually.

Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	<u><i>JSON IrisData</i></u>
Response	<u><i>JSON ResponseStatus</i></u>

/ISAPI/PTZCtrl/channels/<ID>/autoPan?format=json&devIndex=<uuid>

Set the auto-scan parameters for a specific camera.

Request URI Definition

Table 5-47 PUT /ISAPI/PTZCtrl/channels/<ID>/autoPan?format=json&devIndex=<uuid>

Method	PUT
Description	Set the auto-scan parameters for a specific camera.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	<u><i>JSON autoPanData</i></u>
Response	<u><i>JSON ResponseStatus</i></u>

/ISAPI/PTZCtrl/channels/<ID>/presets?format=json&devIndex=<uuid>

Add presets for a specific camera.

Request URI Definition

Table 5-48 POST /ISAPI/PTZCtrl/channels/<ID>/presets?format=json&devIndex=<uuid>

Method	POST
Description	Add presets for a specific camera.
Query	format: determine the format of request or response message.

	devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	<u><i>JSON_PTZPreset</i></u>
Response	<u><i>JSON_ResponseStatus</i></u>

/ISAPI/PTZCtrl/channels/<ID>/presets/<ID>?format=json&devIndex=<uuid>

Delete a configured preset of a specific camera.

Request URI Definition

Table 5-49 DELETE /ISAPI/PTZCtrl/channels/<ID>/presets/<ID>?format=json&devIndex=<uuid>

Method	DELETE
Description	Delete a configured preset of a specific camera.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	None
Response	<u><i>JSON_ResponseStatus</i></u>

Remarks

The first <ID> is the camera No., and the second <ID> is the configured preset No.

/ISAPI/PTZCtrl/channels/<ID>/presets/<ID>/goto?format=json&devIndex=<uuid>

Call a configured preset of a specific camera.

Request URI Definition

Table 5-50 PUT /ISAPI/PTZCtrl/channels/<ID>/presets/<ID>/goto?format=json&devIndex=<uuid>

Method	PUT
Description	Call a configured preset of a specific camera.

Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	None
Response	<u>JSON ResponseStatus</u>

Remarks

The first <ID> is the camera No., and the second <ID> is the preset No.

/ISAPI/PTZCtrl/channels/<ID>/position3D?format=json&devIndex=<uuid>

Set the 3D positioning parameters for a specific camera.

Request URI Definition

Table 5-51 PUT /ISAPI/PTZCtrl/channels/<ID>/position3D?format=json&devIndex=<uuid>

Method	PUT
Description	Set the 3D positioning parameters for a specific camera.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	<u>JSON Position3D</u>
Response	<u>JSON ResponseStatus</u>

5.3.4 Audio and Video Configurations of Channels

/ISAPI/Streaming/channels/<ID>?format=json&devIndex=<uuid>

Get or set the streaming parameters of a camera.

Request URI Definition

Table 5-52 GET /ISAPI/Streaming/channels/<ID>?format=json&devIndex=<uuid>

Method	GET
Description	Get the streaming parameters of a specific camera.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	None.
Response	<u><i>JSON StreamingChannel</i></u>

Table 5-53 PUT /ISAPI/Streaming/channels/<ID>?format=json&devIndex=<uuid>

Method	PUT
Description	Set the streaming parameters of a specific camera.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	<u><i>JSON StreamingChannel</i></u>
Response	<u><i>JSON ResponseStatus</i></u>

Remarks

The <ID> in the request URI refers to the channel No.

/ISAPI/System/Video/inputs/channels?format=json&devIndex=<uuid>

Get the video input parameters of multiple cameras in a batch.

Request URI Definition

Table 5-54 GET /ISAPI/System/Video/inputs/channels?format=json&devIndex=<uuid>

Method	GET
Description	Get the video input parameters of multiple cameras in a batch.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	None.
Response	<u><i>JSON VideoInputChannelLst</i></u>

/ISAPI/System/Video/inputs/channels/<ID>?format=json&devIndex=<uuid>

Get or set the video input parameters of a specific camera.

Request URI Definition

Table 5-55 GET /ISAPI/System/Video/inputs/channels/<ID>?format=json&devIndex=<uuid>

Method	GET
Description	Get the video input parameters of a specific camera.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	None.
Response	<u><i>JSON VideoInputChannel</i></u>

Table 5-56 PUT /ISAPI/System/Video/inputs/channels/<ID>?format=json&devIndex=<uuid>

Method	PUT
Description	Set the video input parameters of a specific camera.
Query	format: determine the format of request or response message.

	devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	<u><i>JSON_VideoInputChannel</i></u>
Response	<u><i>JSON_ResponseStatus</i></u>

5.3.5 OSD Configurations

**/ISAPI/System/Video/inputs/channels/<ID>/overlays?
format=json&devIndex=<uuid>**

Get or set the OSD parameters.

Request URI Definition

**Table 5-57 GET /ISAPI/System/Video/inputs/channels/<ID>/overlays?
format=json&devIndex=<uuid>**

Method	GET
Description	Get the OSD parameters of a specific camera.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	None.
Response	<u><i>JSON_VideoOverlay</i></u>

**Table 5-58 PUT /ISAPI/System/Video/inputs/channels/<ID>/overlays?
format=json&devIndex=<uuid>**

Method	PUT
Description	Set the OSD parameters of a specific camera.
Query	format: determine the format of request or response message.

	devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	<i><u>JSON_VideoOverlay</u></i>
Response	<i><u>JSON_ResponseStatus</u></i>

/ISAPI/System/Video/inputs/channels/<ID>/overlays/text? format=json&devIndex=<uuid>

Get or set multiple texts overlay parameters in a batch for a specified camera.

Request URI Definition

**Table 5-59 GET /ISAPI/System/Video/inputs/channels/<ID>/overlays/text?
format=json&devIndex=<uuid>**

Method	GET
Description	Get the multiple texts overlay parameters of a camera.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	None
Response	<i><u>JSON_TextOverlayList</u></i>

**Table 5-60 PUT /ISAPI/System/Video/inputs/channels/<ID>/overlays/text?
format=json&devIndex=<uuid>**

Method	PUT
Description	Set the multiple texts overlay parameters of a camera.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".

Request	<u><i>JSON_TextOverlayList</i></u>
Response	<u><i>JSON_ResponseStatus</i></u>

Remarks

The **ID** in the request URI is the channel No.

**/ISAPI/System/Video/inputs/channels/<ID>/overlays/text/<ID>?
format=json&devIndex=<uuid>**

Get or set the text overlay parameters of a channel.

Request URL Definition

**Table 5-61 GET /ISAPI/System/Video/inputs/channels/<ID>/overlays/text/<ID>?
format=json&devIndex=<uuid>**

Method	GET
Description	Get the text overlay parameters of a channel.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	None
Response	<u><i>JSON_TextOverlay</i></u>

**Table 5-62 PUT /ISAPI/System/Video/inputs/channels/<ID>/overlays/text/<ID>?
format=json&devIndex=<uuid>**

Method	PUT
Description	Set the text overlay parameters of a channel.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".

Request	<u><i>JSON_TextOverlay</i></u>
Response	<u><i>JSON_ResponseStatus</i></u>

Remarks

The first <ID> in the request URL is the channel No., and the second <ID> in the OSD text No.

**/ISAPI/System/Video/inputs/channels/<ID>/overlays/channelNameOverlay?
format=json&devIndex=<uuid>**

Operations about channel name overlay of a channel.

Request URL Definition

**Table 5-63 GET /ISAPI/System/Video/inputs/channels/<ID>/overlays/channelNameOverlay?
format=json&devIndex=<uuid>**

Method	GET
Description	Get the channel name overlay configuration parameters of a channel.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	None.
Response	<u><i>JSON_channelNameOverlay</i></u>

**Table 5-64 PUT /ISAPI/System/Video/inputs/channels/<ID>/overlays/channelNameOverlay?
format=json&devIndex=<uuid>**

Method	PUT
Description	Set the channel name overlay parameters of a channel.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".

Request	<i><u>JSON_channelNameOverlay</u></i>
Response	<i><u>JSON_ResponseStatus</u></i>

Remarks

The **ID** in the request URL is the channel No.

**/ISAPI/System/Video/inputs/channels/<ID>/overlays/dateTime?
format=json&devIndex=<uuid>**

Get or set the time overlay parameters of a channel.

Request URI Definition

**Table 5-65 GET /ISAPI/System/Video/inputs/channels/<ID>/overlays/dateTime?
format=json&devIndex=<uuid>**

Method	GET
Description	Get the time overlay parameters of a channel.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	None
Response	Succeeded: <i><u>JSON_DateTimeOverlay</u></i> Failed: <i><u>JSON_ResponseStatus</u></i>

**Table 5-66 PUT /ISAPI/System/Video/inputs/channels/<ID>/overlays/dateTime?
format=json&devIndex=<uuid>**

Method	PUT
Description	Set the time overlay parameters of a channel.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".

Request	<u><i>JSON DateTimeOverlay</i></u>
Response	<u><i>JSON ResponseStatus</i></u>

Remarks

- The <ID> in the request URI is the channel No.
- You can also call URI (/ISAPI/System/Video/inputs/channels/<channelID>/overlays/dateTimeOverlay?format=json&devIndex=<uuid>) to implement the function.

5.3.6 Channel Color Configurations

/ISAPI/Image/channels/<ID>/color?format=json&devIndex=<uuid>

Get or set the color parameters of a channel.

Request URI Definition

Table 5-67 GET /ISAPI/Image/channels/<ID>/color?format=json&devIndex=<uuid>

Method	GET
Description	Get the color parameters, e.g.,brightness, hue, contrast, saturation, and so on, of a specific channel.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	None
Response	<u><i>JSON Color</i></u>

Table 5-68 PUT /ISAPI/Image/channels/<ID>/color?format=json&devIndex=<uuid>

Method	PUT
Description	Set the color parameters, e.g.,brightness, hue, contrast, saturation, and so on, of a specific channel.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and

	generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	<i><u>JSON Color</u></i>
Response	<i><u>JSON ResponseStatus</u></i>

Remarks

The <ID> in the request URI is the channel No.

5.3.7 Event Configurations

/ISAPI/Event/triggers/<ID>?format=json&devIndex=<uuid>

Operations about alarm/event linkage actions.

Request URI Definition

Table 5-69 GET /ISAPI/Event/triggers/<ID>?format=json&devIndex=<uuid>

Method	GET
Description	Get the list of alarm/event linkage actions.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	None
Response	<i><u>JSON EventTrigger</u></i>

Table 5-70 PUT /ISAPI/Event/triggers/<ID>?format=json&devIndex=<uuid>

Method	PUT
Description	Set the linkage actions for alarms/events.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".

Request	<u><i>JSON_EventTrigger</i></u>
Response	<u><i>JSON_ResponseStatus</i></u>

Table 5-71 DELETE /ISAPI/Event/triggers/<ID>?format=json&devIndex=<uuid>

Method	DELETE
Description	Remove the linkage actions of alarms/events.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	None
Response	<u><i>JSON_ResponseStatus</i></u>

Remarks

- The **ID** in the URI is the alarm/event ID, which is defined as "**eventType**-channel No.", e.g., the motion detection of channel No.2 is "VMD-2". It is same with the node **id** in the *JSON_EventTrigger* message.
- The following linkage actions are supported: alarm output, channel record, audible warning, email linkage, alarm triggered video wall display, upload to alarm center, upload to FTP, wireless siren linkage, PTZ linkage, and so on.

/ISAPI/Event/schedules/motionDetections?format=json&devIndex=<uuid>

Get or set the arming schedules of motion detection alarm for multiple channels.

Request URI Definition

Table 5-72 GET /ISAPI/Event/schedules/motionDetections?format=json&devIndex=<uuid>

Method	GET
Description	Get the arming schedules of motion detection alarm for multiple channels.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and

	generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	None.
Response	Succeeded: <u>JSON MotionDetectionScheduleList</u> Failed: <u>JSON ResponseStatus</u>

Table 5-73 PUT /ISAPI/Event/schedules/motionDetections?format=json&devIndex=<uuid>

Method	PUT
Description	Set the arming schedules of motion detection alarm for multiple channels.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	<u>JSON MotionDetectionScheduleList</u>
Response	<u>JSON ResponseStatus</u>

/ISAPI/Event/schedules/motionDetections/VMD_video<channelID>? format=json&devIndex=<uuid>

Get or set the motion detection arming schedule of a specified channel.

Request URI Definition

**Table 5-74 GET /ISAPI/Event/schedules/motionDetections/VMD_video<channelID>?
format=json&devIndex=<uuid>**

Method	GET
Description	Get the motion detection arming schedule of a specified channel.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".

Request	None.
Response	Succeeded: <u><i>JSON_Schedule</i></u> Failed: <u><i>JSON_ResponseStatus</i></u>

Table 5-75 PUT /ISAPI/Event/schedules/motionDetections/VMD_video<channelID>?format=json&devIndex=<uuid>

Method	PUT
Description	Set the motion detection arming schedule of a specified channel.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	<u><i>JSON_Schedule</i></u>
Response	<u><i>JSON_ResponseStatus</i></u>

Remarks

If the video input channel number is 2, the **ID** in the URI should be VMD_video2.

/ISAPI/System/Video/inputs/channels/<ID>/motionDetection? format=json&devIndex=<uuid>

Get or set the motion detection alarm parameters of a specified video input channel.

Request URI Definition

Table 5-76 GET /ISAPI/System/Video/inputs/channels/<ID>/motionDetection?format=json&devIndex=<uuid>

Method	GET
Description	Get the motion detection alarm parameters of a specified video input channel.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".

Request	None.
Response	Succeeded: <u><i>JSON_MotionDetection</i></u> Failed: <u><i>JSON_ResponseStatus</i></u>

**Table 5-77 PUT /ISAPI/System/Video/inputs/channels/<ID>/motionDetection?
format=json&devIndex=<uuid>**

Method	PUT
Description	Set the motion detection alarm parameters for a specific video input channel.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	<u><i>JSON_MotionDetection</i></u>
Response	<u><i>JSON_ResponseStatus</i></u>

/ISAPI/Event/schedules/tamperDetections?format=json&devIndex=<uuid>

Get or set arming schedules of video tampering alarm for multiple channels.

Request URI Definition

Table 5-78 GET /ISAPI/Event/schedules/tamperDetections?format=json&devIndex=<uuid>

Method	GET
Description	Get the arming schedules of video tampering alarm for multiple channels.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	None.
Response	<u><i>JSON_TamperDetectionScheduleList</i></u>

Table 5-79 PUT /ISAPI/Event/schedules/tamperDetections?format=json&devIndex=<uuid>

Method	PUT
Description	Set the arming schedules of video tampering alarm for multiple channels.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	<u><i>JSON_TamperDetectionScheduleList</i></u>
Response	<u><i>JSON_ResponseStatus</i></u>

**/ISAPI/Event/schedules/tamperDetections/tamperdetection_video<channelID>?
format=json&devIndex=<uuid>**

Get or set the arming schedule of video tampering detection for a specified channel.

Request URL Definition

**Table 5-80 GET /ISAPI/Event/schedules/tamperDetections/tamperdetection_video<channelID>?
format=json&devIndex=<uuid>**

Method	GET
Description	Get the arming schedule of video tampering detection for a specified channel.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	None.
Response	Succeeded: <u><i>JSON_Schedule</i></u> Failed: <u><i>JSON_ResponseStatus</i></u>

Table 5-81 PUT /ISAPI/Event/schedules/tamperDetections/tamperdetection_video<channelID>?format=json&devIndex=<uuid>

Method	PUT
Description	Set the arming schedule of video tampering detection for a specified channel.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	<i><u>JSON Schedule</u></i>
Response	<i><u>JSON ResponseStatus</u></i>

Remarks

If the video input channel is 2, the **ID** in the URI should be Tamperdetection_video2.

/ISAPI/System/Video/inputs/channels/<ID>/tamperDetection? format=json&devIndex=<uuid>

Get or set the video tampering alarm parameters of a specified video input channel.

Request URI Definition

Table 5-82 GET /ISAPI/System/Video/inputs/channels/<ID>/tamperDetection?format=json&devIndex=<uuid>

Method	GET
Description	Get the video tampering alarm parameters of a specified video input channel.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	None.
Response	Succeeded: <i><u>JSON TamperDetection</u></i> Failed: <i><u>JSON ResponseStatus</u></i>

**Table 5-83 PUT /ISAPI/System/Video/inputs/channels/<ID>/tamperDetection?
format=json&devIndex=<uuid>**

Method	PUT
Description	Set the video tampering alarm parameters of a specified video input channel.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	<i><u>JSON_TamperDetection</u></i>
Response	<i><u>JSON_ResponseStatus</u></i>

/ISAPI/Event/schedules/videolosses?format=json&devIndex=<uuid>

Get or set the arming schedules of video loss alarm for multiple channels.

Request URI Definition

Table 5-84 GET /ISAPI/Event/schedules/videolosses?format=json&devIndex=<uuid>

Method	GET
Description	Get the arming schedules of video loss alarm for multiple channels.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	None.
Response	<i><u>JSON_videolossScheduleList</u></i>

Table 5-85 PUT /ISAPI/Event/schedules/videolosses?format=json&devIndex=<uuid>

Method	PUT
Description	Set the arming schedules of video loss alarm for multiple channels.
Query	format: determine the format of request or response message.

	devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	<u><i>JSON_videolossScheduleList</i></u>
Response	<u><i>JSON_ResponseStatus</i></u>

/ISAPI/Event/schedules/videolosses/Videoloss_video<channelID>? format=json&devIndex=<uuid>

Operations about arming schedule of video loss detection of a specific channel.

Request URL Definition

**Table 5-86 GET /ISAPI/Event/schedules/videolosses/Videoloss_video<channelID>?
format=json&devIndex=<uuid>**

Method	GET
Description	Get the arming schedule of video loss detection of the specific channel.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	None.
Response	<u><i>JSON_Schedule</i></u>

**Table 5-87 PUT /ISAPI/Event/schedules/videolosses/Videoloss_video<channelID>?
format=json&devIndex=<uuid>**

Method	PUT
Description	Set the arming schedule for video loss detection of the specific channel.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and

	generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	<u><i>JSON_Schedule</i></u>
Response	<u><i>JSON_ResponseStatus</i></u>

Remarks

If the video input channel number is 2, the **ID** in the URI should be Videoloss_video2.

5.3.8 Capture Manually

/ISAPI/Streaming/channels/<channelID>/picture?format=json&devIndex=<uuid>

Manually capture pictures in the video stream of a specific channel.

Request URI Definition

Table 5-88 GET /ISAPI/Streaming/channels/<channelID>/picture?format=json&devIndex=<uuid>

Method	GET
Description	Manually capture picture in the video stream of a specific channel.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	None
Response	<u><i>JSON_PictureData</i></u>

Remarks

- This URI is valid only when the event/alarm subscription is enabled and the subscription exists.
- If the channel number configured in the URI does not exist, you will not be able to capture pictures via this URI.
- You can use the returned URL to download the captured picture via the API <http://<ipAddress>:<port>/<videoUrl>> .

5.3.9 Wake Up

/ISAPI/System/wakeUp?format=json&devIndex=<uuid>

Wake up the device from sleep mode.

Request URI Definition

Table 5-89 PUT /ISAPI/System/wakeUp?format=json&devIndex=<uuid>

Method	PUT
Description	Wake up the device from sleep mode.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	None
Response	<i><u>JSON_ResponseStatus</u></i>

Remarks

After waking up the device, you should wait for the device reconnection before any other operations of the device. The reconnection takes about 1 minute.

5.3.10 Import Face Pictures to Library

/ISAPI/Intelligent/FDLib/pictureUpload?format=json&devIndex=<uuid>

Import face pictures to the face picture library.

Request URI Definition



Note

Before importing face pictures to the face picture library, you need to get the FDID via API (/ISAPI/Intelligent/FDLib/search?format=json).

Table 5-90 POST /ISAPI/Intelligent/FDLib/pictureUpload?format=json&devIndex=<uuid>

Method	POST
Description	Import face pictures to the face picture library.  Note The face pictures in binary format are supported. The binary picture data is after the JSON message. Refer to the example of request message for details.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	<i><u>JSON_PictureUploadData</u></i>
Response	Succeeded: <i><u>JSON_MaskInfo</u></i> Failed: <i><u>JSON_ResponseStatus</u></i>

5.4 Access Control Device Operations

Access Control Capability

Function	API
Get the access control capability	<i><u>/ISAPI/AccessControl/capabilities?</u></i> <i><u>format=json&devIndex=<uuid></u></i>

Person Management

Function	API
Delete persons and the linked permissions	<i><u>/ISAPI/AccessControl/UserInfoDetail/Delete?</u></i> <i><u>format=json&devIndex=<uuid></u></i>
Get progress of searching for or deleting person information and permissions	<i><u>/ISAPI/AccessControl/UserInfoDetail/DeleteProcess?</u></i> <i><u>format=json&devIndex=<uuid></u></i>
Add persons	<i><u>/ISAPI/AccessControl/UserInfo/Record?</u></i> <i><u>format=json&devIndex=<uuid></u></i>

Function	API
Edit a person's information	<u>/ISAPI/AccessControl/UserInfo/Modify?format=json&devIndex=<uuid></u>
Get the total number of persons	<u>/ISAPI/AccessControl/UserInfo/Count?format=json&devIndex=<uuid></u>
Search for person details	<u>/ISAPI/AccessControl/UserInfo/Search?format=json&devIndex=<uuid></u>

Card Management

Functions	APIs
Add a card	<u>/ISAPI/AccessControl/CardInfo/Record?format=json&devIndex=<uuid></u>
Edit a card's information	<u>/ISAPI/AccessControl/CardInfo/Modify?format=json&devIndex=<uuid></u>
Delete cards' information	<u>/ISAPI/AccessControl/CardInfo/Delete?format=json&devIndex=<uuid></u>
Get the total number of cards	<u>/ISAPI/AccessControl/CardInfo/Count?format=json?format=json</u>
Search for card details	<u>/ISAPI/AccessControl/CardInfo/Search?format=json&devIndex=<uuid></u>

Face Management

Function	API
Add a face data record by uploading binary data	<u>/ISAPI/Intelligent/FDLib/FaceDataRecord?format=json&devIndex=<uuid></u>
Delete face data records	<u>/ISAPI/Intelligent/FDLib/FDSearch/Delete?format=json&devIndex=<uuid></u>
Get the total number of face data records	<u>/ISAPI/Intelligent/FDLib/Count?format=json&devIndex=<uuid></u>
Search for face data records	<u>/ISAPI/Intelligent/FDLib/FDSearch?format=json&devIndex=<uuid></u>
Set the face data	<u>/ISAPI/Intelligent/FDLib/FDSetUp?format=json&devIndex=<uuid></u>

Fingerprint Management

Function	API
Get fingerprint configuration capability	<u>/ISAPI/AccessControl/FingerPrintCfg/capabilities?format=json&devIndex=<uuid></u>
Get or set parameters of a card reader	<u>/ISAPI/AccessControl/CardReaderCfg/<ID>?format=json&devIndex=<uuid></u>
Apply fingerprint parameters	<u>/ISAPI/AccessControl/FingerPrintDownload?format=json&devIndex=<uuid></u>
Get the applying progress of fingerprint parameters	<u>/ISAPI/AccessControl/FingerPrintProgress?format=json&devIndex=<uuid></u>
Edit fingerprint parameters	<u>/ISAPI/AccessControl/FingerPrintModify?format=json&devIndex=<uuid></u>
Search for fingerprint information	<u>/ISAPI/AccessControl/FingerPrintUpload?format=json&devIndex=<uuid></u>
Get fingerprint deleting capability	<u>/ISAPI/AccessControl/FingerPrint/Delete/capabilities?format=json&devIndex=<uuid></u>
Delete fingerprint information	<u>/ISAPI/AccessControl/FingerPrint/Delete?format=json&devIndex=<uuid></u>
Get the fingerprint deleting progress	<u>/ISAPI/AccessControl/FingerPrint/DeleteProcess?format=json&devIndex=<uuid></u>
Get capability of fingerprint collection	<u>/ISAPI/AccessControl/CaptureFingerPrint/capabilities?format=json&devIndex=<uuid></u>
Collect fingerprint data	<u>/ISAPI/AccessControl/CaptureFingerPrint?format=json&devIndex=<uuid></u>

Door Configurations and Control

Function	API
Remotely control door	<u>/ISAPI/AccessControl/RemoteControl/door/<ID>?format=json&devIndex=<uuid></u>
Get working status of access control devices	<u>/ISAPI/AccessControl/AcsWorkStatus?format=json&devIndex=<uuid></u>
Get or set door parameters	<u>/ISAPI/AccessControl/Door/param/<doorID>?format=json&devIndex=<uuid></u>

User Permission Schedule

Function	API
Get or set the person permission holiday group	<u>/ISAPI/AccessControl/UserRightHolidayGroupCfg/<holidayGroupID>?format=json&devIndex=<uuid></u>
Get or set the person permission holiday schedule	<u>/ISAPI/AccessControl/UserRightHolidayPlanCfg/<holidayPlanID>?format=json&devIndex=<uuid></u>
Get or set the person permission schedule template	<u>/ISAPI/AccessControl/UserRightPlanTemplate/<planTemplateID>?format=json&devIndex=<uuid></u>
Get or set the person permission week schedule	<u>/ISAPI/AccessControl/UserRightWeekPlanCfg/<weekPlanID>?format=json&devIndex=<uuid></u>

Face Recognition

Function	API
Get or set face recognition mode	<u>/ISAPI/AccessControl/FaceRecognizeMode?format=json&devIndex=<uuid></u>

History Events

Function	API
Search for history access control events	<u>/ISAPI/AccessControl/AcsEvent?format=json&devIndex=<uuid></u>

Alarm Listening

Function	API
Configure alarm listening servers	<u>/ISAPI/Event/notification/httpHosts?format=json&devIndex=<uuid></u>

5.4.1 Access Control Capability

[/ISAPI/AccessControl/capabilities?format=json&devIndex=<uuid>](#)

Get access control capability.

Request URI Definition

Table 5-91 GET /ISAPI/AccessControl/capabilities?format=json&devIndex=<uuid>

Method	GET
Description	Get access control capability.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	None
Response	Succeeded: <i><u>JSON AccessControlCap</u></i> Failed: <i><u>JSON ResponseStatus</u></i>

5.4.2 Person Management

/ISAPI/AccessControl/UserInfoDetail/Delete?format=json&devIndex=<uuid>

Delete persons and the linked permissions.

Request URI Definition

Table 5-92 GET /ISAPI/AccessControl/UserInfoDetail/Delete?format=json&devIndex=<uuid>

Method	GET
Description	Delete persons and the linked permissions.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	<i><u>JSON UserInfoDetail</u></i>
Response	<i><u>JSON ResponseStatus</u></i>

Remarks

- If the person information is deleted, the linked card information, fingerprint information, and face information will also be deleted.
- When calling this API succeeded, it does not mean that the persons are deleted. Only when the value of "**progressValue**" returned by calling corresponding URI is "success", it indicates deleting succeeded.

/ISAPI/AccessControl/UserInfoDetail/DeleteProcess? format=json&devIndex=<uuid>

Get progress of searching for or deleting person information and permissions.

Request URI Definition

**Table 5-93 GET /ISAPI/AccessControl/UserInfoDetail/DeleteProcess?
format=json&devIndex=<uuid>**

Method	GET
Description	Get progress of searching for or deleting person information and permissions.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	None
Response	Succeeded: <u><i>JSON UserInfoDetailDeleteProcess</i></u> Failed: <u><i>JSON ResponseStatus</i></u>

Remarks

You can repeatedly call this API to get the progress until the value of "**status**" returned in the message *JSON UserInfoDetailDeleteProcess* is "success" or "failed".

/ISAPI/AccessControl/UserInfo/Record?format=json&devIndex=<uuid>

Add persons.

Request URI Definition

Table 5-94 POST /ISAPI/AccessControl/UserInfo/Record?format=json&devIndex=<uuid>

Method	POST
Description	Add persons.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	<i><u>JSON UserInfo</u></i>
Response	Succeeded: <i><u>JSON UserInfoOutList</u></i> Failed: <i><u>JSON ResponseStatus</u></i>

Remarks

Up to 30 persons can be added for one time.

/ISAPI/AccessControl/UserInfo/Modify?format=json&devIndex=<uuid>

Edit a person' information.

Request URI Definition

Table 5-95 PUT /ISAPI/AccessControl/UserInfo/Modify?format=json&devIndex=<uuid>

Method	PUT
Description	Edit a person' information.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	<i><u>JSON Edit UserInfo</u></i>
Response	<i><u>JSON ResponseStatus</u></i>

/ISAPI/AccessControl/UserInfo/Count?format=json&devIndex=<uuid>

Get the total number of persons.

Request URI Definition

Table 5-96 GET /ISAPI/AccessControl/UserInfo/Count?format=json&devIndex=<uuid>

Method	GET
Description	Get the total number of persons.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	None
Response	Succeeded: <u><i>JSON_UserInfoCount</i></u> Failed: <u><i>JSON_ResponseStatus</i></u>

/ISAPI/AccessControl/UserInfo/Search?format=json&devIndex=<uuid>

Search for person details.

Request URI Definition

Table 5-97 POST /ISAPI/AccessControl/UserInfo/Search?format=json&devIndex=<uuid>

Method	POST
Description	Search for person details.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	<u><i>JSON_UserInfoSearchCond</i></u>
Response	Succeeded: <u><i>JSON_UserInfoSearch</i></u> Failed: <u><i>JSON_ResponseStatus</i></u>

5.4.3 Card Management

/ISAPI/AccessControl/CardInfo/Record?format=json&devIndex=<uuid>

Add a card.

Request URI Definition

Table 5-98 POST /ISAPI/AccessControl/CardInfo/Record?format=json&devIndex=<uuid>

Method	POST
Description	Add a card.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	<u><i>JSON_CardInfo</i></u>
Response	<u><i>JSON_ResponseStatus</i></u>

/ISAPI/AccessControl/CardInfo/Modify?format=json&devIndex=<uuid>

Edit a card's information.

Request URI Definition

Table 5-99 PUT /ISAPI/AccessControl/CardInfo/Modify?format=json&devIndex=<uuid>

Method	PUT
Description	Edit a card's information.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	<u><i>JSON_CardInfo</i></u>
Response	<u><i>JSON_ResponseStatus</i></u>

Remarks

The employee ID and card No. cannot be edited. If you want to edit the card No., you should delete the card first and add a new card again.

/ISAPI/AccessControl/CardInfo/Delete?format=json&devIndex=<uuid>

Delete cards' information.

Request URI Definition

Table 5-100 PUT /ISAPI/AccessControl/CardInfo/Delete?format=json&devIndex=<uuid>

Method	PUT
Description	Delete cards' information.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	<u><i>JSON_CardInfoDelCond</i></u>
Response	<u><i>JSON_ResponseStatus</i></u>

/ISAPI/AccessControl/CardInfo/Count?format=json?format=json

Get the total number of cards.

Request URI Definition

Table 5-101 GET /ISAPI/AccessControl/CardInfo/Count?format=json?format=json

Method	GET
Description	Get the total number of cards.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".

Request	None
Response	Succeeded: <u><i>JSON_CardInfoCount</i></u> Failed: <u><i>JSON_ResponseStatus</i></u>

/ISAPI/AccessControl/CardInfo/Search?format=json&devIndex=<uuid>

Search for card details.

Request URI Definition

Table 5-102 POST /ISAPI/AccessControl/CardInfo/Search?format=json&devIndex=<uuid>

Method	POST
Description	Search for card details.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	<u><i>JSON_CardInfoSearchCond</i></u>
Response	Succeeded: <u><i>JSON_CardInfoSearch</i></u> Failed: <u><i>JSON_ResponseStatus</i></u>

5.4.4 Face Management

/ISAPI/Intelligent/FDLib/FaceDataRecord?format=json&devIndex=<uuid>

Add a face data record by uploading binary data.

Request URI Definition

Table 5-103 POST /ISAPI/Intelligent/FDLib/FaceDataRecord?format=json&devIndex=<uuid>

Method	POST
Description	Add a face data record by uploading binary data.
Query	format: determine the format of request or response message.

	devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	<u><i>JSON_FaceInfo</i></u>
Response	<u><i>JSON_ResponseStatus</i></u>

Remarks

You can directly input the binary picture data after the JSON message when setting the face picture of a person.

/ISAPI/Intelligent/FDLib/FDSearch/Delete?format=json&devIndex=<uuid>

Delete face data records, and it supports deleting in a batch.

Request URI Definition

Table 5-104 PUT /ISAPI/Intelligent/FDLib/FDSearch/Delete?format=json&devIndex=<uuid>

Method	PUT
Description	Delete face data records.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	<u><i>JSON_FaceInfoDelCond</i></u>
Response	<u><i>JSON_ResponseStatus</i></u>

/ISAPI/Intelligent/FDLib/Count?format=json&devIndex=<uuid>

Get the total number of face data records.

Request URI Definition

Table 5-105 GET /ISAPI/Intelligent/FDLib/Count?format=json&devIndex=<uuid>

Method	GET
Description	Get the total number of face data records.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	None
Response	Succeeded: <u><i>JSON_FaceInfoCount</i></u> Failed: <u><i>JSON_ResponseStatus</i></u>

/ISAPI/Intelligent/FDLib/FDSearch?format=json&devIndex=<uuid>

Search for face data records.

Request URI Definition

Table 5-106 POST /ISAPI/Intelligent/FDLib/FDSearch?format=json&devIndex=<uuid>

Method	POST
Description	Search for face data records.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	<u><i>JSON_FaceInfoSearchCond</i></u>
Response	Succeeded: <u><i>JSON_FaceInfoSearch</i></u> Failed: <u><i>JSON_ResponseStatus</i></u>

/ISAPI/Intelligent/FDLib/FDSetUp?format=json&devIndex=<uuid>

Set the face data.

Request URI Definition

Table 5-107 PUT /ISAPI/Intelligent/FDLib/FDSetUp?format=json&devIndex=<uuid>

Method	PUT
Description	Set the face data.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	<i><u>JSON FPID</u></i>
Response	<i><u>JSON ResponseStatus</u></i>

Remarks

- You can directly input the binary picture data after the JSON message when setting the face picture of a person.
- For importing the face picture to a face picture library, if face picture of the person ID (FPID) does not exist, a new data will be created, if face picture of the person ID (FPID) exists, then the face data will be covered.

5.4.5 Fingerprint Management

/ISAPI/AccessControl/FingerPrintCfg/capabilities?format=json&devIndex=<uuid>

Get fingerprint configuration capability.

Request URI Definition

Table 5-108 GET /ISAPI/AccessControl/FingerPrintCfg/capabilities?
format=json&devIndex=<uuid>

Method	GET
Description	Get fingerprint configuration capability.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and

	generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	None
Response	Succeeded: <i><u>JSON_Cap_FingerPrintCfg</u></i> Failed: <i><u>JSON_ResponseStatus</u></i>

/ISAPI/AccessControl/CardReaderCfg/<ID>?format=json&devIndex=<uuid>

Get or set parameters of a card reader.

Request URI Definition

Table 5-109 GET /ISAPI/AccessControl/CardReaderCfg/<ID>?format=json&devIndex=<uuid>

Method	GET
Description	Get parameters of a card reader.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	None
Response	Succeeded: <i><u>JSON_CardReaderCfg</u></i> Failed: <i><u>JSON_ResponseStatus</u></i>

Table 5-110 PUT /ISAPI/AccessControl/CardReaderCfg/<ID>?format=json&devIndex=<uuid>

Method	PUT
Description	Set parameters of a fingerprint and card reader.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	<i><u>JSON_CardReaderCfg</u></i>
Response	<i><u>JSON_ResponseStatus</u></i>

Remarks

The <ID> in the request URI refers to the fingerprint and card reader No., which starts from 1.

/ISAPI/AccessControl/FingerPrintDownload?format=json&devIndex=<uuid>

Apply fingerprint parameters.

Request URI Definition

Table 5-111 POST /ISAPI/AccessControl/FingerPrintDownload?format=json&devIndex=<uuid>

Method	POST
Description	Apply fingerprint parameters.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	<i>JSON_FingerPrintCfg</i>
Response	<i>JSON_ResponseStatus</i>

Remarks

When the message ***JSON_ResponseStatus*** is returned, it does not indicate the fingerprint parameters are applied, you should call related URI to get the applying progress of fingerprint parameters for checking the actual applying status.

/ISAPI/AccessControl/FingerPrintProgress?format=json&devIndex=<uuid>

Get applying progress of fingerprint parameters.

Request URI Definition

Table 5-112 GET /ISAPI/AccessControl/FingerPrintProgress?format=json&devIndex=<uuid>

Method	GET
Description	Get applying progress of fingerprint parameters.
Query	format: determine the format of request or response message.

	devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	None
Response	Succeeded: <u><i>JSON_FingerPrintStatus</i></u> Failed: <u><i>JSON_ResponseStatus</i></u>

Remarks

During the applying, you should repeatedly call this API to get the progress until the value of node "totalStatus" in the response message *JSON_FingerPrintStatus* is 1

/ISAPI/AccessControl/FingerPrintModify?format=json&devIndex=<uuid>

Edit the fingerprint parameters.

Request URI Definition

Table 5-113 POST /ISAPI/AccessControl/FingerPrintModify?format=json&devIndex=<uuid>

Method	POST
Description	Edit the fingerprint parameters.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	<u><i>JSON_FingerPrintModify</i></u>
Response	<u><i>JSON_ResponseStatus</i></u>

Remarks

This API is only used to edit the fingerprint parameters without applying.

/ISAPI/AccessControl/FingerPrintUpload?format=json&devIndex=<uuid>

Search for fingerprint information.

Request URI Definition

Table 5-114 POST /ISAPI/AccessControl/FingerPrintUpload?format=json&devIndex=<uuid>

Method	POST
Description	Search for fingerprint information.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	<i><u>JSON_FingerPrintCond</u></i>
Response	Succeeded: <i><u>JSON_FingerPrintInfo</u></i> Failed: <i><u>JSON_ResponseStatus</u></i>

Remarks

When searching for the information of all fingerprints linked with the specific employee ID (person ID), if succeeded, you should repeatedly call this API to get the search results for multiple times until the value of node "**status**" in the message ***JSON_FingerPrintInfo*** is "NoFp".

/ISAPI/AccessControl/FingerPrint/Delete/capabilities? format=json&devIndex=<uuid>

Get fingerprint deleting capability.

Request URI Definition

**Table 5-115 GET /ISAPI/AccessControl/FingerPrint/Delete/capabilities?
format=json&devIndex=<uuid>**

Method	GET
Description	Get fingerprint deleting capability.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".

Request	None
Response	Succeeded: <u><i>JSON_Cap_FingerPrintDelete</i></u> Failed: <u><i>JSON_ResponseStatus</i></u>

/ISAPI/AccessControl/FingerPrint/Delete?format=json&devIndex=<uuid>

Delete fingerprint information.

Request URI Definition

Table 5-116 PUT /ISAPI/AccessControl/FingerPrint/Delete?format=json&devIndex=<uuid>

Method	PUT
Description	Delete fingerprint information.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	<u><i>JSON_FingerPrintDelete</i></u>
Response	<u><i>JSON_ResponseStatus</i></u>

Remarks

When the message *JSON_ResponseStatus* is returned, it does not indicate the fingerprint parameters are deleted, you should call related URI to get the deleting progress of fingerprint information for checking the actual deleting status.

/ISAPI/AccessControl/FingerPrint/DeleteProcess?format=json&devIndex=<uuid>

Get the fingerprint deleting progress.

Request URI Definition

Table 5-117 GET /ISAPI/AccessControl/FingerPrint/DeleteProcess?format=json&devIndex=<uuid>

Method	GET
Description	Get the fingerprint deleting progress.

Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	None
Response	Succeeded: <u><i>JSON_FingerPrintDeleteProcess</i></u> Failed: <u><i>JSON_ResponseStatus</i></u>

Remarks

During the deleting, you should repeatedly call this API to get the progress until the value of node "status" in the response message *JSON_FingerPrintDeleteProcess* is "success" or "failed".

/ISAPI/AccessControl/CaptureFingerPrint/capabilities? format=json&devIndex=<uuid>

Get capability of fingerprint collection.

Request URI Definition

**Table 5-118 GET /ISAPI/AccessControl/CaptureFingerPrint/capabilities?
format=json&devIndex=<uuid>**

Method	GET
Description	Get capability of fingerprint collection.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	None
Response	Succeeded: <u><i>JSON_Cap_CaptureFingerPrint</i></u> Failed: <u><i>JSON_ResponseStatus</i></u>

/ISAPI/AccessControl/CaptureFingerPrint?format=json&devIndex=<uuid>

Collect fingerprint data.

Request URI Definition

Table 5-119 POST /ISAPI/AccessControl/CaptureFingerPrint?format=json&devIndex=<uuid>

Method	POST
Description	Collect fingerprint data.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	<i><u>JSON_CaptureFingerPrintCond</u></i>
Response	Succeeded: <i><u>JSON_CaptureFingerPrint</u></i> Failed: <i><u>JSON_ResponseStatus</u></i>

5.4.6 Door Configurations and Control

/ISAPI/AccessControl/RemoteControl/door/<ID>?format=json&devIndex=<uuid>

Remotely control door.

Request URI Definition

Table 5-120 PUT /ISAPI/AccessControl/RemoteControl/door/<ID>?format=json&devIndex=<uuid>

Method	PUT
Description	Remotely control door.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	<i><u>JSON_RemoteControlDoor</u></i>
Response	<i><u>JSON_ResponseStatus</u></i>

Remarks

The <ID> in the request URI refers to the door No., its value is between 1 and 65535; when the value is 65535, it indicates all doors.

/ISAPI/AccessControl/AcsWorkStatus?format=json&devIndex=<uuid>

Get working status of access control devices.

Request URI Definition

Table 5-121 GET /ISAPI/AccessControl/AcsWorkStatus?format=json&devIndex=<uuid>

Method	GET
Description	Get working status of access control devices.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	None
Response	Succeeded: <u><i>JSON AcsWorkStatus</i></u> Failed: <u><i>JSON ResponseStatus</i></u>

/ISAPI/AccessControl/Door/param/<doorID>?format=json&devIndex=<uuid>

Get or set the door parameters

Request URI Definition

Table 5-122 GET /ISAPI/AccessControl/Door/param/<doorID>?format=json&devIndex=<uuid>

Method	GET
Description	Get the door parameters.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".

Request	None
Response	Succeeded: <u><i>JSON_DoorParam</i></u> Failed: <u><i>JSON_ResponseStatus</i></u>

Table 5-123 PUT /ISAPI/AccessControl/Door/param/<doorID>?format=json&devIndex=<uuid>

Method	PUT
Description	Set the door parameters.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	<u><i>JSON_DoorParam</i></u>
Response	<u><i>JSON_ResponseStatus</i></u>

Remarks

The <doorID> in the URI refers to the door ID, which starts from 1.

5.4.7 User Permission Schedule

**/ISAPI/AccessControl/UserRightHolidayGroupCfg/<holidayGroupID>?
format=json&devIndex=<uuid>**

Get or set the person permission holiday group.

Request URI Definition

**Table 5-124 GET /ISAPI/AccessControl/UserRightHolidayGroupCfg/<holidayGroupID>?
format=json&devIndex=<uuid>**

Method	GET
Description	Get the person permission holiday group.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and

	generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	None
Response	Succeeded: <u><i>JSON UserRightHolidayGroupCfg</i></u> Failed: <u><i>JSON ResponseStatus</i></u>

**Table 5-125 PUT /ISAPI/AccessControl/UserRightHolidayGroupCfg/<holidayGroupID>?
format=json&devIndex=<uuid>**

Method	PUT
Description	Set the person permission holiday group.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	<u><i>JSON UserRightHolidayGroupCfg</i></u>
Response	<u><i>JSON ResponseStatus</i></u>

Remarks

The <holidayGroupID> in the URI refers to the holiday group number, which starts from 1.

/ISAPI/AccessControl/UserRightHolidayPlanCfg/<holidayPlanID>? format=json&devIndex=<uuid>

Get or set the person permission holiday schedule.

Request URI Definition

**Table 5-126 GET /ISAPI/AccessControl/UserRightHolidayPlanCfg/<holidayPlanID>?
format=json&devIndex=<uuid>**

Method	GET
Description	Get the person permission holiday schedule.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and

	generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	None
Response	Succeeded: <i>JSON UserRightHolidayPlanCfg</i> Failed: <i>JSON ResponseStatus</i>

Table 5-127 PUT /ISAPI/AccessControl/UserRightHolidayPlanCfg/<holidayPlanID>?format=json&devIndex=<uuid>

Method	PUT
Description	Set the person permission holiday schedule.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	<i>JSON UserRightHolidayPlanCfg</i>
Response	<i>JSON ResponseStatus</i>

Remarks

The <holidayPlanID> in the URI refers to the holiday schedule number, which starts from 1.

/ISAPI/AccessControl/UserRightPlanTemplate/<planTemplateID>?format=json&devIndex=<uuid>

Get or set the person permission schedule template.

Request URI Definition

Table 5-128 GET /ISAPI/AccessControl/UserRightPlanTemplate/<planTemplateID>?format=json&devIndex=<uuid>

Method	GET
Description	Get the person permission schedule template.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and

	generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	None
Response	Succeeded: <i>JSON UserRightPlanTemplate</i> Failed: <i>JSON ResponseStatus</i>

Table 5-129 PUT /ISAPI/AccessControl/UserRightPlanTemplate/<planTemplateID>?format=json&devIndex=<uuid>

Method	PUT
Description	Set the person permission schedule template.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	<i>JSON UserRightPlanTemplate</i>
Response	<i>JSON ResponseStatus</i>

Remarks

The <planTemplateID> in the URI refers to the schedule template number, which starts from 1.

/ISAPI/AccessControl/UserRightWeekPlanCfg/<weekPlanID>?format=json&devIndex=<uuid>

Get or set the person permission week schedule.

Request URI Definition

Table 5-130 GET /ISAPI/AccessControl/UserRightWeekPlanCfg/<weekPlanID>?format=json&devIndex=<uuid>

Method	GET
Description	Get the person permission week schedule.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and

	generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	None
Response	Succeeded: <i>JSON <u>UserRightWeekPlanCfg</u></i> Failed: <i>JSON <u>ResponseStatus</u></i>

Table 5-131 PUT /ISAPI/AccessControl/UserRightWeekPlanCfg/<weekPlanID>?format=json&devIndex=<uuid>

Method	PUT
Description	Set the person permission week schedule.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	<i>JSON <u>UserRightWeekPlanCfg</u></i>
Response	<i>JSON <u>ResponseStatus</u></i>

Remarks

The <weekPlanID> in the URI refers to the week schedule number, which starts from 1.

5.4.8 Face Recognition

/ISAPI/AccessControl/FaceRecognizeMode?format=json&devIndex=<uuid>

Get or set face recognition mode.

Request URI Definition

Table 5-132 GET /ISAPI/AccessControl/FaceRecognizeMode?format=json&devIndex=<uuid>

Method	GET
Description	Get face recognition mode.
Query	format: determine the format of request or response message.

	devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	None
Response	Succeeded: <u><i>JSON FaceRecognizeMode</i></u> Failed: <u><i>JSON ResponseStatus</i></u>

Table 5-133 PUT /ISAPI/AccessControl/FaceRecognizeMode?format=json&devIndex=<uuid>

Method	PUT
Description	Set face recognition mode.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	<u><i>JSON FaceRecognizeMode</i></u>
Response	<u><i>JSON ResponseStatus</i></u>

Remarks

- When switching the face recognition modes, all permissions linked with the face will be cleared.
- After switching the face recognition modes, you should reboot the device, otherwise, error code "7" with sub status code "autoReboot" will be returned.

5.4.9 History Events

/ISAPI/AccessControl/AcsEvent?format=json&devIndex=<uuid>

Search for history access control events.

Request URI Definition

Table 5-134 POST /ISAPI/AccessControl/AcsEvent?format=json&devIndex=<uuid>

Method	POST
Description	Search for history access control events.

Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	<u>JSON_AcsEventCond</u>
Response	Succeeded: <u>JSON_AcsEvent</u> Failed: <u>JSON_ResponseStatus</u>

5.4.10 Alarm Listening

/ISAPI/Event/notification/httpHosts?format=json&devIndex=<uuid>

Get or set parameters of all alarm listening servers, add alarm listening server(s) for directly receiving alarm from device, and delete all listening servers.

Request URI Definition

Table 5-135 GET /ISAPI/Event/notification/httpHosts?format=json&devIndex=<uuid>

Method	GET
Description	Get parameters of all alarm listening servers.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	None
Response	Succeeded: <u>JSON_HttpHostNotificationList</u> Failed: <u>JSON_ResponseStatus</u>

Table 5-136 POST /ISAPI/Event/notification/httpHosts?format=json&devIndex=<uuid>

Method	POST
Description	Add alarm listening server(s) for directly receiving alarm from device.
Query	format: determine the format of request or response message.

	devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	<i>JSON_HttpHostNotificationList</i>
Response	<i>JSON_ResponseStatus</i>

Table 5-137 PUT /ISAPI/Event/notification/httpHosts?format=json&devIndex=<uuid>

Method	PUT
Description	Set parameters of all alarm listening servers.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	<i>JSON_HttpHostNotificationList</i>
Response	<i>JSON_ResponseStatus</i>

Table 5-138 DELETE /ISAPI/Event/notification/httpHosts?format=json&devIndex=<uuid>

Method	DELETE
Description	Delete all alarm listening servers.
Query	format: determine the format of request or response message. devIndex: a uuid or guid (a 32-byte or 128-bit random number) for locating the connected or lower-level devices, which is unique and generated by operating system when adding device, and its format is "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx".
Request	None
Response	<i>JSON_ResponseStatus</i>

5.5 Alarm Subscription

Function	API
Get subscribed alarm/event types of a specific device or add types to be subscribed	<u>/ISAPI/Event/notification/subscribeDeviceMgmt/<ID>/devIndex/<uuid>?format=json</u>
Get the alarm/event subscription status	<u>/ISAPI/Event/notification/subscribeDeviceMgmt/<ID>/queryStatus?format=json</u>
Enable subscribing to alarms/events	<u>/ISAPI/Event/notification/subscribeDeviceMgmt?format=json</u>
Cancel subscribed alarm/event types of a specific device	<u>/ISAPI/Event/notification/unSubscribeDeviceMgmt/<ID>/devIndex/<uuid>?format=json</u>
Disable alarm/event subscription	<u>/ISAPI/Event/notification/unSubscribeDeviceMgmt/<ID>?format=json</u>

5.5.1 /ISAPI/Event/notification/subscribeDeviceMgmt?format=json

Enable subscribing alarm/event.

Request URI Definition

Table 5-139 POST /ISAPI/Event/notification/subscribeDeviceMgmt?format=json

Method	POST
Description	Enable subscribing alarm/event.
Query	format: determine the format of request or response message.
Request	<u>JSON_SubscribeDeviceMgmt</u>
Response	Succeeded: <u>JSON_SubscribeDeviceMgmtRsp</u> + <u>JSON_EventNotificationAlert_AlarmEventInfo</u> or <u>JSON_EventNotificationAlert_HeartbeatInfo</u> Failed: <u>JSON_ResponseStatus</u>

Remarks

- The *JSON_EventNotificationAlert_AlarmEventInfo* and *JSON_EventNotificationAlert_HeartbeatInfo* are uploaded repeatedly.
- When the value of **eventMode** is "all", subscribing for all devices is supported, while subscribing for a single device is no longer supported.
- When the value of **eventMode** is "all", Hik Device Gateway will subscribe event and alarm of all added devices. For the devices added later, it will subscribe their event and alarm automatically without further operation.

5.5.2 /ISAPI/Event/notification/subscribeDeviceMgmt/<ID>/devIndex/<uuid>?format=json

Get subscribed alarm/event types of the specific device or add types to be subscribed.

Request URI Definition

Table 5-140 GET /ISAPI/Event/notification/subscribeDeviceMgmt/<ID>/devIndex/<uuid>?format=json

Method	GET
Description	Get the subscribed alarm/event types of the specific device.
Query	format: determine the format of request or response message.
Request	None.
Response	Succeeded: <i><u>JSON_SubscribeDevEvent</u></i> Failed: <i><u>JSON_ResponseStatus</u></i>

Table 5-141 POST /ISAPI/Event/notification/subscribeDeviceMgmt/<ID>/devIndex/<uuid>?format=json

Method	POST
Description	Add alarm/event types of the specific device to be subscribed.
Query	format: determine the format of request or response message.
Request	<i><u>JSON_SubscribeDevEvent</u></i>
Response	<i><u>JSON_ResponseStatus</u></i>

Remarks

The **ID** in the URI is the subscription ID, and the **uuid** is the device ID.

5.5.3 /ISAPI/Event/notification/subscribeDeviceMgmt/<ID>/queryStatus?format=json

Get the alarm/event subscription status.

Request URI Definition

Table 5-142 GET /ISAPI/Event/notification/subscribeDeviceMgmt/<ID>/queryStatus?format=json

Method	GET
Description	Get the alarm/event subscription status.
Query	format : determine the format of request or response message.
Request	None.
Response	Succeeded: <u><i>JSON_SubscribeQueryStatusList</i></u> Failed: <u><i>JSON_ResponseStatus</i></u>

Remarks

The <ID> in the request URI refers to the subscription ID.

5.5.4 /ISAPI/Event/notification/unSubscribeDeviceMgmt/<ID>/devIndex/<uuid>?format=json

Cancel subscribing the alarm/event types of a specific device.

Request URI Definition

Table 5-143 DELETE /ISAPI/Event/notification/unSubscribeDeviceMgmt/<ID>/devIndex/<uuid>?format=json

Method	DELETE
Description	Cancel subscribing the alarm/event types of a specific device.
Query	format : determine the format of request or response message.
Request	None.
Response	<u><i>JSON_ResponseStatus</i></u>

Remarks

The **ID** in the URI is the subscription ID, which is returned by device; the **uuid** is the device ID.

5.5.5 /ISAPI/Event/notification/unSubscribeDeviceMgmt/<ID>?format=json

Disable alarm/event subscription.

Request URI Definition

Table 5-144 DELETE /ISAPI/Event/notification/unSubscribeDeviceMgmt/<ID>?format=json

Method	DELETE
Description	Disable alarm/event subscription.
Query	format : determine the format of request or response message.
Request	None.
Response	<u><i>JSON_ResponseStatus</i></u>

Remarks

The **ID** in the URI is the subscription ID, which is returned by the device gateway.

Appendix A. Request and Response Messages

A.1 JSON_AccessControlCap

JSON message about access control capability

```
{
  "AccessControlCap": {
    "maxPeopleRecordNum": 6000,
    /*int, maximum number of persons*/
    "maxCardRecordNum": 10000,
    /*int, maximum number of cards*/
    "maxFaceRecordNum": 6000,
    /*int, maximum number of face records*/
    "RemoteControlDoorCap": {
      "doorNo": {
        "@min": 1,
        /*int, minimum door No.*/
        "@max": 1
        /*int, maximum door No.*/
      },
      "cmd": {
        "@opt": "open,close,alwaysOpen,alwaysClose,visitorCallLadder,householdCallLadder"
        /* string, command*/
      }
    },
    "FaceRecognizeMode": {
      "mode": {
        /*optional, string, face recognition mode*/
        "@opt": "normalMode,deepMode",
      },
    }
    "cardNo": {
      /*required, string, card No.*/
      "@min": 1,
      "@max": 32
    },
    "cardType": {
      "@opt": "normalCard,patrolCard,hijackCard,superCard"
    },
    "employeeNo": {
      /*required, string, employee ID (person ID)*/
      "@min": 1,
      "@max": 20
    },
    "userType": {
      "@opt": "normal,visitor,blackList"
    },
    "name": {
```



```
"@min": 1,
"@max": 128
},
"Valid": {
  "enable": true,false,
  "beginTime": {
    "@min": 1,
    "@max": 32
  },
  "endTime": {
    "@min": 1,
    "@max": 32
  },
  "timeRangeBegin": "1970-01-01T00:00:00",
  "timeRangeEnd": "2037-12-31T23:59:59",
  "timeType": {
    "@opt": "local"
  }
}
}
```

A.2 JSON_AcsEvent

JSON message about results of search for history access control events

```
{
  "AcsEvent": {
    "searchID": "",
    /*required, string, search ID, which is used to check whether the current search requester is the same as the previous
    one; if they are the same, the search record will be stored in the device to speed up the next search*/
    "responseStatusStrg": "",
    /*required, string, search status description: "OK"-search ended, "MORE"-searching, "NO MATCH"-no matched data
    found*/
    "numOfMatches": ,
    /*required, integer32, number of records returned for one time*/
    "totalMatches": ,
    /*required, integer32, number of matched records returned for one time*/
    "deviceSerial": "D39983214",
    /*optional, string, device serial No.*/
    "MatchList": [{
    /*this node will be returned when the value of "totalMatches" is larger than 0*/
      "major": ,
      /*required, int, macro definition value of major type, refer to Access Control Event Types for details, the value is a
      decimal number when transmitting, 0-all/
      "minor": ,
      /*required, int, macro definition value of minor type, refer to Access Control Event Types for details, the value is a
      decimal number when transmitting, 0-all/
      "time": "",
      /*required, string, time (UTC time), e.g., "2016-12-12T17:30:08+08:00"*/
```

```
"netUser": "",
/*optional, string, network user name*/
"remoteHostAddr": "",
/*optional, string, remote server address*/
"cardNo": "",
/*optional, string, card No.*/
"cardType": ,
/*optional, int, card Type: 1-normal card, 2-disabled card, 3-blocklist card, 4-patrol card, 5-duress card, 6-supper card,
7-visitor card, 8-dismiss card*/
"whiteListNo": ,
/*optional, int, allowlist number*/
"reportChannel": ,
/*optional, int, uploading channel: 1-arm to upload, 2-upload to center group 1, 3-upload to center group 2*/
"cardReaderKind": ,
/*optional, reader type: 1-IC reader, 2-ID card reader, 3-QR code reader, 4-fingerprint and card reader*/
"cardReaderNo": ,
/*optional, int, reader No.*/
"doorNo": ,
/*optional, int, door No. (floor No.)*/
"verifyNo": ,
/*optional, int, multiple authentication No.*/
"alarmInNo": ,
/*optional, int, alarm input No.*/
"alarmOutNo": ,
/*optional, int, alarm output No.*/
"caseSensorNo": ,
/*optional, int, event trigger No.*/
"RS485No": ,
/*optional, int, RS-485 channel No.*/
"multiCardGroupNo": ,
/*optional, int, group No.*/
"accessChannel": ,
/*optional, int, Turnstile No.*/
"deviceNo": ,
/*optional, int, device No.*/
"distractControlNo": ,
/*optional, int, distributed controller No.*/
"employeeNoString": "",
/*optional, string, employee ID (person ID)*/
"localControllerID": ,
/*optional, int, distributed access controller No.: 0-main controller, 1 to 64-distributed controller*/
"InternetAccess": ,
/*optional, int, network interface ID: 1-upstream interface 1, 2-upstream interface 2, 3-downstream interface 1*/
"type": ,
/*optional, int, zone type: 0-instant zone, 1-24-hour zone, 2-delayed zone, 3-internal zone, 4-key zone, 5-fire zone, 6-
perimeter zone, 7-24-hour silent zone, 8-24-hour auxiliary zone, 9-24-hour vibration zone, 10-panic door open zone,
11-panic door closed zone, 255-none*/
"MACAddr": "",
/*optional, string, Mac address*/
"swipeCardType": ,
/*optional, int, authentication type: 0-invalid, 1-QR code*/
"serialNo": ,
```

```
/*optional, int, event serial No.*/
"channelControllerID": ,
/*optional, int, lane controller ID: 1-main controller, 2-sub controller*/
"channelControllerLampID": ,
/*optional, int, lamp board ID of lane controller, which is between 1 and 255*/
"channelControllerIRAdaptorID": ,
/*optional,int, IR adaptor ID of lane controller, which is between 1 and 255*/
"channelControllerIREmitterID": ,
/*optional, int, active infrared intrusion detector ID of lane controller, which is between 1 and 255*/
"userType": "",
/*optional, string, person type: "normal"-normal person (resident), "visitor", "blacklist"-person in blacklist,
"administrators"*/
"currentVerifyMode": "",
/*optional, string, authentication mode: "cardAndPw"-card+password, "card", "cardOrPw"-card or password, "fp"-
fingerprint, "fpAndPw"-fingerprint+password, "fpOrCard"-fingerprint or card, "fpAndCard"-fingerprint+card,
"fpAndCardAndPw"-fingerprint+card+password, "faceOrFpOrCardOrPw"-face or fingerprint or card or password,
"faceAndFp"-face+fingerprint, "faceAndPw"-face+password, "faceAndCard"-face+card, "face", "employeeNoAndPw"-
employee ID+password, "fpOrPw"-fingerprint or password, "employeeNoAndFp"-employee ID+fingerprint,
"employeeNoAndFpAndPw"-employee ID+fingerprint+password, "faceAndFpAndCard"-face+fingerprint+card,
"faceAndPwAndFp"-face+password+fingerprint, "employeeNoAndFace"-employee ID+face, "faceOrfaceAndCard"-face
or face+card, "fpOrface"-fingerprint or face, "cardOrfaceOrPw"-card or face or password, "cardOrFace"-card or face,
"cardOrFaceOrFp"-card or face or fingerprint*/
"currentVerifyMode": "",
/*optional, string, authentication mode: "cardAndPw"-card+password, "card", "cardOrPw"-card or password, "fp"-
fingerprint, "fpAndPw"-fingerprint+password, "fpOrCard"-fingerprint or card, "fpAndCard"-fingerprint+card,
"fpAndCardAndPw"-fingerprint+card+password, "faceOrFpOrCardOrPw"-face or fingerprint or card or password,
"faceAndFp"-face+fingerprint, "faceAndPw"-face+password, "faceAndCard"-face+card, "face", "employeeNoAndPw"-
employee No.+password, "fpOrPw"-fingerprint or password, "employeeNoAndFp"-employee No.+fingerprint,
"employeeNoAndFpAndPw"-employee No.+fingerprint+password, "faceAndFpAndCard"-face+fingerprint+card,
"faceAndPwAndFp"-face+password+fingerprint, "employeeNoAndFace"-employee No.+face, "faceOrfaceAndCard"-
face or face+card, "fpOrface"-fingerprint or face, "cardOrfaceOrPw"-card or face or password, "cardOrFpOrPw"-card
or fingerprint or password*/
"QRCodeInfo": "test",
/*optional, string, QR code informaiton*/
"thermometryUnit": "",
/*optional, string, temperature unit: "celsius"-Celsius (default), "fahrenheit"-Fahrenheit, "kelvin"-Kelvin*/
"currTemperature": ,
/*optional, float, face temperature which is accurate to one decimal place*/
"isAbnomalTemperature": ,
/*optional, boolean, whether the face temperature is abnormal: true=yes, false=no*/
"RegionCoordinates":{
/*optional, face temperature's coordinates*/
"positionX": ,
/*optional, int, normalized X-coordinate which is between 0 and 1000*/
"positionY":
/*optional, int, normalized Y-coordinate which is between 0 and 1000*/
},
"mask": "unknown",
/*optional, string, whether the person is wearing a mask*/
"pictureURL": "test",
/*optional, string, picture URL*/
"filename": "",
```

```
/*optional, string, file name, each picture should has an unique name*/
  "attendanceStatus": "",
/*optional, string, attendance status: "undefined", "checkIn"-check in, "checkOut"-check out, "breakOut"-break out,
"breakIn"-break in, "overtimeIn"-overtime in, "overTimeOut"-overtime out*/
  "label": "",
/*optional, string, custom attendance name*/
  "statusValue":
/*optional, integer, status value*/
  "helmet": "unknown",
/*optional, string, whether the person wears a hard hat: "unknown", "yes", "no"*/
  "visibleLightPicUrl": "test",
/*optional, string, visible light picture URL*/
  "thermalPicUrl": "test",
/*optional, string, thermal imaging picture URL*/
  "appType": "attendance",
/*optional, string, app type: "attendance" (attendance application), "signIn" (check-in application)*/
},
{...}
]]
}
```

A.3 JSON_AcsEventCond

JSON message about conditions of search for history access control events

Basic Message

```
{
  "AcsEventCond": {
    "searchID": "",
/*required, string, search ID, which is used to check whether the current search requester is the same as the previous
one; if they are the same, the search record will be stored in the device to speed up the next search*/
    "searchResultPosition": ,
/*required, integer32, the end position of search result in result list; in a single search, if you cannot get all the records
in the result list, you can mark the end position and get the following records after the marked position in the next
search*/
    "maxResults":
/*required, integer32, the maximum number of search results, which is defined by the device capability, will be
returned if the value of maxResults reaches the limit; in this case, the device will not return error*/
  }
}
```

Advanced Message

```
{
  "AcsEventCond": {
    "searchID": "",
/*required, string, search ID, which is used to check whether the current search requester is the same as the previous
one; if they are the same, the search record will be stored in the device to speed up the next search*/
    "searchResultPosition": ,
```

```
/*required, integer32, the end position of search result in result list; in a single search, if you cannot get all the records
in the result list, you can mark the end position and get the following records after the marked position in the next
search*/
"maxResults": ,
/*required, integer32, the maximum number of search results, which is defined by the device capability, will be
returned if the value of maxResults reaches the limit; in this case, the device will not return error*/
"major": ,
/*optional, int, macro definition value of major type, refer to Access Control Event Types for details, the value is a
decimal number when transmitting, 0-all (default)/
"minor": ,
/*optional, int, macro definition value of minor type, refer to Access Control Event Types for details, the value is a
decimal number when transmitting, 0-all (default)/
"startTime": "",
/*optional, string, start time (UTC time), e.g., "2016-12-12T17:30:08+08:00"*/
"endTime": "",
/*optional, string, end time (UTC time), e.g., "2017-12-12T17:30:08+08:00"*/
"cardNo": "",
/*optional, string, card No.*/
"name": "",
/*optional, string, card holder name*/
"picEnable": ,
/*optional, boolean, whether the event contains picture: false-no, true-yes*/
"employeeNoString": ""
/*optional, string, employee ID (person ID)*/
}
}
```

See Also

Major & Minor Types of Access Control Event

A.4 JSON_AcsWorkStatus

JSON message about working status of access control device

```
{
  "AcsWorkStatus": {
    "doorStatus": [ , , , ],
    /*array, door status: 1-sleepy, 2-remain open, 3-remain closed, 4-open or closed; e.g., [1,2,1,2] indicates that the door
    1 is sleepy, door 2 is remain open, door 3 is sleepy, door 4 is remain open*/
  }
}
```

A.5 JSON_autoPanData

autoPanData message in JSON format

```
{
  "autoPanData":{
```

```
"autoPan":  
/*required, int, auto-scan speed, range: from 0 to 100*/  
}  
}
```

A.6 JSON_Cap_CaptureFingerPrint

JSON message about fingerprint collection capability

```
{  
  "CaptureFingerPrint": {  
    "CaptureFingerPrintCond": {  
      "fingerNo": {  
/*finger No.*/  
        "max": 10,  
        "min": 1  
      }  
    },  
    "fingerData": {  
/*fingerprint data*/  
      "max": 768,  
      "min": 1  
    },  
    "fingerNo": {  
      "max": 10,  
      "min": 1  
    },  
    "fingerPrintQuality": {  
/*fingerprint quality*/  
      "max": 100,  
      "min": 1  
    }  
  }  
}
```

A.7 JSON_Cap_FingerPrintCfg

JSON message about fingerprint configuration capability

```
{  
  "FingerPrintCfg": {  
    "employeeNo": {  
/*required, string, employee ID (person ID) linked with the fingerprint*/  
      "@min": ,  
      "@max":  
    },  
    "enableCardReader": {  
/*required, array, information of fingerprint and card readers to be applied*/  
      "@min": ,  

```

```
"@max":
},
"fingerPrintID": {
/*required, int, fingerprint No., which is from 1 to 10*/
"@min": 1,
"@max": 10
},
"fingerType": {
/*required, string, fingerprint type: "normalFP"-normal fingerprint, "hijackFP"-duress fingerprint, "patrolFP"- patrol
fingerprint, "superFP"-super fingerprint, "dismissingFP"-dismiss fingerprint*/
"@opt": "normalFP,hijackFP,patrolFP,superFP,dismissingFP"
},
"leaderFP": {
/*optional, array, No. list of doors that can be opened by first fingerprint*/
"@min": 1,
"@max": 32
},
"checkEmployeeNo": "true,false",
/*optional, boolean, whether to enable employee ID (person ID) verification: false, true (recommended); by default,
the employee ID (person ID) will be verified; if the node is set to false, the card information applying will speed up*/
"StatusList": {
/*optional, status list*/
"id": {
/*optional, int, fingerprint and card reader No.*/
"@min": ,
"@max":
},
"cardReaderRecvStatus": {
/*optional, int, fingerprint and card reader status: 0-failed, 1-succeeded, 2-the fingerprint module does not exist, 3-try
again or low fingerprint quality, 4-memory full, 5-the fingerprint already exists, 6-the fingerprint ID already exists, 7-
invalid fingerprint ID, 8-configuring this fingerprint module is not required, 10-the reader version is too low to support
employee ID*/
"@opt": "0,1,2,3,4,5,6,7,8,10"
},
"errorMsg": {
/*optional, string, error information of applying, when the value of node "cardReaderRecvStatus" is 5, it indicates that
employee ID (person ID) linked with the fingerprint already exists*/
"@min": ,
"@max":
},
},
"totalStatus": {
/*required, int, applying status: 0-applying, 1-applying completed*/
"@opt": "0,1"
},
"isSupportFingerCover": true
/*whether it supports overwriting the existing fingerprint when applying fingerprint: true; if it does not support, this
node will not be returned*/
"isSupportSetUp": true,
/*whether it supports setting fingerprint parameters: true; if it does not support, this node will not be returned*/
}
}
```

A.8 JSON_Cap_FingerPrintDelete

JSON message about fingerprint deleting capability

```
{
  "FingerPrintDelete" : {
    "mode": {
      /*required, string, deleting mode: "byEmployeeNo"-delete by employee ID (person ID); "byCardReader"-delete by
      fingerprint and card reader*/
      "@opt": "byEmployeeNo,byCardReader"
    },
    "EmployeeNoDetail": {
      /*optional, parameters of deleting by employee ID (person ID), this node is valid only when the value of node "mode"
      is "byEmployeeNo"*/
      "employeeNo": {
        /*optional, string, employee ID (person ID) linked with the fingerprint*/
        "@min": ,
        "@max":
      },
      "enableCardReader": {
        /*optional, array, No. list of fingerprint and card reader with fingerprints to be deleted*/
        "@min": ,
        "@max":
      },
      "fingerPrintID": {
        /*optional, array, finger No. list of fingerprint to be deleted*/
        "@min": ,
        "@max":
      },
    },
    "CardReaderDetail": {
      /*optional, parameters of deleting by fingerprint and card reader, this node is valid only when the value of node
      "mode" is "byCardReader"*/
      "cardReaderNo": {
        /*optional, int, fingerprint and card reader No.*/
        "@min": ,
        "@max":
      },
      "clearAllCard": true,false,
      /*optional, boolean, whether to delete the fingerprint information of all cards: false-no (delete by employee ID), true-
      yes (delete fingerprint information of all employee IDs*/
      "employeeNo": {
        /*optional, string, employee ID (person ID) linked with the fingerprint, this node is valid when the value of
        "clearAllCard" is false*/
        "@min": ,
        "@max":
      },
    }
  }
}
```


A.9 JSON_CaptureFingerPrint

JSON message about collected fingerprint data

```
{
  "CaptureFingerPrint": {
    /*fingerprint data, which should be encoded by Base64*/
    "fingerData": "",
    /*collected fingerprint data, e.g., "MzAxMlEpJkiUCWM5FUicbHs5FjiYBWE5JjiUgKYpFniED2/dJFjM4W
    +hFEjEY5GVJDj0X5c1FFiQ2K7ZJXikC7s1JEiUUtCdFGicRs39JYiQGSLxZFZicE+dVJliYHg/
    VJDi0cQ95JTioejE5FWiUbUJhFDiUbUFVfViMcw4JJTioACN1JDi85zQNJUiI7b2pFHiY4K1xFYjQ7RbxJTjIAxdxFrhw6BxBJriQ
    CDVJJjjAgN4JJciYPt41JVISyvp9FGhoOO2xFaiQJQHmJWiQmQ8OJYiIJwc+FoiQLv
    +IjnioIyCaJFhoKw7iJfigixqqJnigMQAxJEiYaklhFFig5FqtlYjAXH5tI2h40oqtIxiAWqXJEziA2KINFEiI3amFE1iQ2KgJFEiAYK3FEyi
    AYgAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
    AAle4zSBMAbQ4QUhcHX09gIDSoBeGhOzBEIBjt9hEjwewGYwNPEqPNF
    +AtYABSiQJYQzsAACEJC6RbALGbLt2WbCDRIApS3BYSMcULRvIdMOKYBA9dHQNhvg0TV19isKwCBf1QEUDSLdglGDHTuxJt
    /F41IKMDgON0QgliKeQRETEC0S/YDiEQMpggJQAAAAAABak="*/
    "fingerNo": 1,
    /*finger No.*/
    "fingerPrintQuality": 72
    /*fingerprint quality*/
  }
}
```

A.10 JSON_CaptureFingerPrintCond

JSON message about conditions of collecting fingerprint data

```
{
  "CaptureFingerPrintCond": {
    "fingerNo": 1
    /*finger No.*/
  }
}
```

A.11 JSON_CardInfo

JSON message about card information

Basic JSON Message

```
{
  "CardInfo" : {
    "employeeNo": "",
    /*required, string, employee ID (person ID)*/
    "cardNo": "1234567890"
    /*required, string, card No.*/
  }
}
```

```
}  
}
```

Advanced JSON Message

```
{  
  "CardInfo" : {  
    "employeeNo": "",  
    /*required, string, employee ID (person ID)*/  
    "cardNo": "1234567890",  
    /*required, string, card No.*/  
    "cardType": "normalCard"  
    /*optional, string, card type:"normalCard"-normal card (default), "patrolCard"-patrol card, "hijackCard"-duress card,  
    "superCard"-supper card, "dismissingCard"-dismiss card, "emergencyCard"-emergency card (for assigning permission  
    to temporary card, but it cannot open door). When this node does not exist, the default card type is normal card*/  
  }  
}
```

A.12 JSON_CardInfoCount

JSON message about total number of cards

```
{  
  "CardInfoCount" : {  
    "cardNumber":  
  }  
}
```

A.13 JSON_CardInfoDelCond

JSON message about conditions of deleting cards

```
{  
  "CardInfoDelCond" : {  
    "EmployeeNoList" : [{  
    /*optional, person ID list*/  
      "employeeNo": ""  
    /*optional, string, employee ID (person ID)*/  
    }],  
    "CardNoList" : [{  
    /*optional, card No, list, it is mutual exclusive with the person ID list, so only one of the nodes should be selected to  
    configure*/  
      "cardNo": ""  
    /*optional, string, card No.*/  
    }]  
  }  
}
```

Remarks

If the nodes "**EmployeeNoList**" and "**CardNoList**" do not exist or are not configured at same time, it indicates that all cards will be deleted.

A.14 JSON_CardInfoSearch

JSON message about results of searching for card details

```
{
  "CardInfoSearch" : {
    "searchID": "",
    /*required, string, search ID, which is used to check whether the current search requester is the same as the previous
    one; if they are the same, the search record will be stored in the device to speed up the next search*/
    "responseStatusStrg": "",
    /*required, string, search status description: "OK"-search ended, "MORE"-searching, "NO MATCH"-no matched data
    found*/
    "numOfMatches": ,
    /*required, integer32, number of records returned for one time*/
    "totalMatches": ,
    /*required, integer32, number of matched records returned for one time*/
    "CardInfo" : [{
      /*optional, person information*/
      "employeeNo": "",
      /*required, string, employee ID (person ID)*/
      "cardNo": "",
      /*required, string, card No.*/
      "cardType": "",
      /*required, string, card type: "normalCard"-normal card, "patrolCard"-patrol card, "hijackCard"-duress card,
      "superCard"-supper card, "dismissingCard"-dismiss card, "emergencyCard"-emergency card (for assigning permission
      to temporary card, but it cannot open door)*/
    }]
  }
}
```

A.15 JSON_CardInfoSearchCond

JSON message about conditions of search for card details

```
{
  "CardInfoSearchCond": {
    "searchID": "",
    /*required, string, search ID, which is used to check whether the current search requester is the same as the previous
    one; if they are the same, the search record will be stored in the device to speed up the next search*/
    "searchResultPosition": ,
    /*required, integer32, the end position of search result in result list; in a single search, if you cannot get all the records
    in the result list, you can mark the end position and get the following records after the marked position in the next
    search*/
  }
}
```

```
"maxResults": ,
/*required, integer32, the maximum number of search results, which is defined by the device capability, will be
returned if the value of maxResults reaches the limit; in this case, the device will not return error*/
"EmployeeNoList" : [{
/*optional, person ID list*/
"employeeNo": ""
/*optional, string, employee ID (person ID)*/
}],
"CardNoList" : [{
/*optional, card No. list, it is mutual exclusive with the person ID list, so only one of the nodes should be selected to
configure*/
"cardNo": ""
/*optional, string, card No.*/
}]
}
}
```

See Also

If the nodes "**EmployeeNoList**" and "**CardNoList**" do not exist or are not configured at same time, it indicates that all cards will be deleted.

A.16 JSON_CardReaderCfg

JSON message about parameters of a fingerprint and card reader

```
{
"CardReaderCfg": {
"enable": ,
/*required, boolean, whether to enable: true, false*/
"okLedPolarity": "",
/*optional, string, polarity of "OK" LED indicator: "cathode", "anode"*/
"errorLedPolarity": "",
/*optional, string, polarity of "ERROR" LED indicator: "cathode", "anode"*/
"buzzerPolarity": "",
/*optional, string, buzzer polarity: "cathode", "anode"*/
"swipeInterval": ,
/*optional, int, time interval between each two repeated authentication (fingerprint, card, face), unit: second*/
"pressTimeout": ,
/*optional, int, pressing timeout, unit: second*/
"enableFailAlarm": ,
/*optional, boolean, whether to enable authentication times exceeded alarm*/
"maxReadCardFailNum": ,
/*optional, int, maximum number of authentication failed attempts*/
"enableTamperCheck": ,
/*optional, boolean, whether it supports tampering detection*/
"offlineCheckTime": ,
/*optional, int, offline detection time, unit: second*/
"fingerPrintCheckLevel": ,
/*optional, int, fingerprint recognition level: 1 (1/10 error rate), 2 (1/100 error rate), 3 (1/1000 error rate), 4 (1/10000
error rate), 5 (1/100000 error rate), 6 (1/1000000 error rate), 7 (1/10000000 error rate), 8 (1/100000000 error rate), 9
```

```
(3/100 error rate), 10 (3/1000 error rate), 11 (3/10000 error rate), 12 (3/100000 error rate), 13 (3/1000000 error rate),
14 (3/10000000 error rate), 15 (3/100000000 error rate), 16 (Automatic Normal), 17 (Automatic Secure), 18
(Automatic More Secure, currently it is not supported by access control device)*/
    "useLocalController": ,
/*optional, boolean, read-only, whether to connect the distributed access controller*/
    "localControllerID": ,
/*optional, int, read-only, distributed access controller No., whose value is between 1 and 64, if the No. is 0, it
indicates that the controller is not registered; this node is valid when the value of "useLocalController" is true*/
    "localControllerReaderID": ,
/*optional, int, read-only, card reader ID of distributed access controller, if the No. is 0, it indicates that the reader is
not registered; this node is valid when the value of "useLocalController" is true*/
    "cardReaderChannel": ,
/*optional, int, communication channel No. of card reader: 0-Wiegand or offline, 1-RS-485A, 2-RS-485B; this node is
valid when the value of "useLocalController" is true*/
    "fingerPrintImageQuality": ,
/*optional, int, fingerprint picture quality: 1-low (V1), 2-medium (V1), 3-high (V1), 4-highest (V1), 5-low (V2), 6-
medium (V2), 7-high (V2), 8-highest (V2)*/
    "fingerPrintContrastTimeOut": ,
/*optional, int, fingerprint comparison timeout, whose value is between 1 and 20, unit: second; if the value is 255, it
indicates the timeout is infinite*/
    "fingerPrintRecognizeInterval": ,
/*optional, int, interval of fingerprint continuous recognition, whose value is between 1 and 10, unit: second; if the
value is 255, it indicates there is no delay during recognition*/
    "fingerPrintMatchFastMode": ,
/*optional, int, fast mode of fingerprint matching, 1-fast mode 1, 2-fast mode 2, 3-fast mode 3, 4-fast mode 4, 5-fast
mode 5, 255-auto*/
    "fingerPrintModuleSensitive": ,
/*optional, int, fingerprint module sensitive level, whose value between 1 and 8*/
    "fingerPrintModuleLightCondition": "",
/*optional, string, light condition of fingerprint module: "outdoor", "indoor"*/
    "faceMatchThresholdN": ,
/*optional, int, threshold of 1 VS N face picture comparison, whose value between 0 and 100*/
    "faceQuality": ,
/*optional, int, face picture quality, whose value is between 0 and 100*/
    "faceRecognizeTimeOut": ,
/*optional, int, face recognition timeout, whose value is between 1 and 20, unit: second; if the value is 255, it
indicates the timeout is infinite*/
    "faceRecognizeInterval": ,
/*optional, int, interval of face continuous recognition, whose value between 1 and 10, unit: second; if the value is
255, it indicates there is no delay during recognition*/
    "cardReaderFunction": ["", ""],
/*optional, array, read-only, recognition object type: "fingerPrint", "face", "fingerVein"; if multiple types are
configured at same time, it indicates all the configured object types are supported to be recognized, e.g.,
["fingerPrint", "face"] supports recognizing by fingerprint and face*/
    "cardReaderDescription": "",
/*optional, read-only, reader description; when the reader is Wiegand reader or the reader is offline, the value of this
node is "Wiegand\485Offline"*/
    "faceImageSensitometry": ,
/*optional, int, read-only, face picture exposure, whose value is between 0 and 65535*/
    "livingBodyDetect": ,
/*optional, boolean, whether to enable face anti-spoofing detection*/
    "faceMatchThreshold1": ,
```

```
/*optional, int, threshold of 1 VS 1 face picture comparison, whose value is between 0 and 100*/
  "buzzerTime": ,
/*optional, int, buzzing time duration, whose value is between 0 and 5999, unit: second; if the value is 0, it indicates
the long buzzing*/
  "faceMatch1SecurityLevel": ,
/*optional, int, security level of 1 VS 1 face picture recognition: 1-general, 2-medium, 3-strong*/
  "faceMatchNSecurityLevel": ,
/*optional, int, security level of 1 VS N face picture recognition: 1-general, 2-medium, 3-strong*/
  "envirMode": "",
/*optional, string, face recognition scene mode: "indoor", "other*/
  "liveDetLevelSet": "",
/*optional, string, threshold level of face anti-spoofing detection: "low", "middle", "high"*/
  "liveDetAntiAttackCntLimit": ,
/*optional, int, anti-attack times of face anti-spoofing detection, whose value is between 1 and 255 (determined by
device capability)*/
  "enableLiveDetAntiAttack": ,
/*optional, boolean, whether to enable anti-attack of face anti-spoofing detection*/
  "supportDelFPByID": ,
/*optional, boolean, read-only, whether the fingerprint and card reader supports deleting fingerprints by finger ID:
true, false*/
  "fingerPrintCapacity": ,
/*optional, int, read-only, fingerprint capacity*/
  "fingerPrintNum": ,
/*optional, int, read-only, number of the added fingerprints*/
  "defaultVerifyMode": "",
/*optional, string, read-only, default authentication mode: "cardAndPw"-card+password, "card", "cardOrPw"-card or
password, "fp"-fingerprint, "fpAndPw"-fingerprint+password, "fpOrCard"-fingerprint or card, "fpAndCard"-fingerprint
+card, "fpAndCardAndPw"-fingerprint+card+password, "faceOrFpOrCardOrPw"-face or fingerprint or card or
password, "faceAndFp"-face+fingerprint, "faceAndPw"-face+password, "faceAndCard"-face+card, "face",
"employeeNoAndPw"-employee ID+password, "fpOrPw"-fingerprint or password, "employeeNoAndFp"-employee ID
+fingerprint, "employeeNoAndFpAndPw"-employee ID+fingerprint+password, "faceAndFpAndCard"-face+fingerprint
+card, "faceAndPwAndFp"-face+password+fingerprint, "employeeNoAndFace"-employee ID+face,
"faceOrfaceAndCard"-face or face+card, "fpOrface"-fingerprint or face, "cardOrfaceOrPw"-card or face or password,
"cardOrFace"-card or face, "cardOrFaceOrFp"-card or face or fingerprint*/
  "faceRecognizeEnable":
/*optional, int, whether to enable face recognition: 1-yes, 2-no, 3-multi-face attendance*/
}
}
```

A.17 JSON_channelNameOverlay

channelNameOverlay message in JSON format

```
{
  "channelNameOverlay": {
    "enabled": ,
/*required, boolean, whether to enable channel name overlay, "true"-enable (default), "false"-disable*/
    "positionY": ,
/*required, int, Y-coordinate of display position*/
    "positionX":
```

```
/*required, int, X-coordinate of display position*/  
}  
}
```

A.18 JSON_CMSearchDescription

CMSearchDescription message in JSON format

```
{  
  "CMSearchDescription": {  
    "searchID": "C7E71364-4560-0001-6EDD-16ED17B01CCD",  
    /*required, string, search ID*/  
    "trackIDList": [{  
      /*Now only the first trackID takes effect*/  
      "trackID": 101  
      /*required, int, channel to be searched and stream type, e.g., "101"-main stream of channel No.1, "202"-sub-stream  
of channel No.2,... */  
    }],  
    "timeSpanList": [{  
      "timeSpan": {  
        /*time span*/  
        "startTime": "2018-03-07T16:00:00Z",  
        /*required, start time*/  
        "endTime": "2018-03-10T15:59:59Z"  
        /*required, start time*/  
      }  
    }],  
    "contentTypeList": [{  
      /*optional*/  
      "contentType": "video"  
      /*optional, file type to be searched: "video"*/  
    }],  
    "maxResults": 40,  
    /*required, int, maximum number of responses can be received; up to 40 are allowed*/  
    "searchResultPostion": 0,  
    /*required, int, position of search result*/  
    "metadataList": [{  
      "metadataDescriptor": "recordType.meta.hikvision.com"  
      /*required, string, metadata descriptor: "recordType.meta.hikvision.com" (all videos),  
"recordType.meta.hikvision.com/CMR" (scheduled recording), "recordType.meta.hikvision.com/MOTION" (motion  
detection), "recordType.meta.hikvision.com/ALARM" (alarm triggering), "recordType.meta.hikvision.com/EDR" (alarm  
or motion detection), "recordType.meta.hikvision.com/ALARMANDMOTION" (alarm and motion detection)*/  
    }]  
  }  
}
```

Remarks

The **searchResultPostion** is a dynamic index, and used for next search according to the current searched results if the number of results is larger than the **maxResults**.

A.19 JSON_CMSearchResult

CMSearchResult message in JSON format

```
{
  "CMSearchResult":{
    "searchID":"812F04E0-4089-11A3-9A0C-0305E82C2906",
    /*required, string, search ID*/
    "responseStatus": true,
    /*required, boolean, whether to return response status: "true"-yes*/
    "responseStatusStrg":"OK",
    /*required, string, response status details*/
    "numOfMatches": 1,
    /*required, integer, number of matched results*/
    "matchList":[{
      "searchMatchItem":{
        "sourceID":"b049902e72-0049-1158-c0d2-7e330680d93c",
        /*required, string, source ID*/
        "trackID": 101,
        /*required, integer, channel to be searched and stream type, e.g., 101-main stream of channel No.1, 202-sub-stream
        of channel No.2,... */
        "timeSpan": {
          /*time span*/
          "startTime": "2018-03-07T16:00:00Z",
          /*required, start time*/
          "endTime": "2018-03-10T15:59:59Z"
          /*required, start time*/
        },
        "mediaSegmentDescriptor":{
          "contentType":"video",
          /*required, file type to be searched, by default, it is "video"*/
          "codecType":"MPEG4-SP",
          /*required, encoding mode*/
          "rateType":"3Mbps",
          /*optional, bit rate, unit: Mbps*/
          "playbackURI": "",
          /*string, playback URI, e.g., rtsp://10.17.132.49/Streaming/tracks/101/?
          starttime=20170314T103201Z&endtime=20170314T104042Z&name=ch01_08000000016000000&size
          =260358144*/
          "lockStatus":"lock",
          /*optional, video file locking status, "lock"-locked, "unlock"-unlocked*/
          "remainLockTime": 12
          /*remaining locking time, it is valid only when the lockStatus is "lock", 0-lock permanently*/
        },
        "metadataMatches":{
          "metadataDescriptor": "recordType.meta.hikvision.com/MOTION"
          /*required, string, metadata descriptor: "recordType.meta.hikvision.com" (all videos),
          "recordType.meta.hikvision.com/CMR" (scheduled recording), "recordType.meta.hikvision.com/MOTION" (motion
          detection), "recordType.meta.hikvision.com/ALARM" (alarm triggering), "recordType.meta.hikvision.com/EDR" (alarm
          or motion detection), "recordType.meta.hikvision.com/ALARMANDMOTION" (alarm and motion detection)*/
        }
      }
    ]
  }
}
```



```
}  
  }  
}  
}
```

A.20 JSON_Color

Color message in JSON format

```
{  
  "Color": {  
    "brightnessLevel": ,  
    /*optional, int, brightness*/  
    "contrastLevel": ,  
    /*optional, int, contrast*/  
    "saturationLevel": ,  
    /*optional, int, saturation*/  
    "hueLevel": ,  
    /*optional, int, hue*/  
    "grayScale": {  
    /*optional, gray scale*/  
      "grayScaleMode": ""  
    /*optional, string, gray scale mode: "indoor", "outdoor"*/  
    }  
  }  
}
```

A.21 JSON_DateTimeOverlay

DateTimeOverlay message in JSON format

```
{  
  "DateTimeOverlay": {  
    "enabled": ,  
    /*required, boolean, whether to enable time overlay, "true"-enable*/  
    "positionY": ,  
    /*required, int, Y-coordinate of display position*/  
    "positionX": ,  
    /*required, int, X-coordinate of display position*/  
    "dateStyle": "",  
    /*optional, string, date display format, opt="YYYY-MM-DD, MM-DD-YYYY, DD-MM-YYYY, CHR-YYYY-MM-DD, CHR-MM-DD-YYYY, CHR-DD-MM-YYYY, CHR-YYYY/MM/DD, CHR-MM/DD/YYYY, CHR-DD/MM/YYYY"*/  
    "timeStyle": "",  
    /*optional, string, time format, opt="12hour, 24hour"*/  
    "displayWeek": ,  
    /*optional, boolean, whether to display week, "true"-display*/  
  }  
}
```

A.22 JSON_DelDevList

JSON message about device list to be deleted

```
{
  "DelDevList":[{
    "Dev":{
      "devIndex": "2cd6716d-767f-4756-ac55-50276a5e3b4a",
/*required, string, device index (UUID)*/
      "status": "fail",
/*required, deleting result: "success, fail"*/
      "subStatusCode": "badParameters"
/*optional, string, sub status code; it is returned when adding failed: "badParameters" (incorrect parameters),
"theDeviceIdDoesNotExist" (the device ID does not exist), "noMemory" (insufficient device memory)*/
    }
  ]
}
```

A.23 JSON_DeviceInfo

JSON message about device information

```
{
  "DeviceInfo":{
    "deviceName": "",
/*required, device name*/
    "deviceId": "",
/*required, read only, device index (UUID)*/
    "deviceDescription": "",
/*optional, device description*/
    "systemContact": "",
/*read only, string, system*/
    "model": "",
/*required, read only, device model*/
    "serialNumber": "",
/*required, read only, serial number*/
    "macAddress": "",
/*required, read only, MAC address*/
    "firmwareVersion": "",
/*required, read only, firmware version*/
    "firmwareReleasedDate": "",
/*optional, read only, firmware release data*/
    "bootVersion": "",
/*required, read only, drive version*/
    "bootReleasedDate": "",
/*optional, read only, drive release data*/
    "hardwareVersion": "",
/*optional, read only, hardware version*/
    "encoderVersion": "",

```

```
/*optional, read only, encoder version*/
  "encoderReleasedDate": "",
/*optional, read only, encoder release data*/
  "decoderVersion": "",
/*optional, read only, decoder version*/
  "decoderReleasedDate": "",
/*optional, read only, decoder release data*/
  "softwareVersion": "",
/*optional, read only, software version*/
  "deviceType": "",
/*required, read only, device type: "IPCamera, IPDome, DVR, HybirdNVR, NVR, DVS, IPZoom, CVR, Gateway,
SecurityCP, DockStation, ACS"*/
  "telecontrolID": ,
/*optional, read only, digital device type*/
  "operationSystem": ""
/*optional, read only, string, operating system information*/
  "devType":
/*optional, read only, int, device type values*/
}
}
```

Remarks

The read-only nodes are invalid when calling URL by PUT method.

A.24 JSON_DeviceInfoList

JSON Message about device information list

```
{
  "DeviceInfoList":{
    "devIndex": "",
/*optional, string type, device UUID, it cannot be edited and it is returned when adding device, e.g.,
"2cd6716d-767f-4756-ac55-50276a5e3b4a"*/
    "deviceName": "",
/*required, device name*/
    "deviceId": "",
/*required, device ID (uuid/guid), read only*/
    "deviceDescription": "",
/*optional, device description*/
    "systemContact": "",
/*system, read only, string*/
    "model": "",
/*required, device model, read only*/
    "serialNumber": "",
/*required, serial No., read only*/
    "macAddress": "",
/*required, MAC address, read only*/
    "firmwareVersion": "",
/*required, firmware version, read only*/
    "firmwareReleasedDate": "",
```

```
/*optional, firmware release data, read only*/
  "bootVersion": "",
/*required, drive version, read only*/
  "bootReleasedDate": "",
/*optional, drive release data, read only*/
  "hardwareVersion": "",
/*optional, hardware version, read only*/
  "encoderVersion": "",
/*optional, encoder version, read only*/
  "encoderReleasedDate": "",
/*optional, encoder release data, read only*/
  "decoderVersion": "",
/*optional, decoder version, read only*/
  "decoderReleasedDate": "",
/*optional, decoder release data, read only*/
  "softwareVersion": "",
/*optional, software version, read only*/
  "deviceType": "",
/*required, device type, "IPCamera, IPDome, DVR, HybirdNVR, NVR, DVS, IPZoom, CVR, Gateway, SecurityCP, ACS"*/
  "telecontrolID": ,
/*optional, digital device type, read only*/
  "operationSystem":""
/*optional, operating system information, string, read only*/
  "devType":
/*optional, integer, read only, device type values*/
}
}
```

A.25 JSON_DeviceInList

JSON message about input parameter list of adding devices

```
{
  "DeviceInList": [{
    "Device":{
      "protocolType": "",
/*required, string, protocol type: "ehome" (ISUP), "ehomeV5" (ISUP 5.0), "ISAPI" (Intelligent Security API)*/
      "EhomeParams":{
/*it is valid when the value of protocolType is "ehome" or "ehomeV5"*/
        "EhomeID": "111",
/*required, string, ISUP (Intelligent Security Uplink Protocol) ID; Up to 31 characters are allowed*/
        "EhomeKey": ""
/*dependent, string, it is valid only when protocolType is "ehomeV5", and the maximum key length is 32 bytes*/
      },
      "ISAPIParams":{
/*it is valid when the value of protocolType is "ISAPI"*/
        "addressingFormatType": "IPv4Address",
/*optional, string, address type: "IPv4Address" (IPv4 address)*/
        "address": "0.0.0.0",
/*required, string, address; The address format depends on the value of addressingFormatType*/
      }
    }
  ]
}
```

```
"portNo": 8000,
/*required, int, device port number; value range: [1,65535]*/
"userName": "admin",
/*required, string, user name; up to 32 characters are allowed*/
"password": ""
/*required, string, password*/
},
"devName": "DeepinMind",
/*optional, string, device name; up to 32 characters are allowed*/
"devType": "encodingDev"
/*required, string, device type: "encodingDev" (encoding device), "AccessControl" (access control device)*/
}
},
{...}
]
}
```

A.26 JSON_DeviceOutList

JSON Message about output parameter list of adding devices

```
{
  "DeviceOutList": [
    {
      "Device":{
        "devIndex":"2cd6716d-767f-4756-ac55-50276a5e3b4a",
        /*optional, string, device index (uuid/guid); it will be returned when the device is added*/
        "devName": "DeepinMind",
        /*optional, string, device name; up to 32 characters are allowed*/
        "protocolType": "",
        /*required, string, protocol type: "ehome" (ISUP), "ehomeV5" (ISUP 5.0), "ISAPI" (Intelligent Security API)*/
        "EhomeParams":{
          /*it is valid when the value of protocolType is "ehome" or "ehomeV5"*/
          "EhomeID":"111"
          /*required, string, ISUP (Intelligent Security Uplink Protocol) ID; Up to 31 characters are allowed*/
        },
        "ISAPIParams":{
          /*it is valid when the value of protocolType is "ISAPI"*/
          "addressingFormatType": "IPV4Address",
          /*optional, string, address type: "IPV4Address" (IPv4 address)*/
          "address":"0.0.0.0",
          /*optional, string, address. The address format depends on the value of addressingFormatType*/
          "portNo": 8000,
          /*optional, int, device port number; value range: [1,65535]*/
        },
        "status":"fail",
        /*required, adding result: "success" (added), "fail" (adding failed)*/
        "subStatusCode":"badParameters"
        /*optional, string, sub status code; it is returned when adding failed: "badParameters" (incorrect parameter),
        "monitorNodeOverLimit" (no more device can be added), "noMemory" (insufficient device memory), "deviceExist"
        */
      }
    }
  ]
}
```

```
(the device already exist)*/
}
},
{...}
]
}
```

A.27 JSON_DevIndexList

JSON message about the device index list

```
{
  "DevIndexList": ["2cd6716d-767f-4756-ac55-50276a5e3b4a",""]
/*required, string, device index (UUID) array*/
}
```

A.28 JSON_DoorParam

JSON message about door parameters

```
{
  "doorName":
/*optional, string, door name*/
}
```

A.29 JSON_downloadRequest

downloadRequest message in JSON format

```
{
  "downloadRequest":{
    "playbackURI": ""
/*required, string, playback URL, e.g., "rtsp://IpAddress/Streaming/tracks/101?
starttime=20180703T224426Z&endtime=20180703T224426Z"*/
  }
}
```

A.30 JSON_Edit_UserInfo

JSON message about one person's information

Basic Message

```
{
  "UserInfo" : {
```

```
"employeeNo": "",
/*required, string, employee ID (person ID)*/
"name": "",
/*optional, string, person name*/
"Valid": {
/*optional, effective period, the earliest date and time can be "1970-01-01T00:00:00", and the latest date and time
can be "2037-12-31T23:59:59"*/
"beginTime": "",
/*optional, start time of the effective period, e.g., for local time, it is "2017-08-01T17:30:08"; for UTC time, it is
"2017-08-01T17:30:08+08:00"*/
"endTime": ""
/*optional, start time of the effective period, e.g., for local time, it is "2017-08-01T17:30:08"; for UTC time, it is
"2017-08-01T17:30:08+08:00"*/
}
}
}
```

Advanced Message

```
{
  "UserInfo": {
    "employeeNo": "",
/*required, string, employee ID (person ID)*/
    "name": "",
/*optional, string, person name*/
    "userType": "",
/*optional, string, person type: "normal"-host (default), "visitor"-visitor, "blackList"-person in blocklist*/
    "Valid": {
/*optional, effective period, the earliest date and time can be "1970-01-01T00:00:00", and the latest date and time
can be "2037-12-31T23:59:59"*/
      "enable": ,
/*optional, boolean, whether to enable effective period: false (always be effective), true (default)*/
      "beginTime": "",
/*optional, start time of the effective period, e.g., for local time, it is "2017-08-01T17:30:08"; for UTC time, it is
"2017-08-01T17:30:08+08:00"*/
      "endTime": "",
/*optional, start time of the effective period, e.g., for local time, it is "2017-08-01T17:30:08"; for UTC time, it is
"2017-08-01T17:30:08+08:00"*/
      "timeType": ""
/*optional, string, time type: "local"-device local time, "UTC"-UTC time*/
    }
  }
}
```

A.31 JSON_EventNotificationAlert_abnormalAcceleration

JSON message about acceleration exception alarm details

Message Field Description

Field Name	Req. or Opt.	Date Type	Description
channelID	Req.	string	Channel No. of the device that triggered alarm.
dateTime	Req.	ISO 8601	Alarm triggered time, which is in ISO 8601 time format (i.e., "yyyy-MM-ddThh:mm:ss").
activePostCount	Req.	int	Number of times that the same alarm has been triggered.
eventType	Req.	string	Type of event that triggers alarm, here it should be "abnormalAcceleration".
eventState	Req.	string	Durative alarm/event status: "active"-valid, "inactive"-invalid, e.g., when a moving target is detected, the alarm/event information will be uploaded continuously until the status is set to "inactive".
eventDescription	Req.	string	Event description.
devIndex	Opt.	string	Device ID (uuid/guid).
channelName	Req.	string	Camera name.
deviceId	Req.	string	Device ID.
deviceSerial	Opt.	string	Device serial No.

Message Example

```
{
  "EventNotificationAlert": {
    "channelID": "1",
    "dateTime": "2018-01-21T12:50:39+08:00",
    "activePostCount": 1,
    "eventType": "abnormalAcceleration",
    "eventState": "active",
    "eventDescription": "abnormal acceleration alarm",
    "devIndex": "2cd6716d-767f-4756-ac55-50276a5e3b4a",
    "channelName": "Ipdome",
    "deviceId": "Abcd1234"
  }
}
```


A.32 JSON_EventNotificationAlert_abnormalDriving

JSON message about details of abnormal driving behavior alarm

Message Field Description

Field Name	Req. or Opt.	Date Type	Description
channelID	Req.	string	Channel No. of the device that triggered alarm.
dateTime	Req.	ISO 8601	Alarm triggered time, which is in ISO 8601 time format (i.e., "yyyy-MM-ddThh:mm:ss").
activePostCount	Req.	int	Number of times that the same alarm has been triggered.
eventType	Req.	string	Type of event that triggers alarm, here it should be "abnormalDriving".
eventState	Req.	string	Durative alarm/event status: "active"-valid, "inactive"-invalid, e.g., when a moving target is detected, the alarm/event information will be uploaded continuously until the status is set to "inactive".
eventDescription	Req.	string	Event description.
devIndex	Opt.	string	Device ID (uuid/guid).
behaviorType	Opt.	string	Behavior type: "fatigue" (fatigue driving), "calling" (talking on phone), "smoking", "eating", "chatting".
angle	Opt.	int	Rollover angle, unit: degrees.
channelName	Req.	string	Camera name.
deviceId	Req.	string	Device ID.
deviceSerial	Opt.	string	Device serial No.
retransmission	Opt.	boolean	Whether the data is replenished.
DBD	Req.	object	Abnormal driving information.

Table A-1 DBD

Field Name	Req. or Opt.	Date Type	Description
PositionInfo	Opt.	object	GPS information.
snapTime	Req.	ISO 8601	Alarm triggered time, which is in ISO 8601 time format (i.e., "yyyy-MM-ddThh:mm:ss").
fatigueDriving	Opt.	string	Whether the driver is in fatigue: "unknown", "no", "yes".
uphone	Opt.	string	Whether the driver is talking on phone: "unknown", "no", "yes".
smoking	Opt.	string	Whether the driver is smoking: "unknown", "no", "yes".
eating	Opt.	string	Whether the driver is eating: "unknown", "no", "yes".
chatting	Opt.	string	Whether the driver is chatting: "unknown", "no", "yes".
uid	Opt.	string	ID of event to be uploaded, its size is 64 bytes.

Message Example

```
{
  "EventNotificationAlert": {
    "channelID": "1",
    "dateTime": "2018-01-21T12:50:39+08:00",
    "activePostCount": 1,
    "eventType": "abnormalDriving",
    "eventState": "active",
    "eventDescription": "abnormal driving behavior alarm",
    "devIndex": "2cd6716d-767f-4756-ac55-50276a5e3b4a",
    "behaviorType": "eating",
    "channelName": "Ipdome",
    "deviceID": "Abcd1234",
    "DBD": {
      "PositionInfo": {
        "longitude": 100,
        "divisionNS": "N",
        "latitude": 100,
        "direction": 100,
        "speed": 100,
        "height": 1
      }
    }
  },
}
```

```

"snapTime": "2004-05-03T17:30:08+08:00",
"snapPictureURL": "http://10.16.96.31:80/picture/Streaming/tracks/103/?
name=ch0001_02000000712100049510400011512&size=11512",
"videoURL": "http://10.16.96.31:80/picture/Streaming/tracks/103/?
name=ch0001_02000000712100049510400011512&size=11512",
"smoking": "unknown",
"uphone": "unknown",
"fatigueDriving": "unknown",
"notLookStraightAhead": "unknown",
"pilotSafebelt": "unknown",
"driverAbsent": "unknown",
"wearingIRInterruptedSunglasses": "unknown",
"pickingUpObject": "unknown",
"eating": "unknown",
"chatting": "unknown",
"notWearingUniform": "unknown",
"uid": "2019052010392202600Kb4s3iQPzLignc1sRUbFT0hR039Y0YzuLt0VLf99yZi"
}
}
}

```

A.33 JSON_EventNotificationAlert_AccessControlEventMsg

JSON message about access control event details

Message Field Description

Field Name	Req. or Opt.	Date Type	Description
channelID	Req.	string	Channel No. of the device that triggered alarm.
dateTime	Req.	ISO 8601	Alarm triggered time, which is in ISO 8601 time format (i.e., "yyyy-MM-ddThh:mm:ss").
activePostCount	Req.	int	Number of times that the same alarm has been triggered.
eventType	Req.	string	Type of event that triggers alarm, here it should be "AccessControllerEvent".
eventState	Req.	string	Durative alarm/event status: "active"-valid, "inactive"-invalid, e.g., when a moving target is detected, the alarm/event

Field Name	Req. or Opt.	Date Type	Description
			information will be uploaded continuously unit the status is set to "inactive".
eventDescription	Req.	string	Event description.
devIndex	Opt.	string	Device ID (uuid/guid).
channelName	Req.	string	Camera name.
deviceId	Req.	string	Device ID. It is uploaded for ISUP 5.0.
deviceSerial	Opt.	string	Device serial No.
ipAddress	Opt.	string	Device IP address, which is uploaded for ISAPI.
portNo	Opt.	int	Device port No., which is uploaded for ISAPI.
AccessControllerEvent	Req.	object	Access control event information.

Table A-2 AccessControllerEvent

Field Name	Req. or Opt.	Date Type	Description
deviceName	Opt.	string	Device name.
majorEventType	Req.	int	Major alarm and event types (the type value should be converted to a decimal number for transmission), see <i>Major & Minor Types of Access Control Event</i> for details.
subEventType	Req.	int	Minor alarm and event types (the type value should be converted to a decimal number for transmission), see <i>Major & Minor Types of Access Control Event</i> for details.
netUser	Opt.	string	User name for network operations.
remoteHostAddr	Opt.	string	Remote host address.
cardNo	Opt.	string	Card No.

Field Name	Req. or Opt.	Date Type	Description
cardType	Opt.	int	Card type: 1 (normal card), 2 (disability card), 3 (blocklist card), 4 (patrol card), 5 (duress card), 6 (super card), 7 (visitor card), 8 (dismiss card).
whiteListNo	Opt.	int	Allowlist No., which is between 1 and 8.
reportChannel	Opt.	int	Alarm/event uploading channel type: 1 (uploading in arming mode), 2 (uploading by central group 1), 3 (uploading by central group 2).
cardReaderKind	Opt.	int	Reader type: 1 (IC card reader), 2 (ID card reader), 3 (QR code scanner), 4 (fingerprint module).
cardReaderNo	Opt.	int	Card reader No.
doorNo	Opt.	int	Door or floor No.
verifyNo	Opt.	int	Multiple authentication No.
alarmInNo	Opt.	int	Alarm input No.
alarmOutNo	Opt.	int	Alarm output No.
caseSensorNo	Opt.	int	Event trigger No.
RS485No	Opt.	int	RS-485 channel No.
multiCardGroupNo	Opt.	int	Group No.
accessChannel	Opt.	int	Turnstile No.
deviceNo	Opt.	int	Device No.
distractControlNo	Opt.	int	Distributed access controller No.
employeeNo	Opt.	string	Employee No. (person ID)
localControllerID	Opt.	int	Distributed access controller No.: 0 (access controller), 1 to 64 (distributed access controller No. 1 to distributed access controller No. 64).
InternetAccess	Opt.	int	Network interface No.: 1 (upstream network interface No.1), 2

Field Name	Req. or Opt.	Date Type	Description
			(upstream network interface No.2), 3 (downstream network interface No.1).
type	Opt.	int	Zone type: 0 (instant zone), 1 (24-hour zone), 2 (delayed zone), 3 (internal zone), 4 (key zone), 5 (fire alarm zone), 6 (perimeter zone), 7 (24-hour silent zone), 8 (24-hour auxiliary zone), 9 (24-hour shock zone), 10 (emergency door open zone), 11 (emergency door closed zone), 255 (none).
MACAddr	Opt.	string	Physical address.
swipeCardType	Opt.	int	Card swiping types: 0 (invalid), 1 (QR code).
serialNo	Opt.	int	Event serial No., which is used to check whether the event loss occurred.
channelControllerID	Opt.	int	Lane controller ID: 1 (main lane controller), 2 (sub lane controller).
channelControllerLampID	Opt.	int	Light board ID of the lane controller, which is between 1 and 255.
channelControllerIRAdaptorID	Opt.	int	IR adaptor ID of the lane controller, which is between 1 and 255.
channelControllerIREmitterID	Opt.	int	Active infrared intrusion detector No. of the lane controller, which is between 1 and 255.
userType	Opt.	string	Person type: "normal" (normal person (resident)), "visitor" (visitor), "blacklist" (person in the blacklist), "administrators" (administrator).
currentVerifyMode	Opt.	string	Current authentication mode of the reader: "cardAndPw" (card +password), "card", "cardOrPw" (card or password), "fp" (fingerprint), "fpAndPw" (fingerprint

Field Name	Req. or Opt.	Date Type	Description
			+password), "fpOrCard" (fingerprint or card), "fpAndCard" (fingerprint +card), "fpAndCardAndPw" (fingerprint+card+password), "faceOrFpOrCardOrPw" (face or fingerprint or card or password), "faceAndFp" (face+fingerprint), "faceAndPw" (face+password), "faceAndCard" (face+card), "face" (face), "employeeNoAndPw" (employee No.+password), "fpOrPw" (fingerprint or password), "employeeNoAndFp" (employee No.+fingerprint), "employeeNoAndFpAndPw" (employee No.+fingerprint +password), "faceAndFpAndCard" (face+fingerprint+card), "faceAndPwAndFp" (face+password +fingerprint), "employeeNoAndFace" (employee No.+face), "faceOrfaceAndCard" (face or face+card), "fpOrface" (fingerprint or face), "cardOrfaceOrPw" (card or face or password), "cardOrFace" (card or face), "cardOrFaceOrFp" (card or face or fingerprint), "cardOrFpOrPw" (card or fingerprint or password).
currentEvent	Opt.	boolean	Whether it is a real-time event: true, false.
frontSerialNo	Opt.	int	The previous event's serial No. If this field does not exist, the platform will check whether the event loss occurred according to the field serialNo . If both the serialNo and frontSerialNo are returned, the platform will check whether the event loss occurred according to

Field Name	Req. or Opt.	Date Type	Description
			both fields. It is mainly used to solve the problem that the serialNo is inconsistent after subscribing events or alarms.
attendanceStatus	Opt.	string	Attendance status: "undefined", "checkIn" (check in), "checkOut" (check out), "breakOut" (start of break), "breakIn" (end of break), "overtimeIn" (start of overtime), "overTimeOut" (end of overtime).
statusValue	Opt.	int	Status value.
pictureURL	Opt.	string	Captured picture URL.
visibleLightURL	Opt.	string	Visible light picture URL.
thermalURL	Opt.	string	Thermal imaging picture URL.
picturesNumber	Opt.	int	Number of pictures, if there is no picture, the value of this node is 0 or it is not returned.
helmet	Opt.	string	Whether the person is wearing a hard hat.
mask	Opt.	string	Whether the person is wearing a mask.
thermometryUnit	Opt.	string	Temperature unit: "celsius", "fahrenheit", "kelvin"; the default unit is "celsius".
currTemperature	Opt.	float	Face temperature, which is accurate to one decimal place.
isAbnomalTemperature	Opt.	boolean	Whether the face temperature is abnormal: true, false.

Remarks

If devices are added via ISAPI, the picture of access control event is uploaded in binary format, see the example of access control event with binary picture data below.

See Also

Major & Minor Types of Access Control Event

Example

Message Example of Access Control Event with Picture URL

```
{
  "EventNotificationAlert": {
    "channelID": "1",
    "dateTime": "2016-12-12T17:30:08+08:00",
    "activePostCount": 1,
    "eventType": "AccessControllerEvent",
    "eventState": "active",
    "eventDescription": "Access Controller Event",
    "devIndex": "2cd6716d-767f-4756-ac55-50276a5e3b4a",
    "channelName": "Camera 01",
    "AccessControllerEvent": {
      "deviceName": "642",
      "majorEventType": 5,
      "subEventType": 80,
      "netUser": "",
      "remoteHostAddr": "172.6.64.7",
      "cardNo": "123",
      "cardType": 1,
      "whiteListNo": 1,
      "reportChannel": 1,
      "cardReaderKind": 1,
      "cardReaderNo": 0,
      "doorNo": 1,
      "verifyNo": 1,
      "alarmInNo": 1,
      "alarmOutNo": 1,
      "caseSensorNo": 1,
      "RS485No": 1,
      "multiCardGroupNo": 1,
      "accessChannel": 1,
      "deviceNo": 123,
      "distractControlNo": 1,
      "employeeNo": "123",
      "localControllerID": 0,
      "InternetAccess": 1,
      "type": 0,
      "MACAddr": "01:17:24:45:D9:F4",
      "swipeCardType": 0,
      "serialNo": 1,
      "channelControllerID": 1,
      "channelControllerLampID": 1,
      "channelControllerIRAdaptorID": 1,
      "channelControllerIREmitterID": 1,
      "userType": "normal",
      "currentVerifyMode": "",
      "currentEvent": true,
      "frontSerialNo": 0,
      "attendanceStatus": "",
      "statusValue": 0,
    }
  }
}
```

```
"pictureURL": "",
"picturesNumber": 0
}
}
}
```

Example

Message Example of Access Control Event with Binary Picture Data

```
{
  "ipAddress": "172.6.64.7",
  "ipv6Address": "",
  "portNo": 80,
  "protocol": "HTTP",
  "macAddress": "01:17:24:45:D9:F4",
  "channelID": 1,
  "dateTime": "2016-12-12T17:30:08+08:00",
  "activePostCount": 1,
  "eventType": "AccessControllerEvent",
  "eventState": "active",
  "eventDescription": "Access Controller Event",
  "deviceId": "test0123",
  "AccessControllerEvent": {
    "deviceName": "",
    "majorEventType": 1,
    "subEventType": 1,
    "inductiveEventType": "",
    "netUser": "",
    "remoteHostAddr": "",
    "cardNo": "",
    "cardType": 1,
    "name": "",
    "whiteListNo": 1,
    "reportChannel": 1,
    "cardReaderKind": 1,
    "cardReaderNo": 1,
    "doorNo": 1,
    "verifyNo": 1,
    "alarmInNo": 1,
    "alarmOutNo": 1,
    "caseSensorNo": 1,
    "RS485No": 1,
    "multiCardGroupNo": 1,
    "accessChannel": 1,
    "deviceNo": 1,
    "distractControlNo": 1,
    "employeeNo": 1,
    "employeeNoString": "",
    "employeeName": "",
    "localControllerID": 1,
    "InternetAccess": 1,
    "type": 1,
  }
}
```

```
"MACAddr": "",
"swipeCardType": 1,
"serialNo": 1,
"channelControllerID": 1,
"channelControllerLampID": 1,
"channelControllerIRAdaptorID": 1,
"channelControllerIREmitterID": 1,
"userType": "normal",
"currentVerifyMode": ,
"currentEvent": true,
"QRCodeInfo": "",
"thermometryUnit": "",
"currTemperature": ,
"isAbnormalTemperature": ,
"RegionCoordinates": {
  "positionX": ,
  "positionY":
},
"remoteCheck": true,
"mask": "",
"frontSerialNo": 1,
"attendanceStatus": "",
"statusValue": 1,
"pictureURL": "",
"picturesNumber": 1,
"unlockType": "password",
"classroomId": "",
"classroomName": "",
"analysisModule": "",
"customInfo": ""
},
}
--MIME_boundary
Content-Disposition: form-data; name="Picture"; filename="Picture.jpg";
Content-Type: image/jpeg
Content-Length: 516876
Content-ID: pictureImage

fefefwageegfqaeg...
--MIME_boundary
Content-Disposition: form-data; name="VisibleLight"; filename="VisibleLight.jpg";
Content-Type: image/jpeg
Content-Length: 516876
Content-ID: visibleLightImage

fefefwageegfqaeg...
--MIME_boundary
Content-Disposition: form-data; name="Thermal"; filename="Thermal.jpg";
Content-Type: image/jpeg
Content-Length: 516876
Content-ID: thermalImage
```

```
fefefwageegfqaeg...  
--MIME_boundary
```

A.34 JSON_EventNotificationAlert_ADAS

JSON message about driver assistance alarm details

Message Field Description

Field Name	Req. or Opt.	Date Type	Description
channelID	Req.	string	Channel No. of the device that triggered alarm.
dateTime	Req.	ISO 8601	Alarm triggered time, which is in ISO 8601 time format (i.e., "yyyy-MM-ddThh:mm:ss").
activePostCount	Req.	int	Number of times that the same alarm has been triggered.
eventType	Req.	string	Type of event that triggers alarm, here it should be "ADAS".
eventState	Req.	string	Durative alarm/event status: "active"-valid, "inactive"-invalid, e.g., when a moving target is detected, the alarm/event information will be uploaded continuously until the status is set to "inactive".
eventDescription	Req.	string	Event description.
devIndex	Opt.	string	Device ID (uuid/guid).
channelName	Req.	string	Camera name.
deviceId	Req.	string	Device ID.
deviceSerial	Opt.	string	Device serial No.
retransmission	Opt.	boolean	Whether the data is replenished.
ADAS	Req.	object	Driver assistance alarm information.

Table A-3 ADAS

Field Name	Req. or Opt.	Date Type	Description
PositionInfo	Opt.	object	GPS information.
snapTime	Req.	ISO 8601	Alarm triggered time, which is in ISO 8601 time format (i.e., "yyyy-MM-ddThh:mm:ss").
frontVehicleCollision	Opt.	string	Collision with front vehicle: "unknown", "yes", "no".
drivingOnLaneLine	Opt.	string	Driving on the lane line: "no", "left", "unknown", "right".
vehicleDistanceDetection	Opt.	string	Whether the vehicle distance is detected: "unknown", "yes", "no".
pedestrianCollision	Opt.	string	Collision with pedestrian: "unknown", "no", "yes".
vehicleBlindSpotDetection	Opt.	string	Blind spot detection: "unknown", "no", "right", "left", "back".
rapidAcceleration	Opt.	string	Rapid acceleration: "unknown", "yes", "no".
rapidDeceleration	Opt.	string	Rapid deceleration: "unknown", "yes", "no".
suddenTurn	Opt.	string	Vehicle sharp turn detection: "unknown", "no", "right", "left".
rollover	Opt.	string	Rollover detection: "unknown", "yes", "no".
notYieldToPedestrian	Opt.	string	Whether the not yield to pedestrian is detected: "unknown", "yes", "no".
trafficSignWarning	Opt.	string	Traffic sign warning: "unknown", "no", "speedLimit", "heightLimit", "widthLimit".
uid	Opt.	string	ID of event to be uploaded, its size is 64 bytes.

Table A-4 PositionInfo

Field Name	Req. or Opt.	Date Type	Description
longitude	Req.	int	Longitude, which equals to the value of (actual degree $\times$ 3600 $\times$ 100 + actual minute $\times$ 60 $\times$ 100 + actual second $\times$ 100).
divisionNS	Req.	string	North and south hemispheres: "N"-north hemisphere, "S"-south hemisphere.
latitude	Req.	int	Latitude, which equals to the value of (actual degree $\times$ 3600 $\times$ 100 + actual minute $\times$ 60 $\times$ 100 + actual second $\times$ 100).
direction	Req.	int	Vehicle direction, which equals to the value of actual direction value (unit: degree, positive in clockwise, for north direction, the value is 0) $\times$ 100, the actual direction value.
speed	Req.	int	Speed, unit: cm/h.
height	Req.	int	Height, unit: cm.

Message Example

```
{
  "EventNotificationAlert": {
    "channelID": "1",
    "dateTime": "2018-01-21T12:50:39+08:00",
    "activePostCount": 1,
    "eventType": "ADAS",
    "eventState": "active",
    "eventDescription": "",
    "devIndex": "2cd6716d-767f-4756-ac55-50276a5e3b4a",
    "channelName": "Ipdome",
    "deviceId": "Abcd1234",
    "ADAS": {
      "PositionInfo": {
        "longitude": 100,
        "divisionNS": "N",
        "latitude": 100,
        "direction": 100,
        "speed": 100,
        "height": 1
      },
      "snapTime": "2004-05-03T17:30:08+08:00",
      "snapPictureURL": "http://10.16.96.31:80/picture/Streaming/tracks/103/?
name=ch0001_02000000712100049510400011512&size=11512",
      "videoURL": "http://10.16.96.31:80/picture/Streaming/tracks/103/?"
```

```
name=ch0001_02000000712100049510400011512&size=11512",
  "frontVehicleCollision": "unknown",
  "drivingOnLaneLine": "unknown",
  "vehicleDistanceDetection": "unknown",
  "pedestrianCollision": "unknown",
  "vehicleBlindSpotDetection": "unknown",
  "rapidAcceleration": "unknown",
  "rapidDeceleration": "unknown",
  "suddenTurn": "unknown",
  "rollover": "unknown",
  "notYieldToPedestrian": "unknown",
  "trafficSignWarning": "unknown",
  "uid": "2019052010392202600Kb4s3iQPzLignc1sRUbFT0hR039Y0YzuLt0VLf99yZi"
}
}
}
```

A.35 JSON_EventNotificationAlert_addressChange

JSON message about details of device address changed alarm

Message Field Description

Field Name	Req. or Opt.	Date Type	Description
dateTime	Req.	string	Alarm triggered time, which is in ISO 8601 time format (i.e., "2017-04-22T15:39:01+08:00").
activePostCount	Req.	int	Number of times that the same alarm has been triggered.
eventType	Req.	string	Event type, here it should be "addressChange".
eventState	Req.	string	Durative alarm/event status: "active"-valid, "inactive"-invalid. E.g., when a moving target is detected, the alarm/event information will be uploaded continuously until the status is set to "inactive".
eventDescription	Req.	string	Event description, here it is "Address change alarm".
devIndex	Opt.	string	Device index (uuid/guid).

Field Name	Req. or Opt.	Date Type	Description
deviceId	Req.	string	Device ID.
deviceSerial	Opt.	string	Device serial No.
NewAddress	Req.	item	Device's new address.
OldAddress	Req.	item	Device's old address.
ipAddress	Req.	string	Device IPv4 address.
portNo	Req.	int	Device port number.

Message Example

```
{
  "EventNotificationAlert": {
    "dateTime": "2018-03-13T19:42:27+08:00",
    "activePostCount": 1,
    "eventType": "addressChange",
    "eventState": "active",
    "eventDescription": "Address change alarm",
    "devIndex": "2cd6716d-767f-4756-ac55-50276a5e3b4a",
    "deviceId": "Abcd1234",
    "NewAddress": {
      "ipAddress": "1.136.111.106",
      "portNo": 80
    },
    "OldAddress": {
      "ipAddress": "192.168.1.1",
      "portNo": 88
    }
  }
}
```

A.36 JSON_EventNotificationAlert_AlarmEventInfo

JSON message about alarm or event information to be uploaded

```
{
  "EventNotificationAlert": {
    "channelID": "",
    /*optional, dep, string, device channel No.*/
    "dateTime": "",
    /*required, alarm/event triggered or occurred time, it must contain time zone information, e.g.,
    "2017-04-22T15:39:01+08:00"*/
    "activePostCount": ,
    /*required, integer, alarm/event frequency*/
    "eventType": "",
  }
}
```



```

/*required, string, alarm/event types, see details in the Event Types*/
"eventState": "",
/*required, string, durative alarm/event status: "active"-valid, "inactive"-invalid, e.g., when a moving target is
detected, the alarm/event information will be uploaded continuously unit the status is set to "inactive"*/
"eventDescription": "",
/*required, string, description*/
"devIndex": "",
/*optional, string, device ID (uuid/guid)*/
"...": "",
/*for different alarm/event types, the nodes are different*/
"channelName": ""
/*required, string, camera name*/
}
}

```

A.37 JSON_EventNotificationAlert_attendedBaggage

JSON message about object removal alarm details

Message Field Description

Field Name	Req. or Opt.	Date Type	Description
channelID	Req.	string	Channel No. of the device that triggered alarm.
dateTime	Req.	ISO 8601	Alarm triggered time, which is in ISO 8601 time format (i.e., "yyyy-MM-ddThh:mm:ss").
activePostCount	Req.	int	Number of times that the same alarm has been triggered.
eventType	Req.	string	Type of event that triggers alarm, here it should be "attendedBaggage".
eventState	Req.	string	Durative alarm/event status: "active"-valid, "inactive"-invalid, e.g., when a moving target is detected, the alarm/event information will be uploaded continuously unit the status is set to "inactive".
eventDescription	Req.	string	Event description, here it is "Attended baggage alarm".
devIndex	Opt.	string	Device ID (uuid/guid).

Field Name	Req. or Opt.	Date Type	Description
channelName	Req.	string	Camera name.
deviceID	Req.	string	Device ID.
deviceSerial	Opt.	string	Device serial No.
DetectionRegionList	Opt.	object	Detection region, this node is valid in expert mode.
DetectionRegionEntry	Req.	list	Detection region information.
regionID	Req.	string	Detection region ID.
sensitivityLevel	Opt.	int	Sensitivity. Value range: [0,100].
RegionCoordinatesList	Opt.	list	Detection region coordinates.
RegionCoordinates			
positionX	Req.	int	X-coordinate of a vertex on the rule frame, which is between 0 and 1000.
positionY	Req.	int	Y-coordinate of a vertex on the rule frame, which is between 0 and 1000.

Message Example

```
{
  "EventNotificationAlert": {
    "channelID": "1",
    "dateTime": "2018-01-21T12:50:39+08:00",
    "activePostCount": 1,
    "eventType": "attendedBaggage",
    "eventState": "active",
    "eventDescription": "",
    "devIndex": "2cd6716d-767f-4756-ac55-50276a5e3b4a",
    "channelName": "Ipdome",
    "deviceID": "Abcd1234",
    "DetectionRegionList": [{
      "DetectionRegionEntry": {
        "regionID": "1",
        "sensitivityLevel": 50,
        "RegionCoordinatesList": [{
          "RegionCoordinates": {
            "positionX": 0,
            "positionY": 0
          },
          "RegionCoordinates": {
```

```

    "positionX":0,
    "positionY":100
  },
  "RegionCoordinates":{
    "positionX":100,
    "positionY":100
  },
  "RegionCoordinates":{
    "positionX":100,
    "positionY":0
  }
}
}
}
}
}

```

A.38 JSON_EventNotificationAlert_audioexception

JSON message about audio exception alarm details

Message Field Description

Field Name	Req. or Opt.	Date Type	Description
channelID	Req.	string	Channel No. of the device that triggered alarm.
dateTime	Req.	ISO 8601	Alarm triggered time, which is in ISO 8601 time format (i.e., "yyyy-MM-ddThh:mm:ss").
activePostCount	Req.	int	Number of times that the same alarm has been triggered.
eventType	Req.	string	Type of event that triggers alarm, here it should be "audioexception".
eventState	Req.	string	Durative alarm/event status: "active"-valid, "inactive"-invalid, e.g., when a moving target is detected, the alarm/event information will be uploaded continuously until the status is set to "inactive".
eventDescription	Req.	string	Event description, here it is "Audio exception alarm".
devIndex	Opt.	string	Device ID (uuid/guid).

Field Name	Req. or Opt.	Date Type	Description
channelName	Req.	string	Camera name.
deviceId	Req.	string	Device ID.
deviceSerial	Opt.	string	Device serial No.
AudioExceptionDetection	Req.	object	Extension field of audio exception.

Table A-5 AudioExceptionDetection

Field Name	Req. or Opt.	Date Type	Description
alarmType	Req.	string	Sub types of audio exception alarm: "audioLost", "decrease", "increase".
audioDecibel	Dependent	int	Decibel. It is valid when the value of alarmType is "decrease" or "increase".

Message Example

```
{
  "EventNotificationAlert": {
    "channelID": "1",
    "dateTime": "2018-01-21T12:50:39+08:00",
    "activePostCount": 1,
    "eventType": "audioexception",
    "eventState": "active",
    "eventDescription": "",
    "devIndex": "2cd6716d-767f-4756-ac55-50276a5e3b4a",
    "channelName": "Ipdome",
    "deviceId": "Abcd1234",
    "AudioExceptionDetection": {
      "alarmType": "decrease",
      "audioDecibel": 10
    }
  }
}
```

A.39 JSON_EventNotificationAlert_batteryStatus

JSON message about battery status details

Message Field Description

Field Name	Req. or Opt.	Date Type	Description
channelID	Req.	string	Channel No. of the device that triggered alarm.
dateTime	Req.	ISO 8601	Alarm triggered time, which is in ISO 8601 time format (i.e., "yyyy-MM-ddThh:mm:ss").
activePostCount	Req.	int	Number of times that the same alarm has been triggered.
eventType	Req.	string	Event type, here it should be "batteryStatus".
eventState	Req.	string	Durative alarm/event status: "active"-valid, "inactive"-invalid, e.g., when a moving target is detected, the alarm/event information will be uploaded continuously unit the status is set to "inactive".
eventDescription	Req.	string	Event description.
devIndex	Opt.	string	Device ID (uuid/guid).
channelName	Req.	string	Camera name.
deviceId	Req.	string	Device ID.
deviceSerial	Opt.	string	Device serial No.
batteryPercentage	Req.	int	Remaining battery percentage, value range: [0, 100]
batteryType	Req.	int	Battery type: 0 (dry battery), 1 (storage battery).
remainingBattery	Opt.	int	Remaining battery, unit: mA.
chargingStatus	Opt.	int	Charging status: 1 (charging), 2 (not charging).

Message Example

```
{
  "EventNotificationAlert": {
    "channelID": "1",
    "dateTime": "2018-01-21T12:50:39+08:00",
```

```

"activePostCount": 1,
"eventType": "batteryStatus",
"eventState": "active",
"eventDescription": "Battery Status",
"devIndex": "2cd6716d-767f-4756-ac55-50276a5e3b4a",
"channelName": "Ipdome",
"deviceId": "Abcd1234",
"batteryPercentage" : 99,
"batteryType" : 1,
"remainingBattery" : 99,
"chargingStatus" : 1
}
}

```

A.40 JSON_EventNotificationAlert_collision

JSON message about collision detection alarm details

Message Field Description

Field Name	Req. or Opt.	Date Type	Description
channelID	Req.	string	Channel No. of the device that triggered alarm.
dateTime	Req.	ISO 8601	Alarm triggered time, which is in ISO 8601 time format (i.e., "yyyy-MM-ddThh:mm:ss").
activePostCount	Req.	int	Number of times that the same alarm has been triggered.
eventType	Req.	string	Type of event that triggers alarm, here it should be "collision".
eventState	Req.	string	Durative alarm/event status: "active"-valid, "inactive"-invalid, e.g., when a moving target is detected, the alarm/event information will be uploaded continuously unit the status is set to "inactive".
eventDescription	Req.	string	Event description.
devIndex	Opt.	string	Device ID (uuid/guid).
threshold	Opt.	int	Acceleration threshold, unit: m/s^2.
acceleration	Opt.	int	Acceleration, unit: m/s^2.

Field Name	Req. or Opt.	Date Type	Description
speed	Opt.	int	Speed, unit: m/s.
channelName	Req.	string	Camera name.
deviceId	Req.	string	Device ID.
deviceSerial	Opt.	string	Device serial No.

Message Example

```
{
  "EventNotificationAlert": {
    "channelID": "1",
    "dateTime": "2018-01-21T12:50:39+08:00",
    "activePostCount": 1,
    "eventType": "collision",
    "eventState": "active",
    "eventDescription": "collision alarm",
    "devIndex": "2cd6716d-767f-4756-ac55-50276a5e3b4a",
    "threshold": 100,
    "acceleration": 100,
    "speed": 100,
    "channelName": "Ipdome",
    "deviceId": "Abcd1234"
  }
}
```

A.41 JSON_EventNotificationAlert_defocus

JSON message about defocus alarm details

Message Field Description

Field Name	Req. or Opt.	Date Type	Description
channelID	Req.	string	Channel No. of the device that triggered alarm.
dateTime	Req.	ISO 8601	Alarm triggered time, which is in ISO 8601 time format (i.e., "yyyy-MM-ddThh:mm:ss").
activePostCount	Req.	int	Number of times that the same alarm has been triggered.

Field Name	Req. or Opt.	Date Type	Description
eventType	Req.	string	Type of event that triggers alarm, here it should be "defocus".
eventState	Req.	string	Durative alarm/event status: "active"-valid, "inactive"-invalid, e.g., when a moving target is detected, the alarm/event information will be uploaded continuously until the status is set to "inactive".
eventDescription	Req.	string	Event description, here it is "Defocus detection alarm".
devIndex	Opt.	string	Device ID (uuid/guid).
channelName	Req.	string	Camera name.
deviceId	Req.	string	Device ID.
deviceSerial	Opt.	string	Device serial No.

Message Example

```
{
  "EventNotificationAlert": {
    "channelID": "1",
    "dateTime": "2018-01-21T12:50:39+08:00",
    "activePostCount": 1,
    "eventType": "defocus",
    "eventState": "active",
    "eventDescription": "",
    "devIndex": "2cd6716d-767f-4756-ac55-50276a5e3b4a",
    "channelName": "IpDome",
    "deviceId": "Abcd1234"
  }
}
```

A.42 JSON_EventNotificationAlert_devSleep

JSON message about details of the device sleep alarm

Message Field Description

Field Name	Req. or Opt.	Date Type	Description
channelID	Req.	string	Channel number of the device that triggered alarm.
dateTime	Req.	ISO 8601	Alarm triggered time, which is in ISO 8601 time format (i.e., "2017-04-22T15:39:01+08:00").
activePostCount	Req.	int	Number of times that the same alarm has been triggered.
eventType	Req.	string	Event type, here it should be "devSleep".
eventState	Req.	string	Durative alarm/event status: "active"-valid, "inactive"-invalid, e.g., when a moving target is detected, the alarm/event information will be uploaded continuously unit the status is set to "inactive".
eventDescription	Req.	string	Event description, here it is "Device sleep alarm".
devIndex	Opt.	string	Device index (uuid/guid)
channelName	Req.	string	Camera name
status	Req.	string	"sleep" (the device is in the sleep mode)
deviceId	Req.	string	Device ID.
deviceSerial	Opt.	string	Device serial No.

Message Example

```
{
  "EventNotificationAlert": {
    "channelID": "1",
    "dateTime": "2018-01-21T12:50:39+08:00",
    "activePostCount": 1,
    "eventType": "devSleep",
    "eventState": "active",
    "eventDescription": "device sleep alarm",
    "devIndex": "2cd6716d-767f-4756-ac55-50276a5e3b4a",
    "channelName": "Ipdome",
```

```
"status": "sleep",
"deviceId": "Abcd1234"
}
}
```

A.43 JSON_EventNotificationAlert_devXmlEvent

JSON message about device event transmission details

Field Name	Req. or Opt.	Date Type	Description
eventType	Req.	String	Event type: "devXmlEvent".
devIndex	Req.	String	Device ID (uuid/guid).
deviceId	Req.	String	Device ID.
deviceSerial	Req.	String	Device serial No.
payload	Req.	String	Original alarm message in XML format uploaded by the device.

Message Example

```
{
  "EventNotificationAlert": {
    "devIndex": "2cd6716d-767f-4756-ac55-50276a5e3b4a",
    "deviceSerial": "Abcd123401",
    "deviceId": "Abcd1234",
    "eventType": "devXmlEvent",
    "payload": "<EventNotificationAlert version=\"2.0\" xmlns=\"http://www.hikvision.com/ver20/XMLSchema\">
      <ipAddress>172.7.17.231</ipAddress>
      <portNo>80</portNo>
      <protocol>HTTP</protocol>
      <macAddress>58:50:ed:93:87:d3</macAddress>
      <channelID>1</channelID>
      <dateTime>2020-10-23T13:49:02+08:00</dateTime>
      <activePostCount>5</activePostCount>
      <eventType>targetCapture</eventType>
      <eventState>active</eventState>
      <eventDescription>Target capture detection</eventDescription>
    </EventNotificationAlert>"
  }
}
```

A.44 JSON_EventNotificationAlert_devStatusChanged

JSON message about details of device status changed alarm

Message Field Description

Field Name	Req. or Opt.	Date Type	Description
channelID	Req.	string	Channel number of the device that triggered alarm.
dateTime	Req.	ISO 8601	Alarm triggered time, which is in ISO 8601 time format (i.e., "yyyy-MM-ddThh:mm:ss").
activePostCount	Req.	int	Number of times that the same alarm has been triggered.
eventType	Req.	string	Event type, here it should be "devStatusChanged".
eventState	Req.	string	Durative alarm/event status: "active"-valid, "inactive"-invalid, e.g., when a moving target is detected, the alarm/event information will be uploaded continuously unit the status is set to "inactive"
eventDescription	Req.	string	Event description, here it is "device status (online/offline) changed".
devIndex	Opt.	string	Device index (uuid/guid).
channelName	Req.	string	Camera name.
status	Req.	string	Device status: "online", "offline".
deviceId	Req.	string	Device ID.
deviceSerial	Opt.	string	Device serial No.

Message Example

```
{
  "EventNotificationAlert":{
    "channelID":"1",
    "dateTime":"2018-01-21T12:50:39+08:00",
    "activePostCount":1,
    "eventType":"devStatusChanged",
    "eventState":"active",
    "eventDescription":"device status about online or offline changed",
    "devIndex":"2cd6716d-767f-4756-ac55-50276a5e3b4a",
    "channelName":"lpdome",
    "status":"online",
    "deviceId":"Abcd1234"
```

```
}  
}
```

A.45 JSON_EventNotificationAlert_diskerror

JSON message about HDD error alarm details

Message Field Description

Field Name	Req. or Opt.	Date Type	Description
channelID	Req.	string	Channel No. of the device that triggered alarm.
dateTime	Req.	ISO 8601	Alarm triggered time, which is in ISO 8601 time format (i.e., "yyyy-MM-ddThh:mm:ss").
activePostCount	Req.	int	Number of times that the same alarm has been triggered.
eventType	Req.	string	Event type, here it should be "diskerror".
eventState	Req.	string	Durative alarm/event status: "active"-valid, "inactive"-invalid, e.g., when a moving target is detected, the alarm/event information will be uploaded continuously unit the status is set to "inactive".
eventDescription	Req.	string	Event description, here it is "diskerror alarm".
devIndex	Opt.	string	Device ID (uuid/guid).
channelName	Req.	string	Camera name.
deviceID	Req.	string	Device ID.
deviceSerial	Opt.	string	Device serial No.

Message Example

```
{  
  "EventNotificationAlert":{  
    "channelID":"1",  
    "dateTime":"2018-01-21T12:50:39+08:00",  
    "activePostCount":1,  
  }  
}
```

```

"eventType":"diskerror",
"eventState":"active",
"eventDescription":"diskerror alarm",
"devIndex":"2cd6716d-767f-4756-ac55-50276a5e3b4a",
"channelName":"Ipdome",
"deviceId": "Abcd1234"
}
}

```

A.46 JSON_EventNotificationAlert_diskfull

JSON message about HDD full alarm details

Message Field Description

Field Name	Req. or Opt.	Date Type	Description
ipAddress	Opt.	string	IPv4 address.
ipv6Address	Opt.	string	IPv6 address.
portNo	Opt.	string	Port number.
protocolType	Opt.	string	Protocol type: "HTTP", "HTTPS".
macAddress	Opt.	string	Device Mac address.
channelID	Req.	string	Channel No. of the device that triggered alarm.
dateTime	Req.	ISO 8601	Alarm triggered time, which is in ISO 8601 time format (i.e., "yyyy-MM-ddThh:mm:ss").
activePostCount	Req.	int	Number of times that the same alarm has been triggered.
eventType	Req.	string	Event type, here it should be "diskfull".
eventState	Req.	string	Durative alarm/event status: "active"-valid, "inactive"-invalid, e.g., when a moving target is detected, the alarm/event information will be uploaded continuously unit the status is set to "inactive".
eventDescription	Req.	string	Event description, here it is "diskfull alarm".

Field Name	Req. or Opt.	Date Type	Description
devIndex	Opt.	string	Device ID (uuid/guid).
channelName	Req.	string	Camera name.
deviceId	Req.	string	Device ID.
deviceSerial	Opt.	string	Device serial No.

Message Example

```
{
  "EventNotificationAlert":{
    "channelID":"1",
    "dateTime":"2018-01-21T12:50:39+08:00",
    "activePostCount":1,
    "eventType":"diskfull",
    "eventState":"active",
    "eventDescription":"diskfull alarm",
    "devIndex":"2cd6716d-767f-4756-ac55-50276a5e3b4a",
    "channelName":"lpdome",
    "deviceId": "Abcd1234"
  }
}
```

A.47 JSON_EventNotificationAlert_faceCapture

JSON message about face capture alarm details

Message Field Description

Field Name	Req. or Opt.	Date Type	Description
channelID	Req.	string	Channel No. of the device that triggered alarm.
dateTime	Req.	ISO 8601	Alarm triggered time, which is in ISO 8601 time format (i.e., "yyyy-MM-ddThh:mm:ss").
activePostCount	Req.	int	Number of times that the same alarm has been triggered.
eventType	Req.	string	Type of event that triggers alarm, here it should be "faceCapture".

Field Name	Req. or Opt.	Date Type	Description
eventState	Req.	string	Durative alarm/event status: "active"-valid, "inactive"-invalid, e.g., when a moving target is detected, the alarm/event information will be uploaded continuously until the status is set to "inactive".
eventDescription	Req.	string	Event description.
devIndex	Opt.	string	Device ID (uuid/guid).
channelName	Req.	string	Camera name.
deviceId	Req.	string	Device ID.
deviceSerial	Opt.	string	Device serial No.
DeviceGPS	Opt.	item	Device GPS information.
faceCapture	Req.	item	Information list of captured face pictures.

Table A-6 DeviceGPS

Field Name	Req. or Opt.	Date Type	Description
divisionEW	Req.	string	East and west hemispheres: "E"-east hemisphere, "W"-west hemisphere.
longitude	Req.	int	Longitude = degree × 3600 × 100 + minute × 60 × 100 + second × 100
divisionNS	Req.	string	North and south hemispheres: "N"-north hemisphere, "S"-south hemisphere.
latitude	Req.	int	Latitude = degree × 3600 × 100 + minute × 60 × 100 + second × 100
direction	Opt.	int	Vehicle direction = actual direction (unit: degree, the north direction is 0, calculate by clockwise) × 100
speed	Opt.	int	Speed, unit: cm/h.

Table A-7 faceCapture

Field Name	Req. or Opt.	Date Type	Description
uid	Opt.	string	ID of event to be uploaded, its size is 64 bytes.
targetAttrs	Req.	item	Target face features.
faces	Req.	item	Face information list.

Table A-8 targetAttrs

Field Name	Req. or Opt.	Date Type	Description
faceTime	Opt.	string	Capture time, e.g., 2020-07-08T11:42:11+08:00.
bkgUrl	Opt.	string	Background picture URL.
thermalURL	Opt.	string	Picture No.

Table A-9 faces

Field Name	Req. or Opt.	Date Type	Description
faceRect	Opt.	item	Face thumbnail frame.
mask	Opt.	item	Whether the person is wearing a mask or not: "yes", "no".
URL	Opt.	string	Face thumbnail URL.
faceScore	Req.	int	Face score, range: [0, 100].
captureEndMark	Req.	boolean	Whether the capture has ended: true, false.
faceSnapThermometryEnabled	Opt.	boolean	Whether the skin-surface temperature measurement is enabled: true, false.
currTemperature	Opt.	float	Skin-surface temperature, range: [-20.0, 150.0], the value is accurate to one decimal place.
isAbnormalTemperature	Opt.	boolean	Whether the skin-surface temperature is abnormal: true, false.
thermometryUnit	Opt.	string	Temperature unit: "celsius" (Celsius), "fahrenheit" (Fahrenheit), "kelvin" (Kelvin).

Message Example

```
{
  "EventNotificationAlert": {
    "channelID": "1",
    "dateTime": "2018-01-21T12:50:39+08:00",
    "activePostCount": 1,
    "eventType": "faceCapture",
    "eventState": "active",
    "eventDescription": "faceCapture",
    "devIndex": "2cd6716d-767f-4756-ac55-50276a5e3b4a",
    "channelName": "Ipdome",
    "deviceID": "Abcd1234",
    "DeviceGPS": {
      "direction": 100,
      "divisionEW": "E",
      "divisionNS": "N",
      "latitude": 100,
      "longitude": 100,
      "speed": 100
    },
    "faceCapture": {
      "targetAttrs": {
        "faceTime": "2020-07-08T11:42:11+08:00",
        "bkgUrl": "",
        "thermalURL": ""
      },
      "faces": {
        "faceRect": {
          "height": 0.172136,
          "width": 0.094395,
          "x": 0.413604,
          "y": 0.681300
        },
        "mask": {
          "value": "yes"
        }
      },
      "URL": "",
      "faceScore": 78,
      "captureEndMark": true,
      "faceSnapThermometryEnabled": true,
      "currTemperature": 36.7,
      "isAbnomalTemperature": false,
      "thermometryUnit": "celsius"
    },
    "uid": "2020070811421881226232701106303825903565841129341197367679221825"
  }
}
```

A.48 JSON_EventNotificationAlert_facedetection

JSON message about face detection alarm details

Message Field Description

Field Name	Req. or Opt.	Date Type	Description
channelID	Req.	string	Channel No. of the device that triggered alarm.
dateTime	Req.	ISO 8601	Alarm triggered time, which is in ISO 8601 time format (i.e., "2017-04-22T15:39:01+08:00").
activePostCount	Req.	int	Number of times that the same alarm has been triggered.
eventType	Req.	string	Event type, here it should be "facedetection".
eventState	Req.	string	Durative alarm/event status: "active"-valid, "inactive"-invalid. E.g., when a moving target is detected, the alarm/event information will be uploaded continuously until the status is set to "inactive".
eventDescription	Req.	string	Event description.
devIndex	Opt.	string	Device index (uuid/guid).
channelName	Req.	string	Camera name.
deviceId	Req.	string	Device ID.
deviceSerial	Opt.	string	Device serial No.

Message Example

```
{
  "EventNotificationAlert": {
    "channelID": "1",
    "dateTime": "2018-01-21T12:50:39+08:00",
    "activePostCount": 1,
    "eventType": "facedetection",
    "eventState": "active",
    "eventDescription": "Face Detection",
    "devIndex": "2cd6716d-767f-4756-ac55-50276a5e3b4a",
    "channelName": "Ipdome",
    "deviceId": "Abcd1234"
  }
}
```

```
}  
}
```

A.49 JSON_EventNotificationAlert_fieldddetection

JSON message about intrusion alarm details

Message Field Description

Field Name	Req. or Opt.	Date Type	Description
channelID	Req.	string	Channel No. of the device that triggered alarm.
dateTime	Req.	ISO 8601	Alarm triggered time, which is in ISO 8601 time format (i.e., "yyyy-MM-ddThh:mm:ss").
activePostCount	Req.	int	Number of times that the same alarm has been triggered.
eventType	Req.	string	Type of event that triggers alarm, here it should be "fieldddetection".
eventState	Req.	string	Durative alarm/event status: "active"-valid, "inactive"-invalid, e.g., when a moving target is detected, the alarm/event information will be uploaded continuously unit the status is set to "inactive".
eventDescription	Req.	string	Event description, here it is "Intrusion alarm".
devIndex	Opt.	string	Device ID (uuid/guid).
channelName	Req.	string	Camera name.
deviceId	Req.	string	Device ID.
deviceSerial	Opt.	string	Device serial No.
DetectionRegionList	Opt.	object	Detection region, this node is valid in expert mode.
DetectionRegionEntry	Req.	list	Detection region information.
regionID	Req.	string	Detection region ID.

Field Name	Req. or Opt.	Date Type	Description
sensitivityLevel	Opt.	int	Sensitivity. Value range: [0,100].
RegionCoordinatesList	Opt.	list	Detection region coordinates.
RegionCoordinates			
positionX	Req.	int	X-coordinate of a vertex on the rule frame, which is between 0 and 1000.
positionY	Req.	int	Y-coordinate of a vertex on the rule frame, which is between 0 and 1000.
detectionTarget	Opt.	string	Target type: "human", "vehicle", "others".
TargetRect	Opt.		Coordinates of target.
X	Req.	float	X-coordinate.
Y	Req.	float	Y-coordinate.
width	Req.	float	Width of target.
height	Req.	float	Height of target.

Sample Message of Intrusion Alarm Details

```
{
  "EventNotificationAlert":{
    "channelID":"1",
    "dateTime":"2018-01-21T12:50:39+08:00",
    "activePostCount":1,
    "eventType":"fielddetection",
    "eventState":"active",
    "eventDescription":"",
    "devIndex":"2cd6716d-767f-4756-ac55-50276a5e3b4a",
    "channelName":"Ipdome",
    "deviceId":"Abcd1234",
    "DetectionRegionList":[{
      "DetectionRegionEntry":{
        "regionID":"1",
        "sensitivityLevel":50,
        "RegionCoordinatesList":[{
          "RegionCoordinates":{
            "positionX":0,
            "positionY":0
          },
          "RegionCoordinates":{
```

```
"positionX":0,
"positionY":100
},
"RegionCoordinates":{
"positionX":100,
"positionY":100
},
"RegionCoordinates":{
"positionX":100,
"positionY":0
}
}},
"detectionTarget":"human",
"TargetRect":{
"X":0,
"Y":1,
"width":0.5,
"height":0.5
}
}
}}
}
```

A.50 JSON_EventNotificationAlert_flowOverrun

JSON message about traffic flow over-limit alarm details

Message Field Description

Field Name	Req. or Opt.	Date Type	Description
channelID	Req.	string	Channel No. of the device that triggered alarm.
dateTime	Req.	ISO 8601	Alarm triggered time, which is in ISO 8601 time format (i.e., "yyyy-MM-ddThh:mm:ss").
activePostCount	Req.	int	Number of times that the same alarm has been triggered.
eventType	Req.	string	Type of event that triggers alarm, here it should be "flowOverrun".
eventState	Req.	string	Durative alarm/event status: "active"-valid, "inactive"-invalid, e.g., when a moving target is detected, the alarm/event

Field Name	Req. or Opt.	Date Type	Description
			information will be uploaded continuously unit the status is set to "inactive".
eventDescription	Req.	string	Event description.
devIndex	Opt.	string	Device ID (uuid/guid).
channelName	Req.	string	Camera name.
deviceId	Req.	string	Device ID.
deviceSerial	Opt.	string	Device serial No.

Message Example

```
{
  "EventNotificationAlert": {
    "channelID": "1",
    "dateTime": "2018-01-21T12:50:39+08:00",
    "activePostCount": 1,
    "eventType": "flowOverrun",
    "eventState": "active",
    "eventDescription": "",
    "devIndex": "2cd6716d-767f-4756-ac55-50276a5e3b4a",
    "channelName": "Ipdome",
    "deviceId": "Abcd1234"
  }
}
```

A.51 JSON_EventNotificationAlert_GPSUpload

JSON message about details of GPS information uploading event

Message Field Description

Field Name	Req. or Opt.	Date Type	Description
channelID	Req.	string	Channel No. of the device that triggered alarm.
dateTime	Req.	ISO 8601	Alarm triggered time, which is in ISO 8601 time format (i.e., "yyyy-MM-ddThh:mm:ss").
activePostCount	Req.	int	Number of times that the same alarm has been triggered.

Field Name	Req. or Opt.	Date Type	Description
eventType	Req.	string	Type of event that triggers alarm, here it should be "GPSUpload".
eventState	Req.	string	Durative alarm/event status: "active"-valid, "inactive"-invalid, e.g., when a moving target is detected, the alarm/event information will be uploaded continuously until the status is set to "inactive".
eventDescription	Req.	string	Event description.
devIndex	Opt.	string	Device ID (uuid/guid).
channelName	Req.	string	Camera name.
deviceId	Req.	string	Device ID.
deviceSerial	Opt.	string	Device serial No.
GPS	Req.	object	GPS information.

Table A-10 GPS

Field Name	Req. or Opt.	Date Type	Description
divisionEW	Req.	string	East and west hemispheres: "E"-east hemisphere, "W"-west hemisphere.
longitude	Req.	int	Longitude = degree $\times$ 3600 $\times$ 100 + minute $\times$ 60 $\times$ 100 + second $\times$ 100
divisionNS	Req.	string	North and south hemispheres: "N"-north hemisphere, "S"-south hemisphere.
latitude	Req.	int	Latitude = degree $\times$ 3600 $\times$ 100 + minute $\times$ 60 $\times$ 100 + second $\times$ 100
direction	Req.	int	Vehicle direction = actual direction (unit: degree, the north direction is 0, calculate by clockwise) $\times$ 100
speed	Req.	int	Speed, unit: cm/h.
satellites	Req.	int	Number of satellites.
precision	Req.	int	Accuracy factor = original value $\times$ 100
height	Req.	int	Height, unit: cm.

Field Name	Req. or Opt.	Date Type	Description
retransFlag	Req.	int	ANR flag: 0 (real-time packet), 1 (ANR packet).
timeZone	Opt.	string	Time zone.
timeZoneldx	Opt.	int	Time zone index No.: 1 (GMT-12:00), 2 (GMT-11:00), 3 (GMT-10:00), ..., 14 (GMT-01:00), 15 (GMT), 16 (GMT+01:00), ..., 34 (GMT-13:00), 35 (GMT+14:00).

Remarks

If the **timeZone** and **timeZoneldx** conflicted, the **timeZoneldx** is in priority

Message Example

```
{
  "EventNotificationAlert": {
    "channelID": "1",
    "dateTime": "2018-01-21T12:50:39+08:00",
    "activePostCount": 1,
    "eventType": "GPSUpload",
    "eventState": "active",
    "eventDescription": "GPS Info",
    "devIndex": "2cd6716d-767f-4756-ac55-50276a5e3b4a",
    "channelName": "Ipdome",
    "deviceId": "Abcd1234",
    "GPS": {
      "divisionEW": "E",
      "longitude": 130345,
      "divisions": "N",
      "latitude": 28650,
      "direction": 2898,
      "speed": 12399,
      "satellites": 12,
      "precision": 3,
      "height": 1,
      "retransFlag": "0",
      "timeZone": "HH::MM",
      "timeZoneldx": 0
    }
  }
}
```

A.52 JSON_EventNotificationAlert_group

JSON message about people gathering alarm details

Message Field Description

Field Name	Req. or Opt.	Date Type	Description
channelID	Req.	string	Channel No. of the device that triggered alarm.
dateTime	Req.	ISO 8601	Alarm triggered time, which is in ISO 8601 time format (i.e., "yyyy-MM-ddThh:mm:ss").
activePostCount	Req.	int	Number of times that the same alarm has been triggered.
eventType	Req.	string	Type of event that triggers alarm, here it should be "group".
eventState	Req.	string	Durative alarm/event status: "active"-valid, "inactive"-invalid, e.g., when a moving target is detected, the alarm/event information will be uploaded continuously until the status is set to "inactive".
eventDescription	Req.	string	Event description, here it is "People gathering detection alarm".
devIndex	Opt.	string	Device ID (uuid/guid).
channelName	Req.	string	Camera name.
deviceId	Req.	string	Device ID.
deviceSerial	Opt.	string	Device serial No.
DetectionRegionList	Opt.	object	Detection region, this node is valid in expert mode.
DetectionRegionEntry	Req.	list	Detection region information.
regionID	Req.	string	Detection region ID.
sensitivityLevel	Opt.	int	Sensitivity. Value range: [0,100].
RegionCoordinatesList	Opt.	list	Detection region coordinates.
RegionCoordinates			

Field Name	Req. or Opt.	Date Type	Description
positionX	Req.	int	X-coordinate of a vertex on the rule frame, which is between 0 and 1000.
positionY	Req.	int	Y-coordinate of a vertex on the rule frame, which is between 0 and 1000.

Message Example

```
{
  "EventNotificationAlert": {
    "channelID": "1",
    "dateTime": "2018-01-21T12:50:39+08:00",
    "activePostCount": 1,
    "eventType": "group",
    "eventState": "active",
    "eventDescription": "",
    "devIndex": "2cd6716d-767f-4756-ac55-50276a5e3b4a",
    "channelName": "Ipdome",
    "deviceId": "Abcd1234",
    "DetectionRegionList": [{
      "DetectionRegionEntry": {
        "regionID": "1",
        "sensitivityLevel": 50,
        "RegionCoordinatesList": [{
          "RegionCoordinates": {
            "positionX": 0,
            "positionY": 0
          },
          "RegionCoordinates": {
            "positionX": 0,
            "positionY": 100
          },
          "RegionCoordinates": {
            "positionX": 100,
            "positionY": 100
          },
          "RegionCoordinates": {
            "positionX": 100,
            "positionY": 0
          }
        ]
      }
    }]
  }
}
```

A.53 JSON_EventNotificationAlert_HeartbeatInfo

JSON message about heartbeat information to be uploaded

Message Field Description

Field Name	Req. or Opt.	Date Type	Description
channelID	Opt.	string	Channel No. of the device that triggered alarm.
dateTime	Req.	string	Alarm triggered time, which is in ISO 8601 time format (i.e., "yyyy-MM-ddThh:mm:ss").
activePostCount	Req.	int	Number of times that the same alarm has been triggered.
eventType	Req.	string	Event type, here it should be "heartBeat".
eventState	Req.	string	Durative alarm/event status: "active"-valid, "inactive"-invalid, e.g., when a moving target is detected, the alarm/event information will be uploaded continuously unit the status is set to "inactive".
eventDescription	Req.	string	Event description, here it is "Heartbeat".

Remarks

You can calculate the arming duration according to the value of node **activePostCount** and the heartbeat interval.

Example

Message Example of Uploading Heartbeat Information

```
{
  "EventNotificationAlert":{
    "channelID":"1",
    "dateTime":"2017-04-22T15:39:01+08:00",
    "activePostCount":2,
    "eventType":"heartBeat",
    "eventState":"active",
    "eventDescription":"Heartbeat"
  }
}
```

```
}  
}
```

A.54 JSON_EventNotificationAlert_highTempAlarm

JSON message about details of temperature too high alarm

Message Field Description

Field Name	Req. or Opt.	Date Type	Description
channelID	Req.	string	Channel No. of the device that triggered alarm.
dateTime	Req.	ISO 8601	Alarm triggered time, which is in ISO 8601 time format (i.e., "yyyy-MM-ddThh:mm:ss").
activePostCount	Req.	int	Number of times that the same alarm has been triggered.
eventType	Req.	string	Type of event that triggers alarm, here it should be "highTempAlarm".
eventState	Req.	string	Durative alarm/event status: "active"-valid, "inactive"-invalid, e.g., when a moving target is detected, the alarm/event information will be uploaded continuously until the status is set to "inactive".
eventDescription	Req.	string	Event description.
devIndex	Opt.	string	Device ID (uuid/guid).
channelName	Req.	string	Camera name.
deviceId	Req.	string	Device ID.
deviceSerial	Opt.	string	Device serial No.

Message Example

```
{  
  "EventNotificationAlert": {  
    "channelID": "1",  
    "dateTime": "2018-01-21T12:50:39+08:00",  
    "activePostCount": 1,  
    "eventType": " highTempAlarm",  
    "eventState": "active",  
  }  
}
```

```

"eventDescription": "high temperature alarm",
"devIndex": "2cd6716d-767f-4756-ac55-50276a5e3b4a",
"channelName": "Ipdome",
"deviceId": "Abcd1234"
}
}

```

A.55 JSON_EventNotificationAlert_IO

JSON message about IO alarm details

Message Field Description

Field Name	Req. or Opt.	Date Type	Description
channelID	Req.	string	Channel No. of the device that triggered alarm
dateTime	Req.	ISO 8601	Alarm triggered time, which is in ISO 8601 time format (i.e., "yyyy-MM-ddThh:mm:ss").
activePostCount	Req.	int	Number of times that the same alarm has been triggered.
eventType	Req.	string	Event type, here it should be "IO".
eventState	Req.	string	Durative alarm/event status: "active"-valid, "inactive"-invalid, e.g., when a moving target is detected, the alarm/event information will be uploaded continuously unit the status is set to "inactive"
eventDescription	Req.	string	Event description, here it is "IO alarm".
devIndex	Opt.	string	Device ID (uuid/guid)
inputIOPortID	Opt.	int	IO input port ID
channelName	Req.	string	Camera name
deviceId	Req.	string	Device ID.
deviceSerial	Opt.	string	Device serial No.
DeviceLocation	Opt.	object	Device position information.

Field Name	Req. or Opt.	Date Type	Description
			For mobile devices, when the IO alarm is triggered, and the current GPS information is valid, then the device position information needs to be uploaded.

Table A-11 DeviceLocation

Field Name	Req. or Opt.	Data Type	Description
longitudeType	Req.	string	Longitude type: "E" (East), "W" (West).
latitudeType	Req.	string	Latitude type: "S" (South), "N" (North).
Longitude	Req.	object	Longitude.
Latitude	Req.	object	Latitude.
speed	Opt.	int	Device speed. Unit: centimeter/hour. This field is valid for mobile devices only.

Table A-12 Longitude or Latitude

Field Name	Req. or Opt.	Data Type	Description
degree	Req.	int	Degrees.
minute	Req.	int	Minutes, the range of value:[0, 59].
sec	Req.	float	Seconds, the range of value: [0, 59.999999]. It is accurate to 6 decimal places.

Message Example

```
{
  "EventNotificationAlert": {
    "DeviceLocation": {
      "Latitude": {
        "degree": 30,
        "minute": 11,
        "sec": 19.0
      },
      "Longitude": {
        "degree": 120,
        "minute": 12,
        "sec": 18.0
      },
      "latitudeAndLongitude": "N 30° 11' 19.000", E 120° 12' 18.000""
    }
  }
}
```

```

"latitudeType": "N",
"longitudeType": "E",
"speed": 0
},
"activePostCount": 1,
"channelID": "0",
"channelName": "",
"deviceId": "Abcd1234",
"dateTime": "2022-02-18T16:09:57+08:00",
"devIndex": "D62BFE88-738E-42AC-85B6-FBA5BA4C6F7F",
"eventDescription": "IO Alarm",
"eventState": "active",
"eventType": "IO",
"inputIOPortID": 1
}
}

```

A.56 JSON_EventNotificationAlert_linedetection

JSON message about line crossing alarm details

Message Field Description

Field Name	Req. or Opt.	Date Type	Description
channelID	Req.	string	Channel No. of the device that triggered alarm.
dateTime	Req.	ISO 8601	Alarm triggered time, which is in ISO 8601 time format (i.e., "yyyy-MM-ddThh:mm:ss").
activePostCount	Req.	int	Number of times that the same alarm has been triggered.
eventType	Req.	string	Type of event that triggers alarm, here it should be "linedetection".
eventState	Req.	string	Durative alarm/event status: "active"-valid, "inactive"-invalid, e.g., when a moving target is detected, the alarm/event information will be uploaded continuously until the status is set to "inactive".
eventDescription	Req.	string	Event description, here it is "Line crossing detection alarm".

Field Name	Req. or Opt.	Date Type	Description
devIndex	Opt.	string	Device ID (uuid/guid).
channelName	Req.	string	Camera name.
deviceId	Req.	string	Device ID.
deviceSerial	Opt.	string	Device serial No.
DetectionRegionList	Opt.	object	Detection region, this node is valid in expert mode.
DetectionRegionEntry	Req.	list	Detection region information.
regionID	Req.	string	Detection region ID.
sensitivityLevel	Opt.	int	Sensitivity. Value range: [0,100].
RegionCoordinatesList	Opt.	list	Detection region coordinates.
RegionCoordinates			
positionX	Req.	int	X-coordinate of a vertex on the rule frame, which is between 0 and 1000.
positionY	Req.	int	Y-coordinate of a vertex on the rule frame, which is between 0 and 1000.
detectionTarget	Opt.	string	Target type: "human", "vehicle", "others".
TargetRect	Opt.		Coordinates of target.
X	Req.	float	X-coordinate.
Y	Req.	float	Y-coordinate.
width	Req.	float	Width of target.
height	Req.	float	Height of target.

Example

Sample Message of Line Crossing Alarm Details

```
{
  "EventNotificationAlert":{
    "channelID":"1",
    "dateTime":"2018-01-21T12:50:39+08:00",
    "activePostCount":1,
    "eventType":"linedetection",
```



```
"eventState":"active",
"eventDescription":"","
"devIndex":"2cd6716d-767f-4756-ac55-50276a5e3b4a",
"channelName":"Ipdome",
"deviceId":"Abcd1234",
"DetectionRegionList":[{"
  "DetectionRegionEntry":{
    "regionID":"1",
    "sensitivityLevel":50,
    "RegionCoordinatesList":[{"
      "RegionCoordinates":{
        "positionX":0,
        "positionY":0
      },
      "RegionCoordinates":{
        "positionX":0,
        "positionY":100
      },
      "RegionCoordinates":{
        "positionX":100,
        "positionY":100
      },
      "RegionCoordinates":{
        "positionX":100,
        "positionY":0
      }
    }
  }
}],
"detectionTarget":"human",
"TargetRect":{
  "X":0,
  "Y":1,
  "width":0.5,
  "height":0.5
}
}
}
```

A.57 JSON_EventNotificationAlert_loitering

JSON message about loitering alarm details

Message Field Description

Field Name	Req. or Opt.	Date Type	Description
channelID	Req.	string	Channel No. of the device that triggered alarm.
dateTime	Req.	ISO 8601	Alarm triggered time, which is in ISO 8601 time format (i.e., "yyyy-MM-ddThh:mm:ss").
activePostCount	Req.	int	Number of times that the same alarm has been triggered.
eventType	Req.	string	Type of event that triggers alarm, here it should be "loitering".
eventState	Req.	string	Durative alarm/event status: "active"-valid, "inactive"-invalid, e.g., when a moving target is detected, the alarm/event information will be uploaded continuously until the status is set to "inactive".
eventDescription	Req.	string	Event description, here it is "Loitering alarm".
devIndex	Opt.	string	Device ID (uuid/guid).
channelName	Req.	string	Camera name.
deviceId	Req.	string	Device ID.
deviceSerial	Opt.	string	Device serial No.
DetectionRegionList	Opt.	object	Detection region, this node is valid in expert mode.
DetectionRegionEntry	Req.	list	Detection region information.
regionID	Req.	string	Detection region ID.
sensitivityLevel	Opt.	int	Sensitivity. Value range: [0,100].
RegionCoordinatesList	Opt.	list	Detection region coordinates.
RegionCoordinates			

Field Name	Req. or Opt.	Date Type	Description
positionX	Req.	int	X-coordinate of a vertex on the rule frame, which is between 0 and 1000.
positionY	Req.	int	Y-coordinate of a vertex on the rule frame, which is between 0 and 1000.

Message Example

```
{
  "EventNotificationAlert": {
    "channelID": "1",
    "dateTime": "2018-01-21T12:50:39+08:00",
    "activePostCount": 1,
    "eventType": "loitering",
    "eventState": "active",
    "eventDescription": "",
    "devIndex": "2cd6716d-767f-4756-ac55-50276a5e3b4a",
    "channelName": "Ipdome",
    "deviceId": "Abcd1234",
    "DetectionRegionList": [{
      "DetectionRegionEntry": {
        "regionID": "1",
        "sensitivityLevel": 50,
        "RegionCoordinatesList": [{
          "RegionCoordinates": {
            "positionX": 0,
            "positionY": 0
          },
          "RegionCoordinates": {
            "positionX": 0,
            "positionY": 100
          },
          "RegionCoordinates": {
            "positionX": 100,
            "positionY": 100
          },
          "RegionCoordinates": {
            "positionX": 100,
            "positionY": 0
          }
        ]
      }
    }]
  }
}
```

A.58 JSON_EventNotificationAlert_mobilePeopleCounting

JSON message about details of mobile people counting event

Message Field Description

Field Name	Req. or Opt.	Date Type	Description
channelID	Req.	string	Channel No. of the device that triggered alarm.
dateTime	Req.	ISO 8601	Alarm triggered time, which is in ISO 8601 time format (i.e., "yyyy-MM-ddThh:mm:ss").
activePostCount	Req.	int	Number of times that the same alarm has been triggered.
eventType	Req.	string	Type of event that triggers alarm, here it should be "mobilePeopleCounting".
eventState	Req.	string	Durative alarm/event status: "active"-valid, "inactive"-invalid, e.g., when a moving target is detected, the alarm/event information will be uploaded continuously until the status is set to "inactive".
eventDescription	Req.	string	Event description.
devIndex	Opt.	string	Device ID (uuid/guid).
channelName	Req.	string	Camera name.
deviceId	Req.	string	Device ID.
deviceSerial	Opt.	string	Device serial No.
MobilePeopleCounting	Req.	object	Mobile people counting data.

Table A-13 MobilePeopleCounting

Field Name	Req. or Opt.	Date Type	Description
doorIndex	Opt.	int	Vehicle door No.
statisticalMethods	Opt.	string	Statistics method: "timeRange"-statistics by time period.

Field Name	Req. or Opt.	Date Type	Description
TimeRange	Opt.	object	Time period, which is valid only when statisticalMethods is set to "timeRange"
enter	Opt.	int	Number of entered persons.
exit	Opt.	int	Number of exited persons.
stay	Opt.	int	Number of persons in the vehicle currently.
level	Opt.	int	Status levels in the vehicle: 0-free, the value range of stay is [0,20); 1-general, the value of stay is [20,30); 2-busy, the value range of stay is [30,50); 3-congestion, the value of stay is larger than and equal to 50.
GPS	Opt.	object	GPS information.
transFlag	Opt.	int	Retransmission flags: "realTime"-real-time data, "retrans"-retransmitted data.

Table A-14 TimeRange

Field Name	Req. or Opt.	Date Type	Description
startTime	Req.	string	Start time of statistics time period, e.g., 2018-01-21T11:50:39+08:00.
stopTime	Req.	string	End time of statistics time period, e.g., 2018-01-21T11:50:39+08:00.

Table A-15 GPS

Field Name	Req. or Opt.	Date Type	Description
longitude	Req.	int	Longitude = degree × 3600 × 100 + minute × 60 × 100 + second × 100
latitude	Req.	int	Latitude = degree × 3600 × 100 + minute × 60 × 100 + second × 100
vehicleSpeed	Opt.	int	Speed, unit: km/h, range of value: [0, 255].
direction	Opt.	int	Its value is calculated by (actual direction value × 100), the actual direction value equals to the angle between the positive direction of y-axis and the actual direction,

Field Name	Req. or Opt.	Date Type	Description
			and the clockwise direction is positive. Unit: degree.

Message Example

```
{
  "EventNotificationAlert": {
    "channelID": "1",
    "dateTime": "2018-01-21T12:50:39+08:00",
    "activePostCount": 1,
    "eventType": "mobilePeopleCounting",
    "eventState": "active",
    "eventDescription": "mobile people counting",
    "channelName": "Ipdome",
    "deviceID": "Abcd1234",
    "MobilePeopleCounting": {
      "doorIndex": 1,
      "statisticalMethods": "timeRange",
      "TimeRange": {
        "startTime": "2018-01-21T11:50:39+08:00",
        "stopTime": "2018-01-21T12:50:30+08:00"
      },
      "enter": 1,
      "exit": 1,
      "stay": 0,
      "level": 0,
      "GPS": {
        "longitude": 1,
        "latitude": 2,
        "vehicleSpeed": 60,
        "direction": 2
      },
      "transFlag": "realTime"
    }
  }
}
```

A.59 JSON_EventNotificationAlert_OverspeedAlarmMsg

JSON message about overspeed alarm details

Message Field Description

Field Name	Req. or Opt.	Date Type	Description
channelID	Req.	string	Channel No. of the device that triggered alarm.
dateTime	Req.	ISO 8601	Alarm triggered time, which is in ISO 8601 time format (i.e., "yyyy-MM-ddThh:mm:ss").
activePostCount	Req.	int	Number of times that the same alarm has been triggered.
eventType	Req.	string	Type of event that triggers alarm, here it should be "overSpeed".
eventState	Req.	string	Durative alarm/event status: "active"-valid, "inactive"-invalid, e.g., when a moving target is detected, the alarm/event information will be uploaded continuously unit the status is set to "inactive".
eventDescription	Req.	string	Event description.
devIndex	Opt.	string	Device ID (uuid/guid).
channelName	Req.	string	Camera name.
deviceId	Req.	string	Device ID.
deviceSerial	Opt.	string	Device serial No.

Message Example

```
{
  "EventNotificationAlert": {
    "channelID": "1",
    "dateTime": "2018-01-21T12:50:39+08:00",
    "activePostCount": 1,
    "eventType": "overSpeed",
    "eventState": "active",
    "eventDescription": "overSpeed alarm",
    "devIndex": "2cd6716d-767f-4756-ac55-50276a5e3b4a",
    "channelName": "Ipdome",
    "deviceId": "Abcd1234"
  }
}
```

A.60 JSON_EventNotificationAlert_PALMismatch

JSON message about details of video standard mismatch alarm

Message Field Description

Field Name	Req. or Opt.	Date Type	Description
channelID	Req.	string	Channel number of the device that triggered alarm.
dateTime	Req.	ISO 8601	Alarm triggered time, which is in ISO 8601 time format (i.e., "2017-04-22T15:39:01+08:00").
activePostCount	Req.	int	Number of times that the same alarm has been triggered.
eventType	Req.	string	Event type, here it should be "PALMismatch".
eventState	Req.	string	Durative alarm/event status: "active"-valid, "inactive"-invalid, e.g., when a moving target is detected, the alarm/event information will be uploaded continuously until the status is set to "inactive".
eventDescription	Req.	string	Event description.
devIndex	Opt.	string	Device index (uuid/guid)
channelName	Req.	string	Camera name
deviceId	Req.	string	Device ID.
deviceSerial	Opt.	string	Device serial No.

Message Example

```
{
  "EventNotificationAlert": {
    "channelID": "1",
    "dateTime": "2018-01-21T12:50:39+08:00",
    "activePostCount": 1,
    "eventType": "PALMismatch",
    "eventState": "active",
    "eventDescription": "PALMismatch alarm",
    "devIndex": "2cd6716d-767f-4756-ac55-50276a5e3b4a",
    "channelName": "Ipdome",
    "deviceId": "Abcd1234"
  }
}
```



```
}  
}
```

A.61 JSON_EventNotificationAlert_parking

JSON message about parking alarm details

Message Field Description

Field Name	Req. or Opt.	Date Type	Description
channelID	Req.	string	Channel No. of the device that triggered alarm.
dateTime	Req.	ISO 8601	Alarm triggered time, which is in ISO 8601 time format (i.e., "yyyy-MM-ddThh:mm:ss").
activePostCount	Req.	int	Number of times that the same alarm has been triggered.
eventType	Req.	string	Type of event that triggers alarm, here it should be "parking".
eventState	Req.	string	Durative alarm/event status: "active"-valid, "inactive"-invalid, e.g., when a moving target is detected, the alarm/event information will be uploaded continuously until the status is set to "inactive".
eventDescription	Req.	string	Event description, here it is "Parking detection alarm".
devIndex	Opt.	string	Device ID (uuid/guid).
channelName	Req.	string	Camera name.
deviceID	Req.	string	Device ID.
deviceSerial	Opt.	string	Device serial No.
DetectionRegionList	Opt.	object	Detection region, this node is valid in expert mode.
DetectionRegionEntry	Req.	list	Detection region information.
regionID	Req.	string	Detection region ID.
sensitivityLevel	Opt.	int	Sensitivity. Value range: [0,100].

Field Name	Req. or Opt.	Date Type	Description
RegionCoordinatesList	Opt.	list	Detection region coordinates.
RegionCoordinates			
positionX	Req.	int	X-coordinate of a vertex on the rule frame, which is between 0 and 1000.
positionY	Req.	int	Y-coordinate of a vertex on the rule frame, which is between 0 and 1000.

Message Example

```
{
  "EventNotificationAlert": {
    "channelID": "1",
    "dateTime": "2018-01-21T12:50:39+08:00",
    "activePostCount": 1,
    "eventType": "parking",
    "eventState": "active",
    "eventDescription": "",
    "devIndex": "2cd6716d-767f-4756-ac55-50276a5e3b4a",
    "channelName": "Ipdome",
    "deviceId": "Abcd1234",
    "DetectionRegionList": [{
      "DetectionRegionEntry": {
        "regionID": "1",
        "sensitivityLevel": 50,
        "RegionCoordinatesList": [{
          "RegionCoordinates": {
            "positionX": 0,
            "positionY": 0
          },
          "RegionCoordinates": {
            "positionX": 0,
            "positionY": 100
          },
          "RegionCoordinates": {
            "positionX": 100,
            "positionY": 100
          },
          "RegionCoordinates": {
            "positionX": 100,
            "positionY": 0
          }
        ]
      }
    }]
  }
}
```

```
}  
}
```

A.62 JSON_EventNotificationAlert_peopleCounting

JSON message about people counting statistics details

Field Name	Req. or Opt.	Date Type	Description
channelID	Req.	string	Channel No. of the device that triggered alarm.
dateTime	Req.	ISO 8601	Alarm triggered time, which is in ISO 8601 time format and contains the time zone information. For example, 2017-04-22T15:39:01+08:00.
activePostCount	Req.	int	Number of times that the same alarm has been triggered.
eventType	Req.	string	Type of event that triggers alarm, here it should be "peopleCounting".
eventState	Req.	string	Durative alarm/event status: "active"-valid, "inactive"-invalid, e.g., when a moving target is detected, the alarm/event information will be uploaded continuously until the status is set to "inactive".
eventDescription	Req.	string	Event description.
devIndex	Opt.	string	Device ID (uuid/guid).
channelName	Req.	string	Camera name.
deviceId	Req.	string	Device ID.
deviceSerial	Opt.	string	Device serial No.
peopleCounting	Req.	Object	People counting statistics.
DetectionRegionList	Opt.	Array	Detection area (DetectionRegionEntry object).
isDataRetransmission	Opt.	bool	Whether the data is replenished.

peopleCounting

Field Name	Req. or Opt.	Date Type	Description
statisticalMethods	Opt.	string	Counting method: "realTime" (real-time counting), "timeRange" (count by time range), "signalTrigger" (alarm input triggered counting).
RealTime	Opt.	Object	Real time, it is required when statisticalMethods is "realTime".
TimeRange	Opt.	Object	Real time range, it is required when statisticalMethods is "timeRange" or "signalTrigger".
enter	Opt.	int	Number of entered people.
exit	Opt.	int	Number of exited people.
pass	Opt.	int	Number of passers-by. People who enter the counting area but do not cross the counting line are counted as passers-by.
duplicatePeople	Opt.	int	Number of duplicated people. It is valid for people counting cameras. If people entered repeatedly within a specified time period, they will be counted as duplicated people.
FlowStatisticsList	Opt.	Array	Flow direction statistics (FlowStatistics object).
countingSceneMode	Opt.	string	People counting scene mode: "entrance", "exit". The value is "entrance" by default.
deDuplicateEnter	Opt.	int	Deduplicated number of entered people.
deDuplicateExit	Opt.	int	Deduplicated number of exited people.
faceLibDuplicatePeopleList	Opt.	Array	Number of duplicated people in the face picture library (faceLibDuplicatePeople object).

TimeRange

Field Name	Req. or Opt.	Date Type	Description
time	Req.	string	People counting time. For example, 2017-04-22T15:39:01+08:00.

RealTime

Field Name	Req. or Opt.	Date Type	Description
startTime	Req.	string	Start time of people counting. For example, 2017-04-22T15:39:01+08:00.
stopTime	Req.	string	End time of people counting. For example, 2017-04-22T15:39:01+08:00.

FlowStatistics

Field Name	Req. or Opt.	Date Type	Description
countingLineID	Opt.	int	Detection line ID.
pass	Opt.	int	Number of passers-by. People who enter the counting area but do not cross the counting line.
movingDirectionList	Opt.	Array	People moving direction (MovingDirection objects).

MovingDirection

Field Name	Req. or Opt.	Date Type	Description
countingLineSegmentID	Opt.	int	Area ID, range: [1,3]. The four points of the broken line are marked as 1, 2, 3, and 4 respectively in counterclockwise direction from the counting direction. Draw two lines parallel to the counting direction respectively starting from point 2 and point 3 to intersect the counting rule area. The formed three areas are marked as 1, 2,

Field Name	Req. or Opt.	Date Type	Description
			and 3 in counterclockwise direction from the counting direction.
count	Opt.	int	Number of people.

faceLibDuplicatePeople

Field Name	Req. or Opt.	Date Type	Description
FDID	Req.	int	Face picture library ID.
name	Req.	int	Face picture library name.

Message Example

```
{
  "EventNotificationAlert": {
    "channelID": "1",
    "dateTime": "2018-01-21T12:50:39+08:00",
    "activePostCount": 1,
    "eventType": "peopleCounting",
    "eventState": "active",
    "eventDescription": "PeopleCounting alarm",
    "channelName": "Ipdome",
    "deviceId": "Abcd1234",
    "peopleCounting": {
      "statisticalMethods": "timeRange",
      "TimeRange": {
        "startTime": "2018-01-21T11:50:39+08:00",
        "stopTime": "2018-01-21T12:50:30+08:00"
      }
    },
    "enter": 1,
    "exit": 1
  }
}
```

A.63 JSON_EventNotificationAlert_PIR

JSON message about PIR (Passive Infrared) alarm details

Message Field Description

Field Name	Req. or Opt.	Date Type	Description
channelID	Req.	string	Channel No. of the device that triggered alarm.
dateTime	Req.	ISO 8601	Alarm triggered time, which is in ISO 8601 time format (i.e., "yyyy-MM-ddThh:mm:ss").
activePostCount	Req.	int	Number of times that the same alarm has been triggered.
eventType	Req.	string	Event type, here it should be "PIR".
eventState	Req.	string	Durative alarm/event status: "active"-valid, "inactive"-invalid, e.g., when a moving target is detected, the alarm/event information will be uploaded continuously unit the status is set to "inactive".
eventDescription	Req.	string	Event description, here it is "PIR alarm".
devIndex	Opt.	string	Device ID (uuid/guid).
channelName	Req.	string	Camera name.
deviceID	Req.	string	Device ID.
deviceSerial	Opt.	string	Device serial No.

Message Example

```
{
  "EventNotificationAlert": {
    "channelID": "1",
    "dateTime": "2018-01-21T12:50:39+08:00",
    "activePostCount": 1,
    "eventType": "PIR",
    "eventState": "active",
    "eventDescription": "PIR alarm",
    "devIndex": "2cd6716d-767f-4756-ac55-50276a5e3b4a",
    "channelName": "Ipdome",
    "deviceID": "Abcd1234"
  }
}
```

A.64 JSON_EventNotificationAlert_rapidMove

JSON message about fast moving alarm details

Message Field Description

Field Name	Req. or Opt.	Date Type	Description
channelID	Req.	string	Channel No. of the device that triggered alarm.
dateTime	Req.	ISO 8601	Alarm triggered time, which is in ISO 8601 time format (i.e., "yyyy-MM-ddThh:mm:ss").
activePostCount	Req.	int	Number of times that the same alarm has been triggered.
eventType	Req.	string	Type of event that triggers alarm, here it should be "rapidMove".
eventState	Req.	string	Durative alarm/event status: "active"-valid, "inactive"-invalid, e.g., when a moving target is detected, the alarm/event information will be uploaded continuously until the status is set to "inactive".
eventDescription	Req.	string	Event description, here it is "Fast moving alarm".
devIndex	Opt.	string	Device ID (uuid/guid).
channelName	Req.	string	Camera name.
deviceId	Req.	string	Device ID.
deviceSerial	Opt.	string	Device serial No.
DetectionRegionList	Opt.	object	Detection region, this node is valid in expert mode.
DetectionRegionEntry	Req.	list	Detection region information.
regionID	Req.	string	Detection region ID.
sensitivityLevel	Opt.	int	Sensitivity. Value range: [0,100].
RegionCoordinatesList	Opt.	list	Detection region coordinates.
RegionCoordinates			

Field Name	Req. or Opt.	Date Type	Description
positionX	Req.	int	X-coordinate of a vertex on the rule frame, which is between 0 and 1000.
positionY	Req.	int	Y-coordinate of a vertex on the rule frame, which is between 0 and 1000.

Message Example

```
{
  "EventNotificationAlert": {
    "channelID": "1",
    "dateTime": "2018-01-21T12:50:39+08:00",
    "activePostCount": 1,
    "eventType": "rapidMove",
    "eventState": "active",
    "eventDescription": "",
    "devIndex": "2cd6716d-767f-4756-ac55-50276a5e3b4a",
    "channelName": "Ipdome",
    "deviceId": "Abcd1234",
    "DetectionRegionList": [{
      "DetectionRegionEntry": {
        "regionID": "1",
        "sensitivityLevel": 50,
        "RegionCoordinatesList": [{
          "RegionCoordinates": {
            "positionX": 0,
            "positionY": 0
          },
          "RegionCoordinates": {
            "positionX": 0,
            "positionY": 100
          },
          "RegionCoordinates": {
            "positionX": 100,
            "positionY": 100
          },
          "RegionCoordinates": {
            "positionX": 100,
            "positionY": 0
          }
        ]
      }
    }]
  }
}
```

A.65 JSON_EventNotificationAlert_regionEntrance

JSON message about region entrance alarm details

Message Field Description

Field Name	Req. or Opt.	Date Type	Description
channelID	Req.	string	Channel No. of the device that triggered alarm.
dateTime	Req.	ISO 8601	Alarm triggered time, which is in ISO 8601 time format (i.e., "yyyy-MM-ddThh:mm:ss").
activePostCount	Req.	int	Number of times that the same alarm has been triggered.
eventType	Req.	string	Type of event that triggers alarm, here it should be "regionEntrance".
eventState	Req.	string	Durative alarm/event status: "active"-valid, "inactive"-invalid, e.g., when a moving target is detected, the alarm/event information will be uploaded continuously unit the status is set to "inactive".
eventDescription	Req.	string	Event description, here it is "Region entrance alarm".
devIndex	Opt.	string	Device ID (uuid/guid).
channelName	Req.	string	Camera name.
deviceId	Req.	string	Device ID.
deviceSerial	Opt.	string	Device serial No.
DetectionRegionList	Opt.	object	Detection region, this node is valid in expert mode.
DetectionRegionEntry	Req.	list	Detection region information.
regionID	Req.	string	Detection region ID.
sensitivityLevel	Opt.	int	Sensitivity. Value range: [0,100].

Field Name	Req. or Opt.	Date Type	Description
RegionCoordinatesList	Opt.	list	Detection region coordinates.
RegionCoordinates			
positionX	Req.	int	X-coordinate of a vertex on the rule frame, which is between 0 and 1000.
positionY	Req.	int	Y-coordinate of a vertex on the rule frame, which is between 0 and 1000.
detectionTarget	Opt.	string	Target type: "human", "vehicle", "others".
TargetRect	Opt.		Coordinates of target.
X	Req.	float	X-coordinate.
Y	Req.	float	Y-coordinate.
width	Req.	float	Width of target.
height	Req.	float	Height of target.

Message Example

```
{
  "EventNotificationAlert":{
    "channelID":"1",
    "dateTime":"2018-01-21T12:50:39+08:00",
    "activePostCount":1,
    "eventType":"regionEntrance",
    "eventState":"active",
    "eventDescription": "",
    "devIndex":"2cd6716d-767f-4756-ac55-50276a5e3b4a",
    "channelName":"Ipdome",
    "deviceID":"Abcd1234",
    "DetectionRegionList":{
      "DetectionRegionEntry":{
        "regionID":"1",
        "sensitivityLevel":50,
        "RegionCoordinatesList":{
          "RegionCoordinates":{
            "positionX":0,
            "positionY":0
          },
          "RegionCoordinates":{
            "positionX":0,
            "positionY":100
          }
        }
      }
    }
  }
}
```

```
    },
    "RegionCoordinates":{
      "positionX":100,
      "positionY":100
    },
    "RegionCoordinates":{
      "positionX":100,
      "positionY":0
    }
  }},
  "detectionTarget":"human",
  "TargetRect":{
    "X":0,
    "Y":1,
    "width":0.5,
    "height":0.5
  }
}
}}
```

A.66 JSON_EventNotificationAlert_regionExiting

JSON message about region exiting alarm details

Message Field Description

Field Name	Req. or Opt.	Date Type	Description
channelID	Req.	string	Channel No. of the device that triggered alarm.
dateTime	Req.	ISO 8601	Alarm triggered time, which is in ISO 8601 time format (i.e., "yyyy-MM-ddThh:mm:ss").
activePostCount	Req.	int	Number of times that the same alarm has been triggered.
eventType	Req.	string	Type of event that triggers alarm, here it should be "regionExiting".
eventState	Req.	string	Durative alarm/event status: "active"-valid, "inactive"-invalid, e.g., when a moving target is detected, the alarm/event information will be uploaded

Field Name	Req. or Opt.	Date Type	Description
			continuously unit the status is set to "inactive".
eventDescription	Req.	string	Event description, here it is "Exiting alarm".
devIndex	Opt.	string	Device ID (uuid/guid).
channelName	Req.	string	Camera name.
deviceId	Req.	string	Device ID.
deviceSerial	Opt.	string	Device serial No.
DetectionRegionList	Opt.	object	Detection region, this node is valid in expert mode.
DetectionRegionEntry	Req.	list	Detection region information.
regionID	Req.	string	Detection region ID.
sensitivityLevel	Opt.	int	Sensitivity. Value range: [0,100].
RegionCoordinatesList	Opt.	list	Detection region coordinates.
RegionCoordinates			
positionX	Req.	int	X-coordinate of a vertex on the rule frame, which is between 0 and 1000.
positionY	Req.	int	Y-coordinate of a vertex on the rule frame, which is between 0 and 1000.

Message Example

```
{
  "EventNotificationAlert":{
    "channelID":"1",
    "dateTime":"2018-01-21T12:50:39+08:00",
    "activePostCount":1,
    "eventType":"regionExiting",
    "eventState":"active",
    "eventDescription": "",
    "devIndex":"2cd6716d-767f-4756-ac55-50276a5e3b4a",
    "channelName":"Ipdome",
    "deviceId":"Abcd1234",
    "DetectionRegionList":[{"
```

```
"DetectionRegionEntry":{
  "regionID":"1",
  "sensitivityLevel":50,
  "RegionCoordinatesList":[{
    "RegionCoordinates":{
      "positionX":0,
      "positionY":0
    },
    "RegionCoordinates":{
      "positionX":0,
      "positionY":100
    },
    "RegionCoordinates":{
      "positionX":100,
      "positionY":100
    },
    "RegionCoordinates":{
      "positionX":100,
      "positionY":0
    }
  ]},
  "detectionTarget":"human",
  "TargetRect":{
    "X":0,
    "Y":1,
    "width":0.5,
    "height":0.5
  }
}
```

A.67 JSON_EventNotificationAlert_rollover

JSON message about rollover detection alarm details

Message Field Description

Field Name	Req. or Opt.	Date Type	Description
channelID	Req.	string	Channel No. of the device that triggered alarm.
dateTime	Req.	ISO 8601	Alarm triggered time, which is in ISO 8601 time format (i.e., "yyyy-MM-ddThh:mm:ss").

Field Name	Req. or Opt.	Date Type	Description
activePostCount	Req.	int	Number of times that the same alarm has been triggered.
eventType	Req.	string	Type of event that triggers alarm, here it should be "rollover".
eventState	Req.	string	Durative alarm/event status: "active"-valid, "inactive"-invalid, e.g., when a moving target is detected, the alarm/event information will be uploaded continuously until the status is set to "inactive".
eventDescription	Req.	string	Event description.
devIndex	Opt.	string	Device ID (uuid/guid).
threshold	Opt.	int	Rollover angle threshold, unit: degrees.
angle	Opt.	int	Rollover angle, unit: degrees.
speed	Opt.	int	Speed, unit: m/s.
channelName	Req.	string	Camera name.
deviceId	Req.	string	Device ID.
deviceSerial	Opt.	string	Device serial No.

Message Example

```
{
  "EventNotificationAlert": {
    "channelID": "1",
    "dateTime": "2018-01-21T12:50:39+08:00",
    "activePostCount": 1,
    "eventType": "rollover",
    "eventState": "active",
    "eventDescription": "rollover alarm",
    "devIndex": "2cd6716d-767f-4756-ac55-50276a5e3b4a",
    "threshold": 10,
    "angle": 10,
    "speed": 100,
    "channelName": "Ipdome",
    "deviceId": "Abcd1234"
  }
}
```

A.68 JSON_EventNotificationAlert_scenechangedetection

JSON message about scene changed alarm details

Message Field Description

Field Name	Req. or Opt.	Date Type	Description
channelID	Req.	string	Channel No. of the device that triggered alarm.
dateTime	Req.	ISO 8601	Alarm triggered time, which is in ISO 8601 time format (i.e., "2017-04-22T15:39:01+08:00").
activePostCount	Req.	int	Number of times that the same alarm has been triggered.
eventType	Req.	string	Event type, here it should be "scenechangedetection".
eventState	Req.	string	Durative alarm/event status: "active"-valid, "inactive"-invalid. E.g., when a moving target is detected, the alarm/event information will be uploaded continuously until the status is set to "inactive".
eventDescription	Req.	string	Event description.
devIndex	Opt.	string	Device index (uuid/guid).
channelName	Req.	string	Camera name.
deviceId	Req.	string	Device ID.
deviceSerial	Opt.	string	Device serial No.

Message Example

```
{
  "EventNotificationAlert": {
    "channelID": "1",
    "dateTime": "2018-01-21T12:50:39+08:00",
    "activePostCount": 1,
    "eventType": "scenechangedetection",
    "eventState": "active",
    "eventDescription": "Scene Change Detection",
```



```
"devIndex": "2cd6716d-767f-4756-ac55-50276a5e3b4a",  
"channelName": "Ipdome",  
"deviceId": "Abcd1234"  
}  
}
```

A.69 JSON_EventNotificationAlert_shelteralarm

JSON message about video tampering alarm details

Message Field Description

Field Name	Req. or Opt.	Date Type	Description
channelID	Req.	string	Channel No. of the device that triggered alarm.
dateTime	Req.	ISO 8601	Alarm triggered time, which is in ISO 8601 time format (i.e., "yyyy-MM-ddThh:mm:ss").
activePostCount	Req.	int	Number of times that the same alarm has been triggered.
eventType	Req.	string	Event type, here it should be "shelteralarm".
eventState	Req.	string	Durative alarm/event status: "active"-valid, "inactive"-invalid, e.g., when a moving target is detected, the alarm/event information will be uploaded continuously until the status is set to "inactive".
eventDescription	Req.	string	Event description, here it is "shelteralarm alarm".
devIndex	Opt.	string	Device ID (uuid/guid).
channelName	Req.	string	Camera name.
deviceId	Req.	string	Device ID.
deviceSerial	Opt.	string	Device serial No.

Message Example

```
{  
  "EventNotificationAlert":{
```

```

"channelID":"1",
"dateTime":"2018-03-13T19:42:27+08:00",
"activePostCount":1,
"eventType":"shelteralarm",
"eventState":"active",
"eventDescription":"shelteralarm alarm",
"devIndex":"2cd6716d-767f-4756-ac55-50276a5e3b4a",
"channelName":"Camera 01",
"deviceId": "Abcd1234"
}
}

```

A.70 JSON_EventNotificationAlert_unattendedBaggage

JSON message about unattended baggage alarm details

Message Field Description

Field Name	Req. or Opt.	Date Type	Description
channelID	Req.	string	Channel No. of the device that triggered alarm.
dateTime	Req.	ISO 8601	Alarm triggered time, which is in ISO 8601 time format (i.e., "yyyy-MM-ddThh:mm:ss").
activePostCount	Req.	int	Number of times that the same alarm has been triggered.
eventType	Req.	string	Type of event that triggers alarm, here it should be "unattendedBaggage".
eventState	Req.	string	Durative alarm/event status: "active"-valid, "inactive"-invalid, e.g., when a moving target is detected, the alarm/event information will be uploaded continuously unit the status is set to "inactive".
eventDescription	Req.	string	Event description, here it is "Unattended baggage alarm".
devIndex	Opt.	string	Device ID (uuid/guid).
channelName	Req.	string	Camera name.
deviceId	Req.	string	Device ID.
deviceSerial	Opt.	string	Device serial No.

Field Name	Req. or Opt.	Date Type	Description
DetectionRegionList	Opt.	object	Detection region, this node is valid in expert mode.
DetectionRegionEntry	Req.	list	Detection region information.
regionID	Req.	string	Detection region ID.
sensitivityLevel	Opt.	int	Sensitivity. Value range: [0,100].
RegionCoordinatesList	Opt.	list	Detection region coordinates.
RegionCoordinates			
positionX	Req.	int	X-coordinate of a vertex on the rule frame, which is between 0 and 1000.
positionY	Req.	int	Y-coordinate of a vertex on the rule frame, which is between 0 and 1000.

Message Example

```
{
  "EventNotificationAlert": {
    "channelID": "1",
    "dateTime": "2018-01-21T12:50:39+08:00",
    "activePostCount": 1,
    "eventType": "unattendedBaggage",
    "eventState": "active",
    "eventDescription": "",
    "devIndex": "2cd6716d-767f-4756-ac55-50276a5e3b4a",
    "channelName": "Ipdome",
    "deviceId": "Abcd1234",
    "DetectionRegionList": [{
      "DetectionRegionEntry": {
        "regionID": "1",
        "sensitivityLevel": 50,
        "RegionCoordinatesList": [{
          "RegionCoordinates": {
            "positionX": 0,
            "positionY": 0
          },
          "RegionCoordinates": {
            "positionX": 0,
            "positionY": 100
          }
        ]
      }
    ]
  }
}
```

```
"RegionCoordinates":{
  "positionX":100,
  "positionY":100
},
"RegionCoordinates":{
  "positionX":100,
  "positionY":0
}
}
}}
}
```

A.71 JSON_EventNotificationAlert_videoloss

JSON message about video loss alarm details

Message Field Description

Field Name	Req. or Opt.	Date Type	Description
channelID	Req.	string	Channel No. of the device that triggered alarm.
dateTime	Req.	ISO 8601	Alarm triggered time, which is in ISO 8601 time format (i.e., "yyyy-MM-ddThh:mm:ss").
activePostCount	Req.	int	Number of times that the same alarm has been triggered.
eventType	Req.	string	Event type, here it should be "videoloss".
eventState	Req.	string	Durative alarm/event status: "active"-valid, "inactive"-invalid, e.g., when a moving target is detected, the alarm/event information will be uploaded continuously unit the status is set to "inactive".
eventDescription	Req.	string	Event description, here it is "videoloss alarm".
devIndex	Opt.	string	Device ID (uuid/guid).
channelName	Req.	string	Camera name.

Field Name	Req. or Opt.	Date Type	Description
deviceId	Req.	string	Device ID.
deviceSerial	Opt.	string	Device serial No.

Message Example

```
{
  "EventNotificationAlert": {
    "channelID": "1",
    "dateTime": "2018-01-21T12:50:39+08:00",
    "activePostCount": 1,
    "eventType": "videoloss",
    "eventState": "active",
    "eventDescription": "videoloss alarm",
    "devIndex": "2cd6716d-767f-4756-ac55-50276a5e3b4a",
    "channelName": "Ipdome",
    "deviceId": "Abcd1234"
  }
}
```

A.72 JSON_EventNotificationAlert_VMD

JSON message about motion detection alarm details

Message Field Description

Field Name	Req. or Opt.	Date Type	Description
channelID	Req.	string	Channel No. of the device that triggered alarm.
dateTime	Req.	ISO 8601	Alarm triggered time, which is in ISO 8601 time format (i.e., "yyyy-MM-ddThh:mm:ss").
activePostCount	Req.	int	Number of times that the same alarm has been triggered.
eventType	Req.	string	Type of event that triggers alarm, here it should be "VMD".
eventState	Req.	string	Durative alarm/event status: "active"-valid, "inactive"-invalid, e.g., when a moving target is detected, the alarm/

Field Name	Req. or Opt.	Date Type	Description
			event information will be uploaded continuously unit the status is set to "inactive".
eventDescription	Req.	string	Event description, here it is "VMD alarm".
devIndex	Opt.	string	Device ID (uuid/guid).
channelName	Req.	string	Camera name.
deviceId	Req.	string	Device ID.
deviceSerial	Opt.	string	Device serial No.
DetectionRegionList	Opt.	object	Detection region, this node is valid in expert mode.
DetectionRegionEntry	Req.	list	Detection region information.
regionID	Req.	string	Detection region ID.
sensitivityLevel	Req.	int	Sensitivity. Value range: [0, 100].
RegionCoordinatesList	Req.	list	Detection region coordinates.
RegionCoordinates	Req.		
positionX	Req.	int	X-coordinate of a vertex on the rule frame, the range of value: [0, 1000].
positionY	Req.	int	Y-coordinate of a vertex on the rule frame, the range of value: [0, 1000].

Message Example (Normal Mode)

```
{
  "EventNotificationAlert":{
    "channelID": "1",
    "dateTime": "2018-03-13T19:42:27+08:00",
    "activePostCount": 1,
    "eventType": "VMD",
    "eventState": "active",
    "eventDescription": "Motion alarm",
    "devIndex": "2cd6716d-767f-4756-ac55-50276a5e3b4a",
    "channelName": "Camera 01",
    "deviceId": "Abcd1234"
  }
}
```

```
}  
}
```

Message Example (Expert Mode)

```
{  
  "EventNotificationAlert": {  
    "channelID": "1",  
    "dateTime": "2018-03-13T20:36:34+08:00",  
    "activePostCount": 1,  
    "eventType": "VMD",  
    "eventState": "active",  
    "eventDescription": "Motion alarm",  
    "DetectionRegionList": {  
      "DetectionRegionEntry": {  
        "regionID": "1",  
        "sensitivityLevel": 50,  
        "RegionCoordinatesList": {  
          "RegionCoordinates": [{  
            "positionX": 216,  
            "positionY": 216  
          },  
          {  
            "positionX": 756,  
            "positionY": 756  
          },  
          {  
            "positionX": 756,  
            "positionY": 756  
          },  
          {  
            "positionX": 216,  
            "positionY": 216  
          }  
        ]  
      }  
    },  
    "devIndex": "2cd6716d-767f-4756-ac55-50276a5e3b4a",  
    "channelName": "Camera 01"  
  }  
}
```

A.73 JSON_EventTrigger

EventTrigger message in JSON format

```
{  
  "EventTrigger": {  
    "id": "",  
    /*required, string, alarm/event ID*/  
  }  
}
```

```
"eventType":"","
/*required, string, alarm/event types, opt="diskfull, diskerror, videoloss, IO, shelteralarm, VMD, PALMismatch,
overSpeed, highTempAlarm, abnormalAcceleration, collision, rollover, abnormalDriving, GPSUpload"*/
"eventDescription":"","
/*optional, string, alarm/event description*/
"inputIOPortID":"","
/*optional, string, alarm input, it is valid when eventType="IO"*/
"dynInputIOPortID":"","
/*optional, string, dynamic alarm input, it is valid when eventType="IO"*/
"videoInputChannelID":"","
/*optional, string, video input channel No., it is valid when eventType="VMD", "videoloss", or "tamperdetection"*/
"dynVideoInputChannelID":"","
/*optional, string, dynamic video input channel No., it is valid when eventType="VMD", "videoloss", or
"tamperdetection"*/
"intervalBetweenEvents": ,
/*optional, int, time interval between two triggered/occurred alarms/events, unit: second*/
"EventTriggerNotificationList":[
{
  "EventTriggerNotification":{
/*optional*/
    "id":"","
/*required, string, linked alarm/event ID*/
    "notificationMethod":"","
/*required, string, linkage actions: "email", "IM", "IO", "syslog", "HTTP", "FTP" (upload to FTP), "beep" (audible
warning), "ptz", "record", "monitorAlarm" (display alarm on the monitor), "center" (upload to the center),
"LightAudioAlarm" (wireless siren)*/
    "notificationRecurrence":"","
/*optional, string, linkage time, opt="beginning, beginningandend, recurring"*/
    "notificationInterval": ,
/*optional, int, linkage time interval, which depends on alarm/event types, unit: milliseconds*/
    "outputIOPortID": ,
/*optional, int, alarm output No., which depends on alarm/event types*/
    "dynOutputIOPortID": ,
/*optional, int, dynamic alarm output No., which depends on alarm/event types*/
    "videoInputID": ,
/*optional, int, video input channel No., which depends on alarm/event types*/
    "dynVideoInputID": ,
/*optional, int, dynamic video input channel No., which depends on alarm/event types*/
    "ptzAction":{
/*optional, PTZ linkage, it is valid when eventType="ptz"*/
    "ptzChannelID":"","
/*required, string, PTZ channel No.*/
    "actionName":"","
/*required, string, PTZ control types, opt="preset, pattern, patrol"*/
    "actionNum":
/*optional, int, number of linkage actions, which depends on ptzAction*/
    }
  }
},.....]
}
```


Remarks

- When the value of **notificationMethod** is "IO", the fields **outputIOPortID** and **dynOutputIOPortID** are required.
- When the value of **notificationMethod** is "record", one of **videoInputID** and **dynVideoInputID** is required.
- When the value of **notificationMethod** is "ptz", the field **ptzAction** is required.

A.74 JSON_ErrorList

JSON message about error list

```
{
  "requestURL": "",
  /*optional, request URL*/
  "ErrorList":[{
    "id": "",
    /*required, string type, error object ID*/
    "statusCode": ,
    /*status code, it will not be returned when succeeded or no exception*/
    "statusString": "",
    /*status description, it will not be returned when succeeded or no exception*/
    "subStatusCode": "",
    /*sub status code, it will not be returned when succeeded or no exception*/
    "errorCode": ,
    /*error code, it is optional when succeeded, and it corresponds to subStatusCode*/
    "errorMsg": "ok"
  /*optional, error information, which corresponds to subStatusCode*/
  }]
}
```

A.75 JSON_FaceInfo

JSON message about face information (supports uploading face picture in binary data)

Basic Message

```
{
  "FaceInfo": {
    "employeeNo": "",
    /*required, string, person ID*/
  }
}
```

Advanced Message

```
{
  "FaceInfo": {
```

```
"employeeNo": "",
/*required, string, person ID*/
"faceLibType": ""
/*required, string, list library type: "infraredFD"-infrared list library, "blackFD"-blocklist library (default), "staticFD"-
static list library*/
}
}
```

Face Information in Picture Binary Data

```
POST or PUT /ISAPI/Intelligent/FDLib/FaceDataRecord?format=json&devIndex=<uuid>
Accept: text/html, application/xhtml+xml,
Accept-Language: en-US
Content-Type: multipart/form-data; boundary=-----7e13971310878
User-Agent: Mozilla/5.0 (compatible; MSIE 9.0; Windows NT 6.1; WOW64; Trident/5.0)
Accept-Encoding: gzip, deflate
Host: 10.10.36.29:8080
Content-Length: 9907
Connection: Keep-Alive
Cache-Control: no-cache

-----7e13971310878
Content-Disposition: form-data; name="FaceDataRecord";
Content-Type: application/json
Content-Length: 9907

{
  "FaceInfo": {
    "employeeNo": "",
/*required, string, person ID*/
    "faceLibType": ""
/*required, string, list library type: "infraredFD"-infrared list library, "blackFD"-blocklist library, "staticFD"-static list
library*/
  }
}
-----7e13971310878
Content-Disposition: form-data; name="FacelImage";
Content-Type: image/jpeg
Content-Length: 9907

.....JFIF.....`.....C.....
..
..... $. ' ",#..(7),01444.'9=82<.342...C. ....
-----7e13971310878--
```

A.76 JSON_FaceInfoCount

JSON message about number of face data records

```
{
  "FaceInfoCount" : {
    "faceNumber":
  }
}
```

A.77 JSON_FaceInfoDelCond

JSON message about conditions of deleting face data records

Basic Message

```
{
  "FaceInfoDelCond": {
    "EmployeeNoList" : [{
/*required, person ID list*/
      "employeeNo": ""
/*required, string, employee ID (person ID)*/
    }]
  }
}
```

Advanced Message

```
{
  "FaceInfoDelCond": {
    "faceLibType": "",
/*optional, string, list library type: "infraredFD"-infrared list library, "blackFD"-blocklist library (default), "staticFD"-static list library*/
    "EmployeeNoList" : [{
/*required, person ID list*/
      "employeeNo": ""
/*required, string, employee ID (person ID)*/
    }]
  }
}
```

A.78 JSON_FaceInfoSearch

JSON message about results of searching for face data records

```
{
  "responseStatusStrg": "",
/*required, string, search status description: "OK"-search ended, "MORE"-searching, "NO MATCH"-no matched data found*/
  "numOfMatches": ,
/*required, integer32, number of records returned for one time*/
  "totalMatches": ,
}
```

```
/*required, integer32, number of matched records returned for one time*/
"MatchList": [{
/*face information*/
"faceURL": "/HikGatewayStorage/pic?589629E5825F7CA45E093A11A6A50C37",
/*required, string, face picture URL (or picture storage URL), the maximum length is 256 bytes*/
"FPID": ""
/*required, string, person ID*/
}]
}
```

A.79 JSON_FaceInfoSearchCond

JSON message about conditions of searching for face data records

```
{
"searchResultPosition": ,
/*required, integer32, the end position of search result in result list; in a single search, if you cannot get all the records
in the result list, you can mark the end position and get the following records after the marked position in the next
search*/
"maxResults": ,
/*required, integer32, the maximum number of search results, which is defined by the device capability, will be
returned if the value of maxResults reaches the limit; in this case, the device will not return error*/
"FPID": "",
/*optional, string, employee ID (person ID); If this field does not exist or is not configured, it indicates to search all face
data records*/
"faceLibType": ""
/*optional, string, face picture library type: "infraredFD" (infrared list library), "blackFD" (blocklist library, default
value), "staticFD" (static list library)*/
}
```

A.80 JSON_FaceRecognizeMode

JSON message about face recognition modes

```
{
"FaceRecognizeMode": {
"mode": "",
/*required, string, face recognition mode: "normalMode"-normal mode, "deepMode"-deep mode*/
}
}
```

A.81 JSON_FingerPrintCfg

JSON message about fingerprint parameters

A.82 JSON_FingerPrintCond

JSON message about conditions of searching for fingerprint information

```
{
  "FingerPrintCond": {
    "searchID": "",
    /*required, string, search ID, which is used to check whether the current search requester is the same as the previous
    one; if they are the same, the search record will be stored in the device to speed up the next search*/
    "employeeNo": "",
    /*required, string, employee ID (person ID) linked with the fingerprint*/
    "cardReaderNo": ,
    /*optional, int, fingerprint and card reader No.*/
    "fingerPrintID": ,
    /*optional, int, fingerprint No., which from 1 to 10*/
  }
}
```

Remarks

When searching for the information of a fingerprint, all nodes should be configured; when searching for the information of all fingerprints linked with the specific employee ID (person ID), only the nodes **searchID** and **employeeNo** are required.

A.83 JSON_FingerPrintDelete

JSON message about conditions of deleting fingerprint information

Basic Message

```
{
  "FingerPrintDelete" : {
    "EmployeeNoDetail": {
      "employeeNo": "",
      /*optional, string, employee ID (person ID) linked with the fingerprint*/
      "fingerPrintID": [ , , ]
      /*optional, array, finger number list of fingerprints to be deleted, e.g., [1,3,5] indicates that the fingerprints of finger
      No.1, finger No.3, and finger No.5 will be deleted*/
    }
  }
}
```

Advanced Message

```
{
  "FingerPrintDelete" : {
    "mode": "",
    /*optional, string, deleting mode: "byEmployeeNo"-delete by employee ID (person ID) (default); "byCardReader"-
    delete by fingerprint and card reader*/
  }
}
```

```
"EmployeeNoDetail": {
/*optional, parameters of deleting by employee ID (person ID), this node is valid only when the value of node "mode"
is "byEmployeeNo"*/
  "employeeNo": "",
/*optional, string, employee ID (person ID) linked with the fingerprint*/
  "enableCardReader": [ , , ],
/*optional, array, number list of fingerprint and card reader with fingerprints to be deleted, e.g., [1,3,5] indicates that
the fingerprints of reader No.1, reader No.3, and reader No.5 will be deleted*/
  "fingerPrintID": [ , , ],
/*optional, array, finger number list of fingerprints to be deleted, e.g., [1,3,5] indicates that the fingerprints of finger
No.1, finger No.3, and finger No.5 will be deleted*/
  },
  "CardReaderDetail": {
/*optional, parameters of deleting by fingerprint and card reader, this node is valid only when the value of node
"mode" is "byCardReader"*/
    "cardReaderNo": ,
/*optional, int, fingerprint and card reader No.*/
    "clearAllCard": ,
/*optional, boolean, whether to delete the fingerprint information of all cards: false-no (delete by employee ID), true-
yes (delete fingerprint information of all employee IDs*/
    "employeeNo": "",
/*optional, string, employee ID (person ID) linked with the fingerprint, this node is valid when the value of
"clearAllCard" is false*/
  }
}
}
```

A.84 JSON_FingerPrintDeleteProcess

JSON message about fingerprint deleting progress

```
{
  "FingerPrintDeleteProcess" : {
    "status": ""
/*required, string, status: "processing", "success", "failed"*/
  }
}
```

A.85 JSON_FingerPrintInfo

JSON message about results of searching for fingerprint information

```
{
  "FingerPrintInfo": {
    "searchID": "",
/*required, string, search ID, which is used to check whether the current search requester is the same as the previous
one; if they are the same, the search record will be stored in the device to speed up the next search*/
    "status": "",
/*required, string, status: "OK"-the fingerprint already exists, "NoFP"-the fingerprint does not exists*/
  }
}
```

```
"FingerPrintList": [{
  "cardReaderNo": ,
  /*required, int, fingerprint and card reader No.*/
  "fingerPrintID": ,
  /*required, int, fingerprint No., which is from 1 to 10*/
  "fingerType": "",
  /*required, string, fingerprint type: "normalFP"-normal fingerprint, "hijackFP"-duress fingerprint, "patrolFP"- patrol
  fingerprint, "superFP"-super fingerprint, "dismissingFP"-dismiss fingerprint*/
  "fingerData": "",
  /*required, string, fingerprint data, which should be encoded by Base64*/
  "leaderFP": [ , ...]
  /*optional, array, No. list of doors that can be opened by first fingerprint, e.g., [1,3,5] indicates that Door No.1, Door
  No.3, and Door No.5 can be opened by first fingerprint*/
}]
}
```

A.86 JSON_FingerPrintModify

JSON message about fingerprint parameters to be edited

```
{
  "FingerPrintModify": {
    "employeeNo": "",
    /*required, string, employee ID (person ID) linked with the fingerprint*/
    "cardReaderNo": ,
    /*required, int, fingerprint and card reader No.*/
    "fingerPrintID": ,
    /*required, int, fingerprint No., whose value is between 1 and 10*/
    "fingerType": "",
    /*optional, string, fingerprint type: "normalFP"-normal fingerprint, "hijackFP"-duress fingerprint, "patrolFP"- patrol
    fingerprint, "superFP"-super fingerprint, "dismissingFP"-dismiss fingerprint*/
    "leaderFP": [ , , ],
    /*optional, array, No. list of doors that can be opened by first fingerprint, e.g., [1,3,5] indicates that Door No.1, Door
    No.3, and Door No.5 can be opened by first fingerprint*/
  }
}
```

A.87 JSON_FingerPrintStatus

JSON message about fingerprint parameters applying progress

```
{
  "FingerPrintStatus": {
    "StatusList" : [{
      /*optional, int, fingerprint and card reader No.*/
      "cardReaderRecvStatus": ,
      /*optional, int, fingerprint and card reader status: 0-failed, 1-succeeded, 2-the fingerprint module does not exist, 3-try
      again or low fingerprint quality, 4-memory full, 5-the fingerprint already exists, 6-the fingerprint ID already exists, 7-
```



```
invalid fingerprint ID, 8-configuring this fingerprint module is not required, 10-the reader version is too low to support
employee ID*/
    "errorMsg": "",
    /*optional, string, error information of applying, when the value of node "cardReaderRecvStatus" is 5, it indicates that
    employee ID (person ID) linked with the fingerprint already exists*/
    },
    "totalStatus": ,
    /*required, int, applying status: 0-applying, 1-applying completed*/
  }
}
```

A.88 JSON_FocusData

FocusData message in JSON format

```
{
  "FocusData":{
    "focus":
    /*required, int, focus data (vector), if the value is negative, it indicates focus in; if the value is positive, it indicates
    focus out, the value is percentage of fast focusing speed, range: from -100 to 100*/
  }
}
```

A.89 JSON_FPID

JSON message about face picture data.

Basic Message

```
{
  "FPID": "xxx"
  /*required, string, person ID*/
}
```

Advanced Message

```
{
  "FPID": "",
  /*required, string, person ID*/
  "faceLibType": ""
  /*optional, string, face picture library type: "infraredFD" (library of face pictures captured in infrared light), "blackFD"
  (list library), "staticFD" (static library). The default value is "blackFD"*/
}
```

A.90 JSON_Edit_DeviceInfo

JSON message about device information

```
{
  "DeviceInfo": {
    "devIndex": "2cd6716d-767f-4756-ac55-50276a5e3b4a",
    /*string, device UUID, it cannot be edited*/
    "protocolType": "ehome",
    /*required, string, protocol type, which should be same with that of the added device: "ehome" (ISUP), "ehomeV5" (ISUP 5.0), "ISAPI" (Intelligent Security API)*/
    "EhomeParams": {
      /*it is valid only when protocolType are "ehome" or "ehomeV5"*/
      "EhomeID": "111",
      /*required, ISUP ID, string, the maximum ID length is 31 bytes*/
      "EhomeKey": "",
      /*dependent, string, it is valid only when protocolType is "ehomeV5", and the maximum key length is 32 bytes*/
    },
    "ISAPIParams": {
      /*it is valid when the value of protocolType is "ISAPI"*/
      "addressingFormatType": "IPv4Address",
      /*optional, string, address type: "IPv4Address" (IPv4 address)*/
      "address": "0.0.0.0",
      /*optional, string, address; The address format depends on the value of addressingFormatType*/
      "portNo": 8000,
      /*optional, int, device port number; value range: [1,65535]*/
      "userName": "admin",
      /*required, string, user name; up to 32 characters are allowed*/
      "password": "hik123"
    },
    /*required, string, password*/
    "devName": "DeepinMind"
  },
  /*required, string, device name, the maximum length is 32 bytes*/
}
```

A.91 JSON_HttpHostNotification

JSON message about parameters of an alarm listening server

```
{
  "HttpHostNotification": {
    "enable": ,
    /*required, boolean, whether to enable alarm listening: true, false*/
    "url": "",
    /*required, string, URI for uploading event to platform, the maximum length is 128 bytes, e.g., /ISAPI/Event/notification/uploadEvent?format=json*/
    "ipAddress": "",
    /*required, string, alarm listening address, the maximum length is 15 bytes, e.g., 127.0.0.1*/
  }
}
```

```
"portNo":  
/*required, int, alarm listening port No.*/  
}  
}
```

A.92 JSON_HttpHostNotificationList

JSON message about parameters of all alarm listening servers

```
{  
  "HttpHostNotificationList": [{  
    "HttpHostNotification": {  
      "id": "1",  
      /*required, string, listening server No.*/  
      "url": "",  
      /*required, string, URI for uploading event to platform, the maximum length is 128 bytes, e.g., /ISAPI/Event/  
notification/uploadEvent?format=json*/  
      "protocolType": "HTTP",  
      /*required, string, protocol type: "HTTP" and "HTTPS"-->  
      "addressingFormatType": "",  
      /*required, string, address type: "ipaddress" (IP address) and "hostname" (domain name)*/  
      "hostName": "",  
      /*dependent, string, domain name*/  
      "ipAddress": "",  
      /*dependent, string, alarm listening address, the maximum length is 15 bytes, e.g., 127.0.0.1*/  
      "portNo": 80,  
      /*required, int, alarm listening port No.*/  
      "SubscribeEvent": {  
        "heartbeat": 30,  
        /*optional, int, heartbeat interval; unit: second; the default value is 30 seconds*/  
        "eventMode": "all",  
        /*optional, string, event subscription mode: "all" (upload all events/alarms), list" (upload specified minorEvent event/  
alarm)*/  
        "minorEvent": "1,38,75,153,181"  
      },  
      /*dependent, string, event minor type*/  
    }  
  }  
}]  
}
```

A.93 JSON_IOPortData

JSON message about alarm input and output information

```
{  
  "IOPortData": {  
    "outputState": ""  
  },  
  /*required, string, output status, opt="high,low" */  
}
```

```
}  
}
```

A.94 JSON_IOPortList

JSON message about alarm input and output list

```
{  
  "IOPortList":{  
    "IOInputPortList":[{  
      "IOInputPort":{  
        "id": ,  
        /*required, int, alarm input No.*/  
        "enable": ,  
        /*required, boolean, whether to enable alarm input*/  
        "triggering": "",  
        /*optional, string, triggering level: "high", "low"*/  
        "name": ""  
      }  
    }  
  ],  
  "IOOutputPortList":[{  
    "IOOutputPort":{  
      "id": ,  
      /*required, int, alarm output No.*/  
      "enable": ,  
      /*required, boolean, whether to enable alarm output*/  
      "name": ""  
    }  
  }  
}]  
}
```

A.95 JSON_IPAddress

JSON message about IP address information

```
{  
  "IPAddress":{  
    "ipVersion": "",  
    /*required, IP address version, string, "v4, v6, dual"*/  
    "addressingType": "",  
    /*required, string, IP address type, "static, dynamic, apipa"*/  
    "ipAddress": "",  
    /*required, string, IPv4 address, e.g., 255.255.255.0*/  
    "subnetMask": "",  
    /*required, subnet mask of IPv4, e.g., 255.255.255.0*/  
    "ipv6Address": "",  
  }  
}
```

```
/*required, IPv6 address, e.g., 2001:DB8:2de::1*/
  "bitMask": ,
/*required, string, prefix size of IPv6 subnet mask, e.g., 48*/
  "DefaultGateway":{
/*default gateway*/
    "ipAddress": "",
    "ipv6Address": ""
  },
  "PrimaryDNS":{
/*major DNS, which depends on IP address type*/
    "ipAddress": "",
    "ipv6Address": ""
  },
  "SecondaryDNS":{
/*minor DNS, which depends on IP address type*/
    "ipAddress": "",
    "ipv6Address": ""
  },
  "Ipv6Mode":{
/*optional, IPv6 mode*/
    "ipV6AddressingType": "",
/*optional, IPv6 address type, which depends on ipVersion, "ra, manual, dhcp", string, ra-advertised by router,
manual: configured manually, dhcp: allocated by server*/
    "ipv6AddressList":[
/*optional, IPv6 address configuration, multiple addresses are available*/
      {
        "v6Address":{
          "id": "",
/*optional, ID of IPv6 address, string*/
          "type": "",
/*optional, IPv6 address type, "ra, manual, dhcp"*/
          "address": "",
/*optional, IPv6 address, string*/
          "bitMask": 60
/*optional, number of IPv6 mask bit, integer*/
        }
      }
    ],
  },
}
```

Note

- If **addressingType**="dynamic", the nodes below this node is not required, and the DHCP service should be enabled; if **addressingType**="static", the IP address should be configured manually, and the default gateway and DNS nodes are optional; if **addressingType**="APIPA", the IP address

will be automatically configured, and the default gateway and DNS nodes are optional, but the DHCP service is not required.

- The validity of nodes **ipAddress** and **ipv6Address** are determined by **ipVersion**. If **ipVersion="v4"**, only the node **ipAddress** is valid; if **ipVersion="v6"**, only the node **ipv6Address** is valid; if **ipVersion="dual"**, both the nodes **ipAddress** and **ipv6Address** are valid.
 - For device gateway, the node **ipv6Mode** is not available.
-

A.96 JSON_IrisData

IrisData message in JSON format

```
{
  "IrisData":{
    "iris":
      /*required, int, iris data, if the value is negative, it indicates the iris is closed; if the value is positive, it indicates the iris
      is open, the value is the percentage of fast iris adjusting speed, range: from -100 to 100*/
    }
  }
```

A.97 JSON_MaskInfo

JSON message about added face picture information

```
{
  "MaskInfo": {
    "PID": "1"
  }
  /*optional, string, picture ID*/
}
```

A.98 JSON_MediaAccessInfo

MediaAccessInfo message in JSON format

```
{
  "MediaAccessInfo":{
    "URL":"..."
  }
  /*the stream URL is returned by the Device Gateway according to the media stream service rule, the stream is
  obtained via the returned URL*/
}
```

A.99 JSON_MotionDetection

MotionDetection message in JSON format

```
{
  "MotionDetection":{
    "enabled": ,
    /*required, boolean, whether to enable motion detection: "true"-enable*/
    "enableHighlight": ,
    /*optional, boolean, whether to enable highlight: "true"-enable*/
    "samplingInterval": ,
    /*optional, int, sampling interval*/
    "startTriggerTime": ,
    /*optional, int, alarm start time; unit: millisecond*/
    "endTriggerTime": ,
    /*optional, int, alarm end time; unit: millisecond*/
    "regionType": "",
    /*required, string, region type: "grid", "roi", "none"*/
    "Grid":{
      /*optional, it is valid when regionType="grid"*/
      "rowGranularity": ,
      /*required, int, line space*/
      "columnGranularity":
      /*required, int, column space*/
    },
    "ROI":{
      /*optional, it is valid when regionType="roi"*/
      "normalizedScreenWidth": ,
      /*required, int, screen width*/
      "normalizedScreenHeight":
      /*required, int, screen height*/
    },
    "MotionDetectionLayout":{
      /*required, layout*/
      "sensitivityLevel": ,
      /*required, sensitivity, ranging from 0 to 100*/
      "layout":{
        "gridMap": "",
        /*optional, depends on regionType, hexadecimal string*/
        "roiMap":{
          /*optional, depends on regionType*/
        }
      }
    }
  }
}
```

A.100 JSON_MotionDetectionScheduleList

MotionDetectionScheduleList message in JSON format

```
{
  "MotionDetectionScheduleList": [{
    "Schedule": {
      "id": "VMD_video2",
      /*required, string, motion detection channel ID*/
      "eventType": "VMD",
      /*optional, string, event type*/
      "inputIOPortID": 1,
      /*optional, depends on eventType, int, alarm input No.*/
      "outputIOPortID": 1,
      /*optional, depends on eventType, int, alarm output No.*/
      "videoInputChannelID": 2,
      /*optional, depends on eventType, int, video input channel No.*/
      "TimeBlockList": [
        /*required, size= "8"*/
        {
          "TimeBlock": {
            "dayOfWeek": 2,
            /*optional, int, days of the week based on ISO8601, "1"-Monday,... */
            "TimeRange": {
              /*required*/
              "beginTime": "",
              /*required, start time based on ISO8601*/
              "endTime": ""
            }
          }
        },.....
      ],
      "HolidayBlockList": [
        /*optional*/
        {
          "TimeBlock": {
            "TimeRange": {
              /*required*/
              "beginTime": "",
              /*required, start time based on ISO8601*/
              "endTime": ""
            }
          }
        },.....
      ]
    }
  ]
}
```


A.101 JSON_NetworkInterface

JSON message about network interface information

```
{
  "NetworkInterface":{
    "id": "",
    /*required, NIC ID*/
    "IPAddress">{...},
    /*required, IP address, see details in the message JSON_IPAddress*/
    "defaultConnection":true,
    /*optional, boolean, the default connection*/
    "macAddress": " c0:56:e3:a5:72:59"
    /*optional, string, MAC address*/
  }
}
```



Note

For NIC, only **id** and **IPAddress** fields are required.

See Also

[JSON_IPAddress](#)

A.102 JSON_NetworkInterfaceList

JSON message about network interface list

```
{
  "NetworkInterfaceList":[{
    "NetworkInterface":{...}
  }]
  /*see details in the message JSON_NetworkInterface*/
}
```

See Also

[JSON_NetworkInterface](#)

A.103 JSON_PictureData

JSON message about captured picture information

```
{
  "PictureData": {
    "url": "/HikGatewayStorage/pic?C02BA69A5C71FB0AFAE44CF6640E00A9"
```

```
}  
}
```

A.104 JSON_PictureUploadData

JSON message about face picture information to be added to face picture library

```
{  
  "PictureUploadData": {  
    "FDID": "1",  
    /*required, string, face picture library ID*/  
    "FaceAppendData": {  
      /*optional, object, additional information of face picture*/  
      "bornTime": "2020-02-02",  
      /*optional, string, date of birth*/  
      "name": "mmmm",  
      /*required, string, person name, length range : [1,64]*/  
      "sex": "male"  
      /*optional, enum, subType:string, gender: "male" (male), "female" (female)*/  
    }  
  }  
}
```

Example

Example of Importing Face Pictures to the Face Picture Library

```
POST /ISAPI/Intelligent/FDLib/pictureUpload?format=json&devIndex=<uuid>  
Accept: text/html, application/xhtml+xml,  
Accept-Language: en-US  
Content-Type: multipart/form-data; boundary=-----7e13971310878  
User-Agent: Mozilla/5.0 (compatible; MSIE 9.0; Windows NT 6.1; WOW64; Trident/5.0)  
Accept-Encoding: gzip, deflate  
Host: 10.10.36.29:8080  
Content-Length: 9907  
Connection: Keep-Alive  
Cache-Control: no-cache
```

```
-----7e13971310878  
Content-Disposition: form-data; name="importImage";  
Content-Type: application/json  
Content-Length: 9907
```

```
{  
  "PictureUploadData": {  
    "FDID": "1"  
  }  
}
```

```
-----7e13971310878  
Content-Disposition: form-data; name="FacelImage";  
Content-Type: image/jpeg  
Content-Length: 9907
```

```
.....JFIF.....`.....C.....  
..  
..... $. ' ",#..(7),01444.'9=82<.342...C. ....  
-----7e13971310878--
```

A.105 JSON_Position3D

Position3D message in JSON format

```
{  
  "Position3D":{  
    "StartPoint":{  
      "positionX": ,  
      /*required, int, X-coordinate of start point, range: from 0 to 255*/  
      "positionY":  
      /*required, int, Y-coordinate of start point, range: from 0 to 255*/  
    },  
    "EndPoint":{  
      "positionX": ,  
      /*required, int, X-coordinate of end point, range: from 0 to 255*/  
      "positionY":  
      /*required, int, Y-coordinate of end point, range: from 0 to 255*/  
    }  
  }  
}
```

A.106 JSON_PTZAux

PTZAux message in JSON format

```
{  
  "PTZAux":{  
    "id": ,  
    /*required, int, auxiliary function No., for different types, the function No. can be repeated*/  
    "type": "",  
    /*required, string, type: "LIGHT", "WIPER", "FAN", "HEATER"*/  
    "status": ""  
    /*required, string, type: "on", "off"*/  
  }  
}
```

A.107 JSON_PTZData

PTZData message in JSON format

```
{  
  "PTZData":{
```

```
"pan": ,
/*optional, int, panning speed, it is required when pan or tilt, range: from -100 to 100*/
"tilt": ,
/*optional, int, tilting speed, it is required when pan or tilt, range: from -100 to 100*/
"zoom":
/*optional, int, zooming speed, it is required when zoom in or out, range: from -100 to 100*/
}
}
```

A.108 JSON_PTZPreset

PTZPreset message in JSON format

```
{
  "PTZPresetList": [{
    "PTZPreset": {
      "enabled": ,
      /*optional, boolean, whether to set preset, "true"-enabled*/
      "id": ,
      /*required, int, preset No.*/
      "presetName": ""
      /*required, string, preset name*/
    }
  }]
}
```

A.109 JSON_RemoteControlDoor

JSON message about door remote control parameters

```
{
  "RemoteControlDoor": {
    "cmd": ""
    /*string, command: "open"-open door, "close"-close door (in control), "alwaysOpen"-remain door open (out of control), "alwaysClose"-remain door closed (out of control), "visitorCallLadder"-visitor calls elevator, "householdCallLadder"-resident calls elevator*/
  }
}
```

A.110 JSON_ResponseStatus

JSON message about response information and status

```
{
  "requestURL": "",
  /*optional, string, request URL*/
  "statusCode": ,
}
```

```
/*required, int, status code*/
"statusString": "",
/*required, string, status description*/
"subStatusCode": "",
/*required, string, sub status code*/
"errorCode": ,
/*optional, int, error code, which correspond to subStatusCode; this field is required when statusCode does not equal to 1*/
"errorMsg": ""
/*optional, string, error code; this field is required when statusCode does not equal to 1*/
"rebootRequired":
/*optional, int, whether the reboot is required: 1=yes (reboot to take effect), other values-no; if reboot is not required, this field may be not returned*/
}
```



Note

See [**Status Codes**](#) for details about the status codes and sub status codes.

A.111 JSON_ResponseStatus_AuthenticationFailed

JSON message about response information and status returned when authentication failed

```
{
  "requestURL": "",
  /*optional, string, URL, e.g., "/ISAPI/Streaming/channels/1"*/
  "statusCode": ,
  /*required, int, status code*/
  "statusString": "",
  /*required, string, status description*/
  "subStatusCode": "",
  /*required, string, sub status code*/
  "errorCode": ,
  /*dependent, error code, it is required when the value of statusCode is not 1, and it corresponds to subStatusCode*/
  "errorMsg": "",
  /*dependent, error description, it is required when the value of statusCode is not 1*/
  "lockStatus": "",
  /*optional, string, locking status: "unlock", "lock"*/
  "retryTimes": ,
  /*optional, int, remaining authentication attempts*/
  "resLockTime":
  /*optional, int, remaining locking time, unit: second*/
}
```

Example

Message Example Returned When Authentication Failed

```
{
  "requestURL": "/ISAPI/Streaming/channels/1",
  "statusCode": 4,
  "statusString": "Invalid Operation",
}
```

```
"subStatusCode": "badAuthorization",
"errorCode": 0x40000003,
"errorMsg": "authentication failed",
"lockStatus": "unlock",
"retryTimes": 5,
"resLockTime": 30
}
```

A.112 JSON_Schedule

Schedule message in JSON format

```
{
  "Schedule": {
    "id": "",
    /*required, string, alarm channel ID*/
    "eventType": "",
    /*optional, string, event type*/
    "inputIOPortID": "",
    /*optional, int, alarm input port No., which is valid only when eventType="IO"*/
    "outputIOPortID": "",
    /*optional, int, alarm output port No., which is valid only when eventType="IO"*/
    "videoInputChannelID": "",
    /*optional, int, depends on eventType, video input channel No.*/
    "TimeBlockList": [
    /*required, size="8"*/
    {
      "TimeBlock": {
        "dayOfWeek": "",
        /*optional, int, days of the week based on ISO8601, "1"-Monday...*/
        "TimeRange": {
        /*required*/
          "beginTime": "",
        /*required, start time based on ISO8601*/
          "endTime": ""
        /*required, end time based on ISO8601*/
        }
      },
      .....
    },
    "HolidayBlockList": [
    /*optional*/
    {
      "TimeBlock": {
        "TimeRange": {
        /*required*/
          "beginTime": "",
        /*required, start time based on ISO8601*/
          "endTime": ""
        /*required, end time based on ISO8601*/
        }
      }
    }
  ]
}
```

```
}  
,.....]  
}  
}
```

A.113 JSON_SearchDescription

JSON message about search conditions of channel list

```
{  
  "SearchDescription": {  
    "position": ,  
    /*required, int, index, which starts from 0*/  
    "maxResult": 100,  
    /*required, int, the maximum number of searched results in a single search*/  
    "Filter": {  
      "key": "",  
      /*optional, search by keywords (fuzzy search: device model/device name/account ID); the maximum length is 64 bytes*/  
      "devType": "encodingDev",  
      /*optional, string, device type: "encodingDev" (encoding device), "AccessControl" (access control device)*/  
      "protocolType": ["ehome"],  
      /*required, string, protocol type: "ehome"-ISUP (Intelligent Security Uplink Protocol), "ehomeV5"-ISUP 5.0, "ISAPI" (Intelligent Security API)*/  
      "devStatus": ["online", "offline", "sleep"],  
      /*optional, string, device status: "online", "offline", "sleep"*/  
      "ISAPIUserBound": [true, false],  
      /*optional, whether the channel has been linked to ISAPI user: true (yes), false (no)*/  
      "ISAPIPortBound": [true, false]  
      /*optional, whether the channel has been linked to ISAPI port: true (yes), false (no)*/  
      "EhomeParams": {  
        "EhomeID": "111"  
      },  
      /*optional, string, search by device number (match whole words); the maximum length is 31 bytes*/  
      "devIndex": "2cd6716d-767f-4756-ac55-50276a5e3b4a"  
    },  
    /*optional, string, search by device UUID (match whole words)*/  
    "devUUID": ""  
  }  
}
```

Note

When the value of **ISAPIUserBound** or **ISAPIPortBound** is [true]/[false]/[true,false], searching by device model is not supported (i.e., the value of **key** cannot be set to device model).

A.114 JSON_SearchResult

JSON message about device search results

```
{
  "SearchResult": {
    "numOfMatches": 10,
    /*optional, int, number of returned records*/
    "totalMatches": 100,
    /*optional, int, total number of matched records*/
    "MatchList": [{
      /*MatchList is returned when totalMatches is larger than 0*/
      "Device": {
        "devIndex": "2cd6716d-767f-4756-ac55-50276a5e3b4a",
        /*required, string, device index (uuid/guid)*/
        "devName": "",
        /*required, string, device name; up to 32 characters are allowed*/
        "devMode": "",
        /*optional, string, device model*/
        "devVersion": "",
        /*optional, string, device version*/
        "devSerial": "",
        /*optional, string, device serial number; the maximum length is 12 bytes*/
        "offlineHint": 0,
        /*optional, int, offline reason: 0 (Waiting for the device to connect to Device Gateway), 1 (Incorrect ISUP 5.0 key), 2
        (ISUP version mismatched), 3 (Network exception)*/
        "devType": "encodingDev",
        /*required, string, device type: "encodingDev" (encoding device), "AccessControl" (access control device)*/
        "protocolType": "ehome",
        /*required, string, protocol type: "ehome" (ISUP), "ehomeV5" (ISUP 5.0), "ISAPI" (Intelligent Security API)*/
        "EhomeParams": {
          /*it is valid when the value of protocolType is "ehome" or "ehomeV5"*/
          "EhomeID": "111"
        },
        /*optional, string, ISUP (Intelligent Security Uplink Protocol) ID; Up to 31 characters are allowed*/
      },
      "ISAPIParams": {
        /*it is valid when the value of protocolType is "ISAPI"*/
        "addressingFormatType": "IPV4Address",
        /*optional, string, address type: "IPV4Address" (IPv4 address)*/
        "address": "0.0.0.0",
        /*optional, string, address; The address format depends on the value of addressingFormatType*/
        "portNo": 8000
        /*optional, int, device port number; value range: [1,65535]*/
      },
      "devStatus": "online",
      /*required, string, device status: "online", "offline", "sleep"*/
      "activeStatus": "true",
      /*dependent, boolean, activation status: "true" (active), "false" (inactive). It is valid when the value of devStatus is
      "online"*/
      "videoChannelNum": 20,
      /*optional, int, the number of video channels*/
      "ISAPIUserBound": true,
      /*optional, boolean, whether the channel has been linked to ISAPI user: true (yes), false (no)*/
      "ISAPIPortBound": false
      /*optional, boolean, whether the channel has been linked to ISAPI port: true (yes), false (no)*/
    }
  ]
}
```



```
,  
{...}  
}  
}
```

A.115 JSON_StreamInfo

JSON message about stream information request

```
{  
  "StreamInfo":{  
    "id": "",  
    /*required, streaming channel No., string*/  
    "streamType": "",  
    /*required, string, stream type: "main"-main stream, "sub"-sub-stream, "third"-third stream*/  
    "method": "",  
    /*required, string, streaming mode: "preview" (live view), "playback", "twoWayAudio" (two-way audio)*/  
    "PlayBackParams":{  
      /*dependent, it is valid only when the method is "playback"*/  
      "startTime": "",  
      /*required, start time, based on ISO8601 date/time, e.g., 2018-03-12T19:10:07*/  
      "endTime": ""  
    }  
  }  
}
```

A.116 JSON_StreamingChannel

StreamingChannel message in JSON format

```
{  
  "StreamingChannel": {  
    "id": "",  
    /*required, string, camera No.*/  
    "channelName": "",  
    /*required, string, camera name*/  
    "enabled": ,  
    /*required, boolean, whether to start getting stream: true (enable)*/  
    "Transport":{  
      /*required*/  
      "maxPacketSize": ,  
      /*optional, int, maximum packet size*/  
      "audioPacketLength": ,  
      /*optional, int, audio packet size*/  
      "audioInboundPacketLength": ,  
      /*optional, int, received audio packet size*/  
      "audioInboundPortNo": ,  
    }  
  }  
}
```

```
/*optional, int, audio receiving port number*/
    "videoSourcePortNo": ,
/*optional, int, port number of video source*/
    "audioSourcePortNo": ,
/*optional, int, port number of audio source*/
    "ControlProtocolList":[{
/*required*/
        "ControlProtocol": {
/*required*/
            "streamingTransport": ""
/*required, string , stream transfer protocol, e.g., HTTP*/
        }
    }],
    "Unicast":{
/*optional, unicast*/
        "enabled": ,
/*required, boolean, whether to enable unicast: true (enable)*/
        "interfaceID": "",
/*optional, string, network interface ID*/
        "rtpTransportType": ""
/*optional, string, RTP transfer type: "RTP/UDP, RTP/TCP"*/
    },
    "Multicast":{
/*optional, multicast*/
        "enabled": "",
/*required, boolean, whether to enable unicast: true (enable)*/
        "userTriggerThreshold": ,
/*optional. int, triggering threshold*/
        "destIPAddress": "",
/*optional, dependent, string, destination IPv4 address*/
        "videoDestPortNo": ,
/*optional, int, destination video port number*/
        "audioDestPortNo": ,
/*optional, int, destination audio port number*/
        "destIPv6Address": "",
/*optional, dependent, string, destination IPv6 address*/
        "ttl":
/*optional, int, saving time of parsed domain name in DNS server (time to live), it is a 8-bit field in the header of IPv4
address packet*/
    },
    "Security":{
/*optional, security*/
        "enabled": ,
/*required, boolean, whether to enable security authentication: true (enable)*/
        "certificateType": ""
/*required, string, certification type, opt="digest, digest/baisc", def="digest" */
    }
},
    "Video":{
/*optional*/
        "enabled": ,
/*required, boolean, whether to enable getting video stream: true (enable)*/
```

```
"videoInputChannelID":"","
/*required, string, video input No.*/
"videoCodecType":"","
/*required, string, video encoding type, opt="MPEG4,MJPEG,3GP,H.264,HK.264,MPNG,SVAC,H.265"*/
"videoScanType",
/*optional, string, video scan mode, opt="progressive,interlaced"*/
"videoResolutionWidth": ,
/*required, int, video resolution (width)*/
"videoResolutionHeight": ,
/*required, int, video resolution (height)*/
"videoResolutionName":"","
/*optional, string, video format, video resolution, opt="3MP,5MP,none"*/
"videoPositionX": ,
/*optional, int, X-coordinate*/
"videoPositionY": ,
/*optional, int, Y-coordinate*/
"videoQualityControlType":"","
/*optional, string, video quality control type, opt="CBR,VBR"*/
"constantBitRate": ,
/*optional, dependent, int, constant bit rate*/
"fixedQuality": ,
/*optional, int, image quality, which is represented by percentage, range: from 0 to 100*/
"vbrUpperCap": ,
/*optional, dependent, int, maximum streaming rate, unit: Kbps*/
"vbrLowerCap": ,
/*optional, dependent, int, minimum streaming rate, unit: Kbps*/
"maxFrameRate": ,
/*required, int, maximum frame rate*/
"keyFrameInterval": ,
/*optional, int, I frame interval, unit: milliseconds*/
"rotationDegree": ,
/*optional, int, rotation degree, range: from 0 to 360*/
"mirrorEnabled": ,
/*optional, boolean, whether to enable mapping: true (enable)*/
"snapshotImageType": "",
/*optional, string, captured picture format, opt="JPEG,GIF,PNG"*/
"Mpeg4Profile": "",
/*optional, dependent, string, complexity of MPEG4 encoding, opt="SP,ASP"*/
"H264Profile": "",
/*optional, dependent, string, complexity of H264 encoding, opt="Baseline,Main,High,Extended"*/
"SVACProfile":"","
/*optional, dependent, string, complexity of SVAC encoding, opt="Baseline,Main,High,Extended"*/
"GovLength": ,
/*optional, int, actual I frame interval*/
"SVC":{
/*scalable video encoding function*/
"enabled": ,
/*required, boolean, whether to enable scalable video encoding function: true (enable)*/
"SVCMode": ""
/*optional, dependent, string, SVC mode, opt="manual,auto"*/
},
"PacketType": "",
```

```
/*optional, string, packet type*/
    "smoothing": ,
/*optional, int, stream smoothness, range: from 1 to 100, 1-SD, 100-smooth*/
    "H265Profile": "",
/*optional, string, complexity of H265 encoding*/
    "SmartCodec":{
/*optional, dependent, Smart264 encoding*/
    "enabled": "true"
/*required, boolean, whether to enable Smart264 encoding*/
    },
    "vbrAverageCap": 1
/*optional, dependent, int, video average bit rate, it is valid when Smart264 encoding is enabled, unit: Kbps*/
    },
    "Audio":{
/*optional*/
    "enabled": ,
/*required, boolean, whether to enable getting audio stream: true (enable)*/
    "audioInputChannelID": "",
/*required, string, audio input No.*/
    "audioCompressionType": "",
/*required, string, audio encoding type, opt="G.711alaw,G.711ulaw,G.726,G.729,G.729a,G.729b,PCM,MP3,AC3,AAC,ADPCM, MP2L2"*/
    "audioInboundCompressionType": "G.711alaw",
/*optional, string, received audio encoding type, opt="G.711alaw,G.711ulaw,G.726,G.729,G.729a,G.729b,PCM,MP3,AC3,AAC,ADPCM,MP2L2"*/
    "audioBitRate": ,
/*optional, int, audio bit rate, unit: Kbps*/
    "audioSamplingRate": ,
/*optional, float, audio sampling rate, unit: KHz*/
    "audioResolution": ,
/*optional, int, audio resolution, unit: bits*/
    "VoiceChanger":{
/*optional*/
    "enabled": true,
/*required, boolean, whether to enable adjusting volume: true (enable)*/
    "level":
/*required, int, level range: from -12 to 12*/
    }
    },
    "enableCABAC": ,
/*optional, boolean, whether to enable improving stream encoding performance: true (enable)*/
    "subStreamRecStatus": ,
/*optional, boolean, whether to support return sub-stream status: true (support)*/
    "customStreamEnable": true
/*optional, boolean, whether to enable stream custom: true (enable)*/
    }
}
```

A.117 JSON_SubscribeDevEvent

JSON message about event or alarm types to be subscribed

```
{
  "SubscribeDevEvent":{
    "devIndex": "2cd6716d-767f-4756-ac55-50276a5e3b4a",
    /*required, string, device ID*/
    "uploadMode": "all"
    /*required, string, device arming mode: "all"-subscribe to all events of device*/
  }
}
```

A.118 JSON_SubscribeDeviceMgmt

JSON message about subscription conditions

```
{
  "SubscribeDeviceMgmt":{
    "eventMode": "device", /*required, string, arming mode: "device"-arm specified devices, "all"-arm all added
devices*/
    /*"DevEventList" depends on "eventMode" as "device"*/
    "DevEventList":[{
      "Dev":{
        "devIndex": "2cd6716d-767f-4756-ac55-50276a5e3b4a",/*required, string, device ID*/
        "uploadMode": "all",/*required, string type, device arming mode: "all"-subscribe to all events of specified
devices*/
      }
    }],
    /*"defenceMode" depends on "eventMode" being "all"*/
    "defenceMode":"all",/*required, string type, arming mode: "all"-subscribe to all events*/
  }
}
```

A.119 JSON_SubscribeDeviceMgmtRsp

JSON message about subscription results

```
{
  "SubscribeDeviceMgmtRsp":{
    "id": "",
    /*optional, integer, subscription ID*/
  }
}
```

A.120 JSON_SubscribeQueryStatusList

JSON message about subscription status

```
{
  "SubscribeQueryStatusList": [{
    "Dev": {
      "devIndex": "",
      /*required, string, device ID (uuid or guid)*/
      "status": ""
      /*required, arming status: "continue"-connected, "abnormalLink"-connection exception, "fail"-subscribing failed*/
    }
  }]
}
```

A.121 JSON_TamperDetection

JSON message about video tampering detection parameters

```
{
  "TamperDetection": {
    "enabled": ,
    /*required, boolean, whether to enable tampering alarm configuration, "true"-enabled*/
    "normalizedScreenSize": {
      /*required, screen size*/
      "normalizedScreenWidth": "",
      /*required, int, screen width*/
      "normalizedScreenHeight": ""
      /*required, int, screen height*/
    },
    "tamperSensitivityLevel": "",
    /*required, int, sensitivity, ranging from 0 to 100*/
    "TamperDetectionRegionList": [{
      "TamperDetectionRegion": {
        "id": "",
        /*required, int, tampering detection region ID*/
        "enabled": true,
        /*optional, boolean, whether to enable the region*/
        "sensitivityLevel": "",
        /*required, int, sensitivity, ranging from 0 to 100*/
        "RegionCoordinatesList": [{
          /*required, tampering detection region*/
          "RegionCoordinates": {
            "positionX": ,
            /*required, int, tampering region x-coordinate*/
            "positionY":
            /*required, int, tampering region y-coordinate*/
          }
        }]
      }
    }]
  }
}
```

```
}  
}  
}  
}
```

Remarks

If devices access the gateway via ISUP 2.0, when the value of **enabled** (whether to enable tampering alarm) is true, the value of **tampersensitivityLevel** should be larger than 0; when the value of **enabled** is false, the value of **tampersensitivityLevel** should be 0.

A.122 JSON_TamperDetectionScheduleList

JSON format about tampering detection schedule list

```
{  
  "TamperDetectionScheduleList": [{  
    "Schedule": {  
      "id": "Tamperdetection_video2",  
      /*required, channel ID*/  
      "eventType": "",  
      /*optional, string, event type*/  
      "inputIOPortID": "",  
      /*optional, int, depends on eventType, alarm input No.*/  
      "outputIOPortID": "",  
      /*optional, depends on eventType alarm output No.*/  
      "videoInputChannelID": "",  
      /*optional, depends on eventType, video input channel No.*/  
      "TimeBlockList": [  
        /*required, size="8"*/  
        {  
          "TimeBlock": {  
            "dayOfWeek": "",  
            /*optional, int, days of the week based on ISO8601, 1="Monday",...*/  
            "TimeRange": {  
              /*required*/  
              "beginTime": "",  
              /*required, start time based on ISO8601*/  
              "endTime": ""  
            }  
          }  
        },.....  
      ],  
      "HolidayBlockList": [  
        /*optional*/  
        {  
          "TimeBlock": {  
            "TimeRange": {  
              /*required */  
              "beginTime": "",  
              /*required, start time based on ISO8601*/  
              "endTime": ""  
            }  
          }  
        },.....  
      ]  
    }  
  ]  
}
```

```
        "beginTime": "",
/*required, start time based on ISO8601*/
        "endTime": ""
/*required, end time based on ISO8601*/
    }
}
},.....
]
}
}}
}
```

A.123 JSON_TextOverlay

TextOverlay message in JSON format

```
{
  "TextOverlay": {
    "id": "",
/*required, string, text No.*/
    "enabled": ,
/*required, boolean, whether to enable text overlay*/
    "displayText": "",
/*required, string, text details*/
    "positionX": ,
/*required, float, X-coordinate of display position*/
    "positionY":
/*required, float, Y-coordinate of display position*/
  }
}
```

A.124 JSON_TextOverlayList

TextOverlayList message in JSON format

```
{
  "TextOverlayList": [{
/*this node is optional when configuring OSD parameters; when switching to GB mode, this node will be automatically
obtained, a backup will be saved by device*/
    "TextOverlay": {
      "id": "",
/*required, string, text No.*/
      "enabled": ,
/*required, boolean, whether to enable text overlay*/
      "displayText": "",
/*required, string, text details*/
      "positionX": ,
/*required, float, X-coordinate of display position*/
      "positionY":
```



```
/*required, float, Y-coordinate of display position*/
}
},...]
}
```

A.125 JSON_Time

JSON message about time synchronization parameters

```
{
  "Time":{
    "timeMode":"manual",
    /*required, string, time synchronization mode: "manual", "NTP"*/
    "localTime":""
  }
  /*dependent, string, local time, e.g., 2019-02-28T10:50:44+08:00*/
}
```

A.126 JSON_Track

JSON message about recording schedule

```
{
  "Track": {
    "id": 201,
    /*required, int, stream channel number; e.g., 201 indicates main stream of channel 2, 202 indicates sub stream of channel 2, 203 indicates picture capture parameters of channel 2*/
    "Channel": 201,
    /*required, int, stream channel number*/
    "Enable": true,
    /*required, boolean, whether to enable recording, "false"-disable, "true"-enable*/
    "Description": "trackType=standard,contentype=video,codecType=H.264-BP,resolution=0x0,framerate=4294967296.0 fps,bitrate=2048 kbps",
    /*required, string, stream description, the device set the parameters according to the description*/
    "TrackGUID": "{00000000-0000-0000-0000-000000000000}",
    /*required, string, video GUID*/
    "Size": 12,
    /*optional, int, video size*/
    "Duration": "PODToH",
    /*optional,string, video duration*/
    "DefaultRecordingMode": "CMR",
    /*required, default recording type, e.g., CMR, EDR, NONE*/
    "LoopEnable": false,
    /*optional, boolean, whether to enable continuous recording*/
    "SrcDescriptor": {
      /*optional, recording source (camera) description*/
      "SrcGUID": "{00000000-0000-0000-0000-000000000000}",
      /*required, camera GUID*/
      "SrcChannel": 1,

```

```
/*required, recording channel*/
  "StreamHint": "video, mp4, 640x480, 20 fps, 6000 kbps",
/*required, stream description*/
  "SrcDriver": "RTSP",
/*required, string, playback protocol type*/
  "SrcType": "H.265",
/*required, encoding type*/
  "SrcUrl": "rtsp://localhost/PSIA/Streaming/channels/101",
/*required, video URL*/
  "SrcUrlMethods": "DESCRIBE, SETUP, PLAY, TEARDOWN",
/*required, method for operating URL*/
  "SrcLogin": "admin"
/*required, string, user name*/
},
  "TrackSchedule": {
    "ExternalScheduleBlockReferences": [{
/*optional*/
      "ScheduleBlockReference": {
        "ScheduleBlockGUID": "{F018AD02-BC04-4520-8BB8-123409AC5678}"
      }
    }],
    "ScheduleBlockList": [{
/*optional, recording schedule list*/
      "ScheduleBlock": {
        "ScheduleBlockGUID": "{00000000-0000-0000-0000-000000000000}",
/*required, string, recording schedule GUID*/
        "ScheduleBlockType": "www.hikvision.com/racm/schedule/ver10",
/*required, string, recording schedule type*/
        "ScheduleAction": [{
          "id": 1,
/*required, int, schedule No.*/
          "ScheduleActionStartTime": {
/*start time*/
            "DayOfWeek": "Monday",
/*required, string, date of the week*/
            "TimeOfDay": "00:00:00"
/*required, string, time*/
          },
          "ScheduleActionEndTime": {
/*end time*/
            "DayOfWeek": "Tuesday",
/*required, string, date of the week*/
            "TimeOfDay": "00:00:00"
/*required, string, time*/
          },
          "ScheduleDSTEnable": ,
/*required, boolean, whether to enable daylight saving time, "false"-disable, "true"-enable*/
          "Description": "nothing",
/*required, string, description*/
          "Actions": {
/*optional, operations*/
            "Record": true,
```

```
/*required, boolean, "true"-enable, "false"-disable*/
  "Log": false,
/*required, boolean, "true"-enable, "false"-disable*/
  "SaveImg": false,
/*required, boolean, "true"-enable, "false"-disable*/
  "ActionRecordingMode": ""
/*required, string, recording type, e.g., "CMR"-continuous recording, "EDR"-recording based on I/O alarm or motion
detection, "ALARM"-recording based on I/O alarm, "MOTION"-recording based on motion detection,
"ALARMANDMOTION"-recording based on alarm and motion detection, "COMMAND"-recording triggered command,
"SMART"-recording based on VCA alarm*/
  }
  }}
}
}}
},
  "CustomExtensionList": [{
/*optional*/
    "CustomExtension": {
      "CustomExtensionName": "",
/*required, string*/
      "enableSchedule": false,
/*required, boolean, whether to enable recording schedule, "true"-enable, "false"-disable*/
      "SaveAudio": false,
/*optional, boolean, whether to record the audio, "false"-disable, "true"-enable*/
      "RedundancyRecord": false,
/*optional, boolean, whether to enable redundant recording, "false"-disable, "true"-enable*/
      "PreRecordTimeSeconds": 5,
/*optional, int, pre-record time, unit: second*/
      "PostRecordTimeSeconds": 5,
/*optional, int, post-record time, unit: second*/
      "HolidaySchedule": {
/*optional, holiday schedule*/
        "ScheduleBlock": {
          "ScheduleBlockGUID": "{00000000-0000-0000-0000-000000000000}",
/*required, holiday schedule GUID*/
          "ScheduleBlockType": ""
/*required, holiday schedule type*/
        }
      }
    }
  ]
}
```

A.127 JSON_TrackList

TrackList message in JSON format

```
{
  "TrackList":[{
```

```
"Track":{
  ...
  /*see the message JSON_Track for details*/
}
}}
}
```

See Also

[JSON_Track](#)

A.128 JSON_TwoWayAudioChannel

JSON message about parameters of a two-way audio channel

```
{
  "TwoWayAudioChannel"{
    "id": ,
    /*required, two-way audio channel No.*/
    "enabled": ,
    /*required, whether to enable two-way audio, boolean type, "true"-enabled*/
    "audioCompressionType": "",
    /*required, audio encoding type: "G.711alaw", "G.711ulaw", "G.726", "G.729", "G.729a", "G.729b", "PCM", "MP3",
    "AC3", "AAC", "ADPCM", "MP2L2"*/
    "audioInboundCompressionType": "",
    /*optional, int, "G.711alaw,G.711ulaw,G.726,G.729,G.729a,G.729b,PCM,MP3,AC3,AAC,ADPCM"*/
    "speakerVolume": ,
    /*optional, int, volume*/
    "microphoneVolume": ,
    /*optional, int, microphone volume*/
    "noisereduce": ,
    /*optional, boolean, whether to enable noise reduction*/
    "audioBitRate": ,
    /*optional, int, bit rate: "8kbps, 16kbps, 32kbps, 64kbps, 128kbps, 192kbps, 40kbps, 48kbps, 56kbps, 80kbps, 96kbps,
    112kbps, 144kbps, 160kbps"*/
    "audioInputType": ""
    /*optional, int, audio input type: "MicIn, LineIn"*/
    "associateVideoInputs":{
    /*optional, related video channel information*/
      "enabled":true,
    /*required, boolean, whether to enable the related channel*/
      "videoInputChannelList":{
    /*required, list of related video channels*/
        "videoInputChannelID":["1",...]
    /*optional, string, related video channel ID*/
      }
    },
    "audioSamplingRate":22
    /*optional, float, audio sampling rate; unit: kHz*/
  }
```

```
}  
}
```

A.129 JSON_UpgradeParams

JSON message about device upgrade parameters

```
{  
  "UpgradeParams":{  
    "FTPServerIP":"","  
    /*required, string, IP address of FTP (File Transfer Protocol) server*/  
    "FTPServerPort": ,  
    /*required, int, port No. of FTP server*/  
    "userName":"","  
    /*required, string, user name of FTP server*/  
    "password":"","  
    /*required, string, password of FTP server*/  
    "file":"","  
    /*required, string, upgrade file name*/  
  }  
}
```

A.130 JSON_UserInfo

JSON message about person information

Basic Message

```
{  
  "UserInfo" : [{  
    "employeeNo": "",  
    /*required, string, employee ID (person ID); the length depends on the device capability*/  
    "name": "",  
    /*optional, string, person name; the length depends on the device capability*/  
    "Valid" : {  
      /*required, effective period, the earliest date and time can be "1970-01-01T00:00:00", and the latest date and time  
      can be "2037-12-31T23:59:59"*/  
      "beginTime": "",  
      /*required, start time of the effective period, e.g., for local time, it is "2017-08-01T17:30:08"; for UTC time, it is  
      "2017-08-01T17:30:08+08:00"*/  
      "endTime": "",  
      /*required, start time of the effective period, e.g., for local time, it is "2017-08-01T17:30:08"; for UTC time, it is  
      "2017-08-01T17:30:08+08:00"*/  
    }  
  }  
}]  
}
```

Advanced Message

```
{
  "UserInfo" : [{
    "employeeNo": "",
    /*required, string, employee ID (person ID); the length depends on the device capability*/
    "name": "",
    /*optional, string, person name; the length depends on the device capability*/
    "userType": "",
    /*optional, string, person type: "normal"-resident (default), "visitor"-visitor, "blackList"-person in blacklist*/
    "Valid": {
    /*required, effective period, the earliest date and time can be "1970-01-01T00:00:00", and the latest date and time
    can be "2037-12-31T23:59:59"*/
      "enable": ,
      /*optional, boolean, whether to enable effective period: false (always be effective), true (default)*/
      "beginTime": "",
      /*required, start time of the effective period, e.g., for local time, it is "2017-08-01T17:30:08"; for UTC time, it is
      "2017-08-01T17:30:08+08:00"*/
      "endTime": "",
      /*required, start time of the effective period, e.g., for local time, it is "2017-08-01T17:30:08"; for UTC time, it is
      "2017-08-01T17:30:08+08:00"*/
      "timeType": ""
    },
    /*optional, string, time type: "local"-device local time, "UTC"-UTC time*/
    "doorRight": "1,3",
    /*optional, string, the ID of door or lock, whose access permission has been applied to the person. When this node
    does not exist, the default door/lock ID is 1. Multiple IDs are separated by the comma*/
    "RightPlan" : [{
    /*optional, door permission schedule (lock permission schedule)*/
      "doorNo": 1,
      /*optional, int, door ID (lock ID). When this node does not exist, the default door/lock ID is 1*/
      "planTemplateNo": "1,3,5"
    /*optional, string, schedule template number. When this node does not exist, the default value is 1*/
    }
      "password": "123456",
    /*optional, string, password*/
    "localUIRight": true
    /*optional, boolean, whether the user has the permission for accessing device's local UI*/
  }
  ]
}
```

A.131 JSON_UserInfoCount

JSON message about person quantity

```
{
  "UserInfoCount" : {
    "userNumber":
```

```
}  
}
```

A.132 JSON_UserInfoDetail

JSON message about person information

```
{  
  "UserInfoDetail" : {  
    "mode": "",  
    /*required, string, deleting mode: "all"-delete all, "byEmployeeNo"-delete by employee ID (person ID)*/  
    "EmployeeNoList" : [{  
      /*optional, person ID list*/  
      "employeeNo": ""  
      /*optional, string, employee ID (person ID); this node is valid only when the value of "mode" is "byEmployeeNo"*/  
    }]  
  }  
}
```

A.133 JSON_UserInfoDetailDeleteProcess

JSON message about progress of person search or deleting

```
{  
  "UserInfoDetailDeleteProcess" : {  
    "status": ""  
    /*required, string, status: "processing", "success", "failed"*/  
  }  
}
```

A.134 JSON_UserInfoOutList

JSON message about added person information list

```
{  
  "UserInfoOutList": {  
    "UserInfoOut": [{  
      "employeeNo": "",  
      /*string, employee ID*/  
      "errorCode": ,  
      "errorMsg": "",  
      "statusCode": ,  
      "statusString": "",  
      "subStatusCode": ""  
    }],  
    {  
      "employeeNo": "",
```

```
"errorCode": ,
"errorMsg": "",
"statusCode": ,
"statusString": "",
"subStatusCode": ""
}}
}
}
```

A.135 JSON_UserInfoSearch

JSON message about results of searching for person details

```
{
  "UserInfoSearch" : {
    "searchID": "",
    /*required, string, search ID, which is used to check whether the current search requester is the same as the previous
    one; if they are the same, the search record will be stored in the device to speed up the next search*/
    "responseStatusStrg": "OK",
    /*required, string, search status description: "OK"-search ended, "MORE"-searching, "NO MATCH"-no matched data
    found*/
    "numOfMatches": 1,
    /*required, integer32, number of records returned for one time*/
    "totalMatches": 1,
    /*required, integer32, number of matched records returned for one time*/
    "UserInfo" : [{
    /*optional, person information*/
      "employeeNo": "test",
      /*required, string, employee ID (person ID)*/
      "name": "test",
      /*optional, string, person name*/
      "userType": "normal",
      /*optional, string, person type: "normal" (resident), "visitor" (visitor), "blackList" (person in blocklist), "patient"
      (patient), "maintenance" (including the cleaning staff and maintenance staff). The maintenance staff can enter the
      room anytime*/
      "closeDelayEnabled": true,
      /*optional, boolean, whether to delay closing the door*/
      "Valid" : {
      /*required, effective period, the earliest date and time can be "1970-01-01T00:00:00", and the latest date and time
      can be "2037-12-31T23:59:59"*/
        "enable": ,
        /*required, boolean, whether to enable effective period: false (always be effective), true*/
        "beginTime": "",
        /*required, start time of the effective period, e.g., for local time, it is "2017-08-01T17:30:08"; for UTC time, it is
        "2017-08-01T17:30:08+08:00"*/
        "endTime": "",
        /*required, start time of the effective period, e.g., for local time, it is "2017-08-01T17:30:08"; for UTC time, it is
        "2017-08-01T17:30:08+08:00"*/
        "timeType": ""
        /*optional, string, time type: "local"-device local time, "UTC"-UTC time*/
      }
    }
  }
}
```



```
}
  "belongGroup": "1,3,5",
  /*optional, string, group, to which the person belongs*/
  "password": "123456",
  /*optional, string, password*/
  "doorRight": "1,3",
  /*optional, string, No. of door or lock that has access permission, e.g., "1,3" indicates having permission to access
  door (lock) No. 1 and No. 3*/
  "RightPlan": [{
  /*optional, array, access permission schedule of the door or lock*/
    "doorNo": 1,
  /*optional, int, door No. (lock ID)*/
    "planTemplateNo": "1,3,5"
  /*optional, string, schedule template No.*/
  }],
  "maxOpenDoorTime": 0,
  /*optional, int, the maximum authentication attempts*/
  "openDoorTime": 0,
  /*optional, int, authenticated attempts*/
  "roomNumber": 123,
  /*optional, int, room number*/
  "floorNumber": 1,
  /*optional, int, floor number*/
  "doubleLockRight": true,
  /*optional, boolean, whether the person has the permission to open the double-locked door: true (yes), false (no)*/
  "localUIRight": true,
  /*optional, boolean, whether the person has the permission to access the device local UI: true (yes), false (no)*/
  "userVerifyMode": "card",
  /*optional, string, person authentication mode: "cardAndPw" (card and password), "card" (card), "cardOrPw" (card or
  password), "fp" (fingerprint), "fpAndPw" (fingerprint and password), "fpOrCard" (fingerprint or card), "fpAndCard"
  (fingerprint and card), "fpAndCardAndPw" (fingerprint, card, and password), "faceOrFpOrCardOrPw" (face or
  fingerprint or card or password), "faceAndFp" (face and fingerprint), "faceAndPw" (face and password), "faceAndCard"
  (face and card), "face" (face), "employeeNoAndPw" (employee No. and password), "fpOrPw" (fingerprint or
  password), "employeeNoAndFp" (employee No. and fingerprint), "employeeNoAndFpAndPw" (employee No.,
  fingerprint, and password), "faceAndFpAndCard" (face, fingerprint, and card), "faceAndPwAndFp" (face, password,
  and fingerprint), "employeeNoAndFace" (employee No. and face), "faceOrfaceAndCard" (face or face + card),
  "fpOrface" (fingerprint or face), "cardOrfaceOrPw" (card or face or password), "cardOrFace" (card or face),
  "cardOrFaceOrFp" (card or face or fingerprint), "cardOrFpOrPw" (card or fingerprint or password). The priority of the
  person authentication mode is higher than that of the card reader authentication mode*/
  "dynamicCode": "123456",
  /*optional, string, dynamic permission code*/
  "callNumbers": ["1-1-1-401"],
  /*optional, array, list of called numbers, the default rule is "X-X-X-X", e.g., "1-1-1-401". This node is the extension of
  the node roomNumber. When both roomNumber and callNumbers exist, the callNumbers has higher priority*/
  "floorNumbers": [1, 2],
  /*optional, array, floor No. list. This node is the extension of floorNumber. When both floorNumber and
  floorNumbers exist, the floorNumbers has higher priority. The objects of floorNumbers should be corresponding to
  that of callNumbers*/
  "numOfFace": 0,
  /*optional, int, number of linked face pictures. If it is not returned, it indicates that this function is not supported*/
  "numOffP": 0,
  /*optional, int, number of linked fingerprints. If it is not returned, it indicates that this function is not supported*/
```

```
"numOfCard": 0,
/*optional, int, number of linked cards. If it is not returned, it indicates that this function is not supported*/
"numOfIris": 0,
/*optional, int, number of linked iris. If it is not returned, it indicates that this function is not supported*/
"gender": "male",
/*optional, string, gender of the person in the face picture: "male", "female", "unknown"*/
"PersonInfoExtends": [{
/*optional, array, extended person information*/
  "id": 1,
/*optional, int, ID of extended person information; range: [1,32]*/
  "value": "test"
/*optional, string, extended content of the person information*/
}],
"groupName": "test",
/*optional, string, group name; length range: [1,64]*/
"age": 0,
/*optional, int, age, value range: [0,120]*/
"PatientInfos": {
/*optional, object, the patient information, it is valid and required when the value of userType is "patient"*/
  "deviceId": "test",
/*optional, string, device ID*/
  "admissionTime": "1970-01-01T00:00:00+08:00",
/*optional, datetime, hospitalized date*/
  "chargeNurse": "test",
/*optional, string, nurse in charge; length range: [0,32]*/
  "chargeDoctor": "test",
/*optional, string, doctor in charge; length range: [0,32]*/
  "nursingLevel": "tertiary",
/*optional, string, nursing level: "unknow" (unknown), "tertiary", "secondary", "primary", "special"*/
  "doctorsAdvice": "test",
/*optional, string, advice from doctor; length range: [0,128]*/
  "allergicHistory": "test"
/*optional, string, allergy, length range:[0,128]*/
},
"TromboneRule": {
/*optional, object, trombone rule*/
  "industryType": "buildings",
/*optional, string, industry/scene type: "buildings", "prison", "medicalTreatment" (medical treatment),
"broadcasting"*/
  "unitType": "indoor",
/*optional, string, device type: "indoor" (indoor unit), "villa" (villa door station), "confirm" (doorphone), "outdoor"
(outdoor unit), "fence" (outer door station), "doorbell", "manage" (main station), "acs" (access control device),
"wardStation" (ward extension), "bedheadExtension" (bedhead extension), "bedsideExtension" (bedbeside extension),
"terminal", "netAudio" (network speaker), "interactive", "amplifier" (speaker)*/
  "SIPVersion": "V10"
/*optional, string, private SIP version; length range: [0,32]*/
},
"ESDType": "handAndFoot"
/*optional, string, ESD detection type: "handAndFoot" (detect both hand and foot), "no" (no detection), "hand"
(detect hand), "foot" (detect foot)*/
}]
```

```
}  
}
```

A.136 JSON_UserInfoSearchCond

JSON message about person details search condition

```
{  
  "UserInfoSearchCond": {  
    "searchID": "",  
    /*required, string, search ID, which is used to check whether the current search requester is the same as the previous  
    one; if they are the same, the search record will be stored in the device to speed up the next search*/  
    "searchResultPosition": 0,  
    /*required, integer32, the end position of search result in result list; in a single search, if you cannot get all the records  
    in the result list, you can mark the end position and get the following records after the marked position in the next  
    search*/  
    "maxResults": 30,  
    /*required, integer32, the maximum number of search results, which is defined by the device capability, will be  
    returned if the value of maxResults reaches the limit; in this case, the device will not return error*/  
    "EmployeeNoList" : [{  
    /*optional, person ID list; if this node does not exist or it is not configured, it indicates that all persons will be searched  
    for*/  
      "employeeNo": ""  
    }]  
  }  
}
```

A.137 JSON_UserRightHolidayGroupCfg

JSON message about parameters of the person permission holiday group

```
{  
  "UserRightHolidayGroupCfg": {  
    /*required, object, parameters of person permission holiday group*/  
    "enable": true,  
    /*required, boolean, whether to enable: true (enable), false (disable)*/  
    "groupName": "test",  
    /*required, string, holiday group name*/  
    "holidayPlanNo": "1,3,5"  
    /*required, string, holiday group schedule number; the node can be empty*/  
  }  
}
```

A.138 JSON_UserRightHolidayPlanCfg

JSON message about person permission holiday schedule

```
{
  "UserRightHolidayPlanCfg": {
    /*required, object, parameters of person permission holiday schedule*/
    "enable": true,
    /*required, boolean, whether to enable the holiday schedule: true (enable), false (disable)*/
    "beginDate": "1970-01-01",
    /*required, date, holiday start time, it is the device local time*/
    "endDate": "1970-01-01",
    /*required, date, holiday end time, it is the device local time*/
    "HolidayPlanCfg": [
      /*optional, array, holiday schedule parameters*/
      {
        "id": 1,
        /*required, int, time period number, value range: [1,8]*/
        "enable": true,
        /*required, boolean, whether to enable: true (enable), false (disable)*/
        "TimeSegment": {
          /*required, object, time period*/
          "beginTime": "00:00:00",
          /*required, string, start time, it is the device local time*/
          "endTime": "00:00:00"
          /*required, string, end time, it is the device local time*/
        }
      }
    ]
  }
}
```

A.139 JSON_UserRightPlanTemplate

JSON message about person permission schedule template

```
{
  "UserRightPlanTemplate": {
    /*required, object, parameters of person permission schedule template*/
    "enable": true,
    /*required, boolean, whether to enable: true (enable), false (disable)*/
    "templateName": "test",
    /*required, string, template name*/
    "weekPlanNo": 1,
    /*required, int, week schedule number*/
    "holidayGroupNo": "1,3,5"
    /*required, string, holiday group number; this filed can be empty*/
  }
}
```

```
}  
}
```

A.140 JSON_UserRightWeekPlanCfg

JSON message about person permission week schedule

```
{  
  "UserRightWeekPlanCfg": {  
    /*optional, object, parameters of person permission week schedule*/  
    "enable": true,  
    /*required, boolean, whether to enable the week schedule: true (enable), false (disable)*/  
    "WeekPlanCfg": [  
      /*optional, array, week schedule parameters*/  
      {  
        "week": "Monday",  
        /*required, string, the day of week: "Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"*/  
        "id": 1,  
        /*required, int, time period number, value range: [1,8]*/  
        "enable": true,  
        /*required, boolean, whether to enable: true (enable), false (disable)*/  
        "TimeSegment": {  
          /*optional, object, time period*/  
          "beginTime": "10:10:00",  
          /*required, string, start time, it is the device local time*/  
          "endTime": "12:10:00"  
          /*required, string, end time, it is the device local time*/  
        }  
      }  
    ]  
  }  
}
```

A.141 JSON_VideoInputChannel

VideoInputChannel message in JSON format

```
{  
  "VideoInputChannel":{  
    "id": "",  
    /*required, string, camera No.*/  
    "inputPort": "",  
    /*required, string, port number*/  
    "videoInputEnabled": ,  
    /*optional, boolean, whether to enable video input, "true"-enable, "false"-disable*/  
    "videoFormat": "",  
    /*optional, string, video standard: "PAL,NTSC"*/  
    "name": "",  
    /*optional, string, camera name*/  
  }  
}
```

```
"portType":""
/*optional, string, port type: "SDI", "OPT", "VGA", "HDMI", "YPbPr"*/
}
}
```

A.142 JSON_VideoInputChannelLst

VideoInputChannelLst message in JSON format

```
{
  "VideoInputChannelList":[{
    "VideoInputChannel":{
      "id": "",
      /*required, string, camera No.*/
      "inputPort": "",
      /*required, port number, string*/
      "videoInputEnabled": ,
      /*optional, boolean, whether to enable video input, "true"-enable, "false"-disable*/
      "videoFormat": "",
      /*optional, video standard, string, opt="PAL,NTSC"*/
      "name": "",
      /*optional, camera name, string*/
      "portType": ""
      /*optional, port type, string, opt="SDI, OPT, VGA, HDMI, YPbPr"*/
    }
  ]
}
```

A.143 JSON_videolossScheduleList

JSON message about video loss arming schedules

```
{
  "videolossScheduleList":[{
    "Schedule":{
      "id": "",
      /*required, string, video loss ID*/
      "eventType": "",
      /*optional, string, event type*/
      "inputIOPortID": ,
      /*optional, depends on eventType, int, alarm input No.*/
      "outputIOPortID": ,
      /*optional, depends on eventType, int, alarm output No.*/
      "videoInputChannelID": ,
      /*optional, depends on eventType, int, video input channel No.*/
      "TimeBlockList":[
      /*required, size="8"*/
      {
        "TimeBlock":{
```

```
    "dayOfWeek": ,
    /*optional, int, days of the week based on ISO8601: 1 (Monday)...*/
    "TimeRange":{
    /*required*/
        "beginTime": "",
    /*required, start time based on ISO8601*/
        "endTime": ""
    /*required, end time based on ISO8601*/
    }
    },.....
    ],
    "HolidayBlockList":[
    /*optional*/
    {
        "TimeBlock":{
            "TimeRange":{
    /*required, */
                "beginTime": "",
    /*required, start time based on ISO8601*/
                "endTime": ""
    /*required, end time based on ISO8601*/
            }
        }
    },.....
    ]
    }
    }}
}
```

A.144 JSON_VideoOverlay

JSON message about OSD parameters

```
{
  "VideoOverlay": {
    "normalizedScreenSize": {
    /*required*/
      "normalizedScreenWidth": 704,
    /*required, read only, int, standard screen width*/
      "normalizedScreenHeight": 576
    /*required, read only, int, standard screen height*/
    },
    "attribute": {
    /*optional*/
      "transparent": ,
    /*required, boolean, whether to enable transparent: false (disable, default), true (enable) */
      "flashing":
    /*required, boolean, whether to enable flashing: false (disable, default), true (enable) */
    },
  },
}
```

```
"fontSize": "",
/*optional, string, font size, opt="adaptive,16*16,32*32,48*48,64*64"*/
"frontColorMode": "",
/*optional, string, font color mode, opt="auto,customize"*/
"alignment": "",
/*optional, string, alignment mode, opt="customize,alignRight,alignLeft"*/
"TextOverlayList": [{
/*this node is optional when configuring OSD parameters; when switching to GB mode, this node will be automatically
obtained, a backup will be saved by device*/
  "TextOverlay": {
    "id": "",
/*required, string, text No.*/
    "enabled": ,
/*required, boolean, whether to enable text overlay*/
    "displayText": "",
/*required, string, text details*/
    "positionX": ,
/*required, float, X-coordinate of display position*/
    "positionY":
/*required, float, Y-coordinate of display position*/
  }
},...],
"DateTimeOverlay": {
  "enabled": ,
/*required, boolean, whether to enable time overlay, true-enable (default), false-disable*/
  "positionY": ,
/*required, int, y-coordinate of display position*/
  "positionX": ,
/*required, int, x-coordinate of display position*/
  "dateStyle": "",
/*optional, string, date format, opt="YYYY-MM-DD, MM-DD-YYYY, DD-MM-YYYY, CHR-YYYY-MM-DD, CHR-MM-DD-YYYY,
CHR-DD-MM-YYYY, CHR-YYYY/MM/DD, CHR-MM/DD/YYYY, CHR-DD/MM/YYYY"*/
  "timeStyle": "",
/*optional, string, time format, opt="12hour, 24hour"*/
  "displayWeek":
/*optional, boolean, whether to display week, true-display*/
},
"channelNameOverlay": {
  "enabled": ,
/*required, boolean, whether to enable channel name overlay, true-enable (default), false-disable*/
  "positionY": ,
/*required, int, y-coordinate of display position*/
  "positionX":
/*required, int, x-coordinate of display position*/
},
}
}
```


Appendix B. Event Types

Heartbeat

eventType	Event Type Name and Details
heartBeat	Heartbeat information, see details in <u><i>JSON_EventNotificationAlert_HeartbeatInfo</i></u>

General Events

eventType	Event Type Name and Details
devStatusChanged	Device status changed alarm, see details in <u><i>JSON_EventNotificationAlert_devStatusChanged</i></u> .
devSleep	Device sleep alarm, see details in <u><i>JSON_EventNotificationAlert_devSleep</i></u> .
IO	IO alarm, see details in <u><i>JSON_EventNotificationAlert_IO</i></u> .
VMD	Motion detection alarm, see details in <u><i>JSON_EventNotificationAlert_VMD</i></u> .
videoloss	Video loss alarm, see details in <u><i>JSON_EventNotificationAlert_videoloss</i></u> .
shelteralarm	Video tampering alarm, see details in <u><i>JSON_EventNotificationAlert_shelteralarm</i></u> .
PALMismatch	Video standard mismatch alarm, see details in <u><i>JSON_EventNotificationAlert_PALMismatch</i></u> .
diskfull	HDD full alarm, see details in <u><i>JSON_EventNotificationAlert_diskfull</i></u> .
diskerror	HDD error alarm, see details in <u><i>JSON_EventNotificationAlert_diskerror</i></u> .
PIR	Passive infrared alarm, see details in <u><i>JSON_EventNotificationAlert_PIR</i></u> .
batteryStatus	Battery status alarm, see details in <u><i>JSON_EventNotificationAlert_batteryStatus</i></u> .
addressChange	Device address changed alarm, see details in <u><i>JSON_EventNotificationAlert_addressChange</i></u> .

Smart Events

eventType	Event Type Name and Details
scenechangedetection	Scene changed alarm, see details in <i><u>JSON_EventNotificationAlert_scenechangedetection</u></i> .
facedetection	Face detection alarm, see details in <i><u>JSON_EventNotificationAlert_facedetection</u></i> .
fielddetection	Intrusion alarm, see details in <i><u>JSON_EventNotificationAlert_fielddetection</u></i> .
linedetection	Line crossing alarm, see details in <i><u>JSON_EventNotificationAlert_linedetection</u></i> .
regionEntrance	Region entrance alarm, see details in <i><u>JSON_EventNotificationAlert_regionEntrance</u></i> .
regionExiting	Region exiting alarm, see details in <i><u>JSON_EventNotificationAlert_regionExiting</u></i> .
unattendedBaggage	Unattended baggage alarm, see details in <i><u>JSON_EventNotificationAlert_unattendedBaggage</u></i> .
attendedBaggage	Object removal alarm, see details in <i><u>JSON_EventNotificationAlert_attendedBaggage</u></i> .
loitering	Loitering alarm, see details in <i><u>JSON_EventNotificationAlert_loitering</u></i> .
group	People gathering alarm, see details in <i><u>JSON_EventNotificationAlert_group</u></i> .
parking	Parking alarm, see details in <i><u>JSON_EventNotificationAlert_parking</u></i> .
rapidMove	Fast moving alarm, see details in <i><u>JSON_EventNotificationAlert_rapidMove</u></i> .
defocus	Defocus alarm, see details in <i><u>JSON_EventNotificationAlert_defocus</u></i> .
audioexception	Audio exception alarm, see details in <i><u>JSON_EventNotificationAlert_audioexception</u></i> .

Mobile Device Events

eventType	Event Type Name and Details
overSpeed	Overspeed alarm, see details in <i>JSON_EventNotificationAlert_OverspeedAlarmMsg</i> .
highTempAlarm	Temperature too high alarm, see details in <i>JSON_EventNotificationAlert_highTempAlarm</i> .
abnormalAcceleration	Acceleration exception alarm, see details in <i>JSON_EventNotificationAlert_abnormalAcceleration</i> .
collision	Collision detection alarm, see details in <i>JSON_EventNotificationAlert_collision</i> .
rollover	Rollover detection alarm, see details in <i>JSON_EventNotificationAlert_rollover</i> .
abnormalDriving	Abnormal driving behavior detection alarm, see details in <i>JSON_EventNotificationAlert_abnormalDriving</i> .
GPSUpload	GPS information uploading event, see details in <i>JSON_EventNotificationAlert_GPSUpload</i> .
mobilePeopleCounting	Mobile people counting event, see details in <i>JSON_EventNotificationAlert_mobilePeopleCounting</i> .
flowOverrun	Traffic flow over-limit alarm, see details in <i>JSON_EventNotificationAlert_flowOverrun</i> .
ADAS	Driver assistance alarm, see details in <i>JSON_EventNotificationAlert_ADAS</i> .
faceCapture	Face capture alarm, see details in <i>JSON_EventNotificationAlert_faceCapture</i> .
peopleCounting	People counting data uploading, see details in <i>JSON_EventNotificationAlert_peopleCounting</i> .

Access Control Event

eventType	Event Type Name and Details
AccessControllerEvent	Access Control Event, see details in <i>JSON_EventNotificationAlert_AccessControlEventMsg</i> .

Device Transmitted Event

Events reported by ISUP5.0 devices, which are not mentioned in above sections can be transmitted via the Device Gateway. Events in JSON format reported by ISUP5.0 devices can be transmitted directly.

Note

The event transmission is supported by ISUP5.0 devices only.

eventType	Event Type Name and Details
devXmlEvent	Transmitted event in XML format, see <u>JSON_EventNotificationAlert_devXmlEvent</u> for details.

Appendix C. Major & Minor Types of Access Control Event

The access control events are classified as four major types, i.e., device alarms, device exceptions, device operations, and device events. Each major type corresponds to multiple minor types.

1-Device Alarm

Minor Type Value	Minor Type Name
1024	Zone Short Circuit Attempts Alarm
1025	Zone Disconnected Alarm
1026	Zone Exception Alarm
1027	Zone Restored
1028	Zone Tampering Alarm
1029	Zone Tampering Restored
1030	Card Reader Tampering Alarm
1031	Card Reader Tampering Restored
1032	Alarm Input Alarm Triggered
1033	Alarm Input Restored
1034	Duress Alarm
1035	No Memory for Offline Event Storage Alarm
1036	Maximum Failed Card Authentications Alarm
1037	SD Card Full Alarm
1038	Capture Linkage Alarm
1039	Secure Door Control Unit Tampering Alarm
1040	Secure Door Control Unit Tampering Restored

2-Device Exception

Minor Type Value	Minor Type Name
39	Network Disconnected
58	RS485 Connection Exception
59	RS485 Connection Restored
1024	Power on
1025	Power off
1026	Watchdog Reset
1027	Low Battery Voltage
1028	Battery Voltage Restored
1029	AC Power Disconnected
1030	AC Power Restored
1031	Network Restored
1032	Flash Reading and Writing Exception
1033	Card Reader Offline
1034	Card Reader Online
1035	Indicator Turns off
1036	Indicator Resumed
1037	Lane Controller Offline
1038	Lane Controller Online
1039	Secure Door Control Unit Offline
1040	Secure Door Control Unit Online
1053	ID Card Reader Disconnected
1054	ID Card Reader Connected
1055	Fingerprint Module Disconnected
1056	Fingerprint Module Connected
1057	Camera Disconnected
1058	Camera Connected
1059	COM Port Disconnected

Minor Type Value	Minor Type Name
1060	COM Port Connected
1061	Device Unauthorized
1062	Face Recognition Terminal Online
1063	Face Recognition Terminal Offline
1064	Local Login Lock
1065	Local Login Unlock

3-Device Operation

Minor Type Value	Minor Type Name
80	Local Login
90	Local Upgrade
112	Remote Login
113	Remote Logout
118	Remote Operation: Get Parameters
119	Remote Operation: Set Parameters
120	Remote Operation: Get Status
121	Remote Arming
122	Remote Disarming
123	Remote Reboot
126	Remote Upgrade
134	Remote Operation: Export Configuration File
135	Remote Operation: Import Configuration File
214	Remote Operation: Enable Alarm Output Manually
215	Remote Operation: Disable Alarm Output Manually
1024	Door Remotely Open
1025	Door Remotely Closed
1026	Remain Open Remotely
1027	Remain Closed Remotely

Minor Type Value	Minor Type Name
1028	Remote: Manual Time Sync
1029	Network Time Protocol Synchronization
1030	Remote Operation: Clear All Card No.
1031	Remote Operation: Restore Defaults
1032	Zone Arming
1033	Zone Disarming
1034	Local Operation: Restore Defaults
1035	Remote Operation: Capture
1036	Edit Network Parameters
1037	Edit GPRS Parameters
1038	Edit Control Center Parameters
1039	Enter Dismiss Code
1049	Remotely Arming
1050	Remotely Disarming
1055	M1 Card Encryption Verification Enabled
1056	M1 Card Encryption Verification Disabled
1057	Opening Door with NFC Card Enabled
1058	Opening Door with NFC Card Disabled

5-Device Event

Minor Type Value	Minor Type Name
1	Valid Card Authentication Completed
2	Card and Password Authentication Completed
3	Card and Password Authentication Failed
4	Card and Password Authentication Timed Out
5	Card and Password Authentication Timed Out
6	No Permission
7	Invalid Card Swiping Time Period

Minor Type Value	Minor Type Name
8	Expired Card
9	Card No. Not Exist
10	Anti-passing Back Authentication Failed
11	Interlocking Door Not Closed
12	Card Not in Multiple Authentication Group
13	Card Not in Multiple Authentication Duration
14	Multiple Authentications: Super Password Authentication Failed
15	Multiple Authentication Completed
16	Multiple Authenticated
17	Open Door with First Card Started
18	Open Door with First Card Stopped
19	Remain Open Started
20	Remain Open Stopped
21	Door Unlocked
22	Door Locked
23	Exit Button Pressed
24	Exit Button Released
25	Door Open (Contact)
26	Door Closed (Contact)
27	Door Abnormally Open (Contact)
28	Door Open Timed Out (Contact)
29	Alarm Output Enabled
30	Alarm Output Disabled
31	Remain Closed Started
32	Remain Closed Stopped
33	Multiple Authentications: Remotely Open Door
34	Multiple Authentications: Super Password Authentication Completed

Minor Type Value	Minor Type Name
35	Multiple Authentications: Repeated Authentication
36	Multiple Authentications Timed Out
37	Doorbell Ring
38	Fingerprint Matched
39	Fingerprint Mismatched
40	Card and Fingerprint Authentication Completed
41	Card and Fingerprint Authentication Failed
42	Card and Fingerprint Authentication Timed Out
43	Card and Fingerprint and Password Authentication Completed
44	Card and Fingerprint and Password Authentication Failed
45	Card and Fingerprint and Password Authentication Timed Out
46	Fingerprint and Password Authentication Completed
47	Fingerprint and Password Authentication Failed
48	Fingerprint and Password Authentication Timed Out
49	Fingerprint Not Exists
50	Card Platform Authentication
51	Call Center
54	Face and Fingerprint Authentication Completed
55	Face and Fingerprint Authentication Failed
56	Face and Fingerprint Authentication Timed Out
57	Face and Password Authentication Completed
58	Face and Password Authentication Failed
59	Face and Password Authentication Timed Out
60	Face and Card Authentication Completed
61	Face and Card Authentication Failed
62	Face and Card Authentication Timed Out
63	Face and Password and Fingerprint Authentication Completed
64	Face and Password and Fingerprint Authentication Failed

Minor Type Value	Minor Type Name
65	Face and Password and Fingerprint Authentication Timed Out
66	Face and Card and Fingerprint Authentication Completed
67	Face and Card and Fingerprint Authentication Failed
68	Face and Card and Fingerprint Authentication Timed Out
69	Employee ID and Fingerprint Authentication Completed
70	Employee ID and Fingerprint Authentication Failed
71	Employee ID and Fingerprint Authentication Timed Out
72	Employee ID and Fingerprint and Password Authentication Completed
73	Employee ID and Fingerprint and Password Authentication Failed
74	Employee ID and Fingerprint and Password Authentication Timed Out
75	Face Authentication Completed
76	Face Authentication Failed
77	Employee ID and Face Authentication Completed
78	Employee ID and Face Authentication Failed
79	Employee ID and Face Authentication Timed Out
80	Face Recognition Failed
81	First Card Authorization Started
82	First Card Authorization Ended
92	Unlocking Exception
93	Unlocking Timed Out
101	Employee ID and Password Authentication Completed
102	Employee ID and Password Authentication Failed
103	Employee ID and Password Authentication Timed Out
104	Face Anti-Spoofing Detection Failed
105	Face and ID Card Authenticated
112	Face and ID Card Authentication Failed

Minor Type Value	Minor Type Name
113	Blocklist Event
114	Valid Message
115	Invalid Message
116	Mac Detection
119	M1 Card Encryption Verification Failed
142	Local Face Modeling Failed
148	Password Authentication Failed Times Exceeded
151	Passwords Mismatched
152	Employee ID Not Exists
153	Combined Authentication Completed
154	Combined Authentication Timed Out
155	Authentication Type Mismatched
156	Authenticated via QR Code
157	Authentication via OQ Code Failed
158	Authenticated via Householder
159	Authenticated via Bluetooth
160	Authentication via Bluetooth Failed
162	Authentication Failed: Invalid M1 Card
163	Verifying CPU Card Encryption Failed
164	Disabling NFC Verification Failed
168	EM Card Recognition Disabled
169	M1 Card Recognition Disabled
170	CPU Card Recognition Disabled
171	ID Card Recognition Disabled
172	Card Password Filling Failed
173	Local Upgrade Failed
174	Remote Upgrade Failed
175	Remote Upgrade of Extension Module Completed

Minor Type Value	Minor Type Name
176	Remote Upgrade of Extension Module Failed
177	Remote Upgrade of Fingerprint Module Completed
178	Remote Upgrade of Fingerprint Module Failed
179	Dynamic Verification Code Verified
180	Dynamic Verification Code Failed
181	Password Verified
182	Consumption Timed Out
183	Error Correction Timed Out
184	The Maximum Sum of Consumption Reached
185	The Maximum Times of Consumption Reached
186	Consumption Confirmation Timed Out
187	The Maximum Number of Blocklists Reached
188	Desfire Card Encryption Authentication Failed
189	Desfire Card Recognition Disabled
190	Iris Verified
191	Iris Verification Failed
192	Iris Anti-Spoofing Failed
193	90% of Maximum Persons Alarm

Appendix D. Status Codes

The status classification of this SDK is based on the status codes of HTTP. 7 kinds of status codes are predefined, including 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid Message Format), 6 (Invalid Message Content), and 7 (Reboot Required). Each kind of status code contains multiple sub status codes.

StatusCode=1

SubStatusCode	Error Code	Description
ok	0x1	Operation completed.
riskPassword	0x10000002	Risky password.

StatusCode=2

SubStatusCode	Error Code	Description
noMemory	0x20000001	Insufficient memory.
upgrading	0x20000003	Upgrading.
networkError	0x20000009	Network error.

StatusCode=3

SubStatusCode	Error Code	Description
deviceError	0x30000001	Device hardware error.
createSocketError	0x30000004	Creating socket failed.
sendRequestError	0x30000006	Sending request failed.
passwordDecodeError	0x30000008	Decrypting password failed.
passwordEncryptError	0x30000009	Encrypting password failed.
pictureUploadFailed	0x3000000B	Uploading picture failed.
connecteDatabaseError	0x3000000E	Connecting to database failed.
internalError	0x30000014	Internal error.
uninitialized	0x3000000C	Uninitialized.

StatusCode=4

SubStatusCode	Error Code	Description
notSupport	0x40000001	Not supported.
lowPrivilege	0x40000002	No permission.
badAuthorization	0x40000003	Authentication failed.
methodNotAllowed	0x40000004	Invalid HTTP method.
notActivated	0x40000007	Inactivated.
hasActivated	0x40000008	Activated.
invalidContent	0x4000000A	Invalid message content.
maxSessionUserLink	0x4000000B	No more user can log in.
loginPasswordError	0x4000000C	Incorrect password.
MgmtLokedError	0x4000000D	Logging in management platform failed. IP is locked.
interfaceOperationError	0x40001002	Operation failed.
openFileError	0x40001014	Opening file failed.
taskPacking	0x40001034	The resource is already occupied.
taskNoRecFile	0x40001039	The video file does not exist.
updateLangUnmatched	0x40001042	Upgrade packet language mismatches.
userMaxNum	0x40001047	No more user can be added.
monitorNodeOverLimit	0x4000104D	No more camera can be added.
deviceExist	0x40001054	The device is already added.
pwdErrorLoginFailed	0x40001055	Login failed. Check the user name and password.
setArmingError	0x40001083	Setting arming information failed.
taskModifyFailed	0x400010B1	Editing task failed.
getDeviceInfoFailed	0x400010BC	Getting device information failed.
noDiskSpace	0x400010E6	Insufficient HDD space.
cannotSameAsOldPassword	0x400010E8	The new password and old password must be different.
originalPassError	0x400010E9	Incorrect old password.

SubStatusCode	Error Code	Description
writeFileError	0x400010EA	Writing file failed.
accessFileDirectoryFailed	0x40001104	Accessing file path failed.
unKnownErrorCode	0x4000111D	Unknown error code.
deviceVervisionNotMatch	0x40001128	Device version mismatches.
theSessionIdDoesNotExist	0x40001135	The session ID does not exist.
theCameraIdDoesNotExist	0x40001137	The camera ID does not exist.
theDeviceIdDoesNotExist	0x4000113B	The device ID does not exist.
gettingResourceNodeInformationFailed	0x40001176	Getting resource node information failed.
noMoreTasksCanBeAdded	0x4000118A	No more task can be added.
theZoneAlreadyExists	0x40001388	The zone already exist.
thePartitionAlreadyExists	0x40001389	The partition(area) already exists.
thePartitionRelatedZone	0x4000138A	The partition(area) has been linked to zone(s). Cancel the linkage to delete the partition(area).

StatusCode=5

SubStatusCode	Error Code	Description
badJsonFormat	0x50000002	Invalid JSON format.
badURLFormat	0x50000003	Invalid URL format.

StatusCode=6

SubStatusCode	Error Code	Description
badParameters	0x60000001	Incorrect parameter.
badXmlContent	0x60000003	Incorrect XML message content.
badPort	0x6000000B	Port number conflicted.
portError	0x6000000C	Invalid port number.
badVersion	0x6000000F	Version mismatches.
requestMemoryNULL	0x6000003F	No memory is requested.

SubStatusCode	Error Code	Description
tokenTimeout	0x600000040	The token timed out.
passwordLenNoMoreThan16	0x60000005F	Up to 16 characters are allowed in the password.
eventCodeExist	0x600000060	The event code already exists.
diskError	0x60001009	HDD error.

StatusCode=7

SubStatusCode	Error Code	Description
rebootRequired	0x700000001	Reboot device to take effect.



See Far, Go Further